Neural Computation

Neural Computation (extended)

Week 1:

Introduction and linear models

Module team

Edgbaston



Dr. Alex Krull
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(Module lead)

Dubai



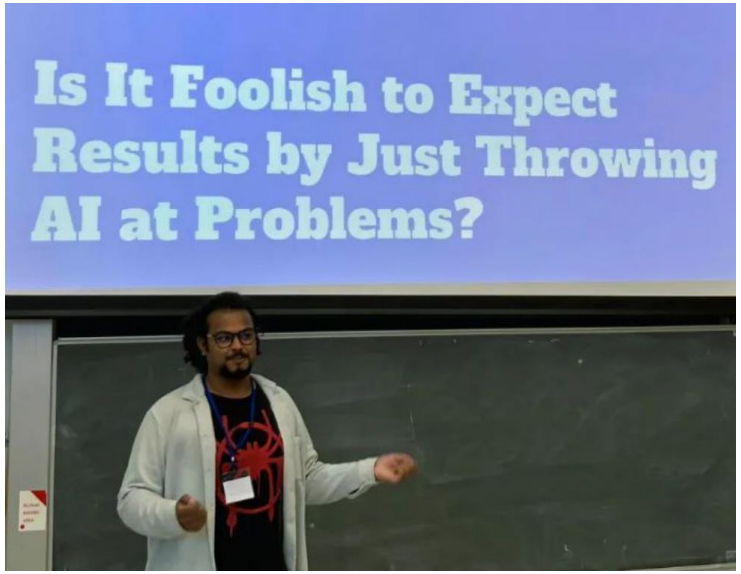
Prof. Kashif Rajpoot
k.m.rajpoot@bham.ac.uk
(Dubai module lead)



Dr. Sauleh Eetemadi
s.eetemadi@bham.ac.uk
(Dubai module lecturer)

Course TAs for Edgbaston

Edgbaston



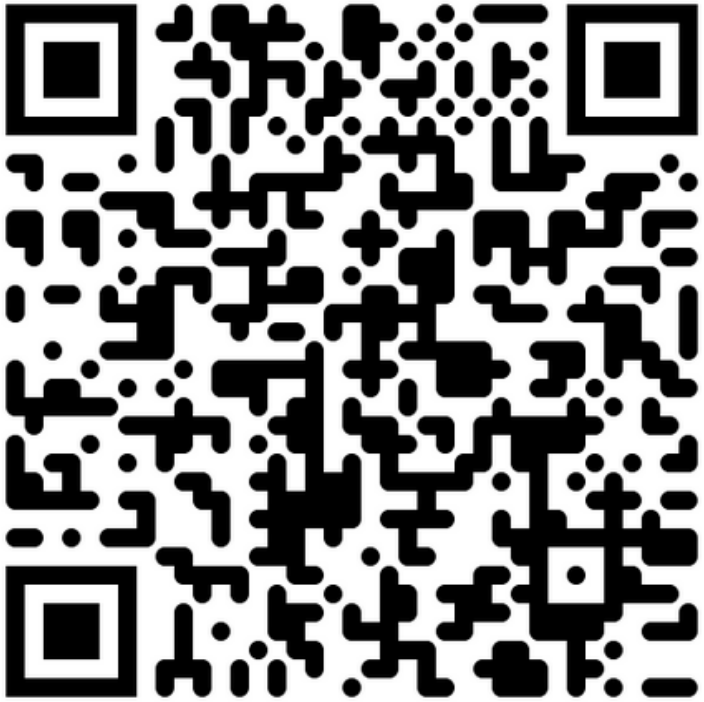
Abhinandan Pal
axp1209@student.bham.ac.uk

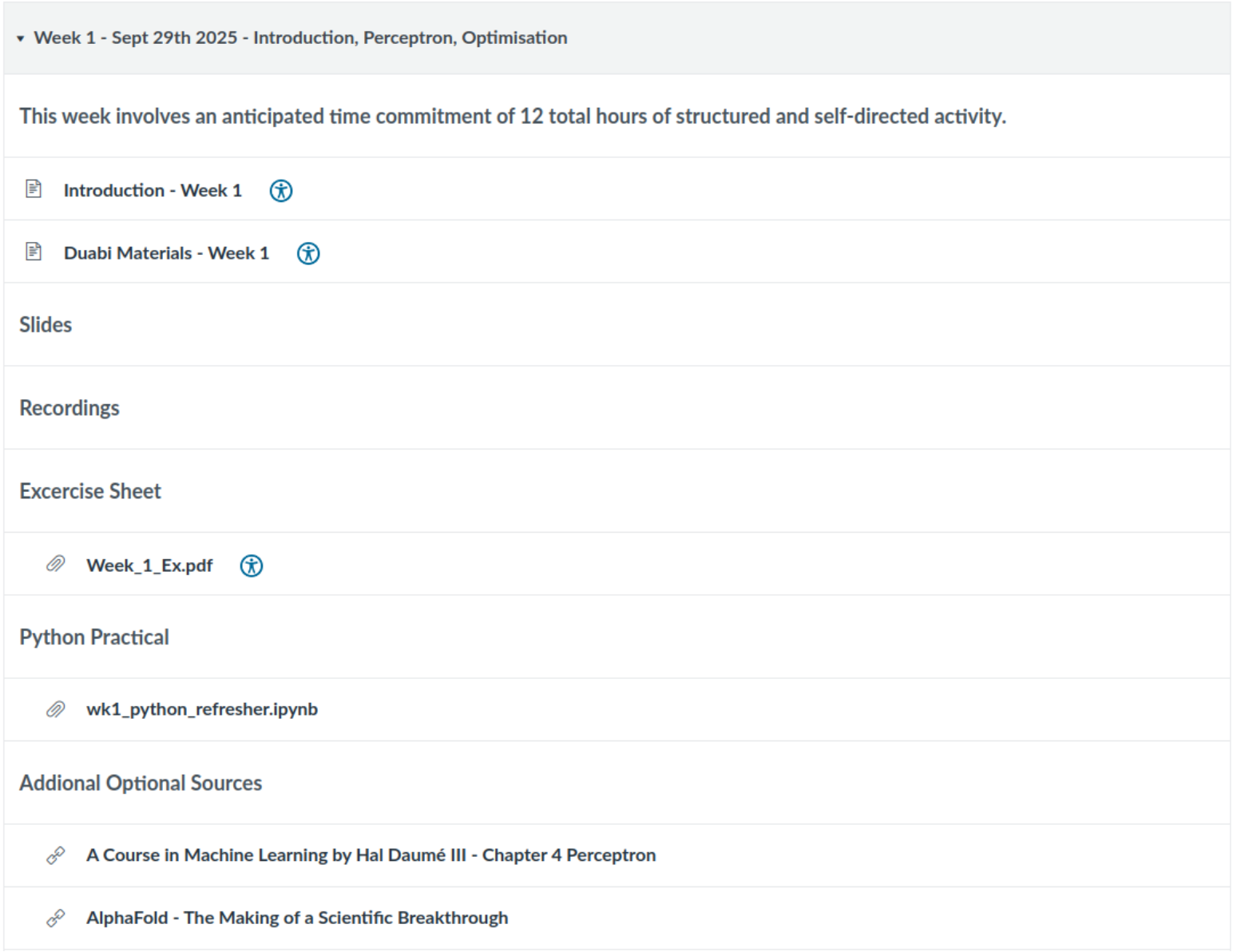


Thomas Muir
tdm554@student.bham.ac.uk

Canvas layout

- https://canvas.bham.ac.uk/courses/83781/modules





Where you can ask your questions

Teams Group:



Office Hour:

- Friday 10am-12noon
- In the In Murray Learning Centre Room 139.
- Email a.f.f.krull@bham.ac.uk to get in.

Common questions

- Any logistic questions, please send to csoffice@contacts.bham.ac.uk

Module plan (tentative)

Week	Date	Topic	Edgbaston lecturer	Dubai lecturer	CA
1	29/9/2025	Intro Perceptron Optimisation problems	Alex	Kashif	
2	6/10/2025	Learning for linear classification and linear regression SGD, BGD, ADAM	Alex	Kashif	
3	13/10/2025	MLP Backprop Underfitting/Overfitting	Alex	Kashif	
4	20/10/2025	CNN: Classification, Segmentation, Image2Image. Regularisations: Batch norm, Layer norm, instance norm, Drop out	Alex	Kashif	
5	27/10/2025	Latent Spaces: AE	Alex	Kashif	CA1 (10%)
6	3/11/2025	Self-supervised: (Contrastive learning)	Alex	Kashif	
7	10/11/2025	Gen Models: VAEs, GAN	Alex	Kashif	
8	17/11/2025	Diffusion	Alex	Kashif	
9	24/11/2025	Autoregressive Models, Sequence models: NGRAMS, RNNs	Alex	Kashif	
10	1/12/2025	Transformers, Vision Transformers	Alex	Sauleh	
11	8/12/2025	Embeddings Multimodal and modern architectures	Alex	Sauleh	CA2 (10%)
Exam May/June 2026 (80%)					

Weekly Sessions (Edgbaston)

Monday:

- Release of
  - Materials
  - Exercise sheet
  - Python practical

Can include
- Last week's material
- this week's Tuesday material

Tuesday:

- Lecture 11:00 TLB-102

Thursday:

- Lecture 14:00 in STH-205

Fridays:

- Tutorial 17:00 STH-205

You solve the exercises here.

We discuss the solutions

Programming environment

- Set up on your PCs/laptops
  - Set up a virtual environment via Anaconda (recommended)
  - Jupyter notebook:
    - Anaconda navigator
    - pip install jupyter
    - Visual Studio Code Jupyter extension (recommended)
  - Install Pytorch via pip
- Online CPU and GPU access: JupyterHub (details to be shared via Canvas and in class)
- Google Colab
  - https://colab.research.google.com/







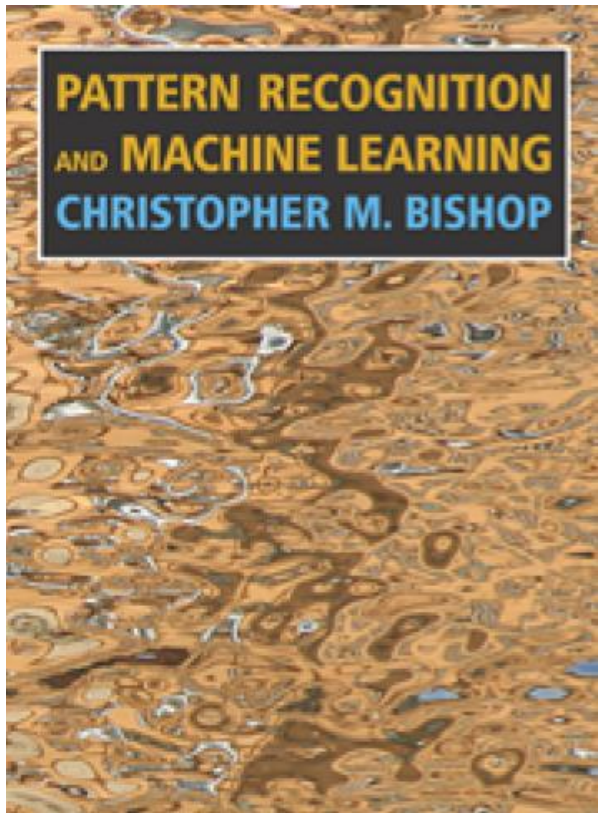


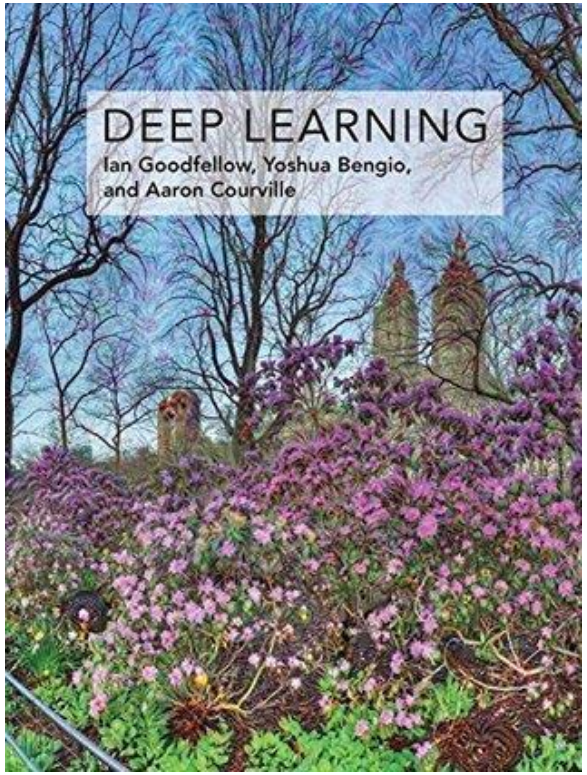




Recommended textbooks

1. Deep learning, Goodfellow et al., MIT Press, 2016 (online and in Library)
2. Pattern Recognition and Machine Learning, Bishop, Springer, 2007 (online and in Library)





Deep Learning

- Table of Contents
- Acknowledgements
- Notation
- 1 Introduction
- Part I: Applied Math and Machine Learning Basics
  - 2 Linear Algebra
  - 3 Probability and Information Theory
  - 4 Numerical Computation
  - 5 Machine Learning Basics
- Part II: Modern Practical Deep Networks
  - 6 Deep Feedforward Networks
  - 7 Regularization for Deep Learning
  - 8 Optimization for Training Deep Models
  - 9 Convolutional Networks
  - 10 Sequence Modeling: Recurrent and Recursive Nets
  - 11 Practical Methodology
  - 12 Applications
- Part III: Deep Learning Research
  - 13 Linear Factor Models
  - 14 Autoencoders
  - 15 Representation Learning
  - 16 Structured Probabilistic Models for Deep Learning
  - 17 Monte Carlo Methods
  - 18 Confronting the Partition Function
  - 19 Approximate Inference
  - 20 Deep Generative Models

Free online courses

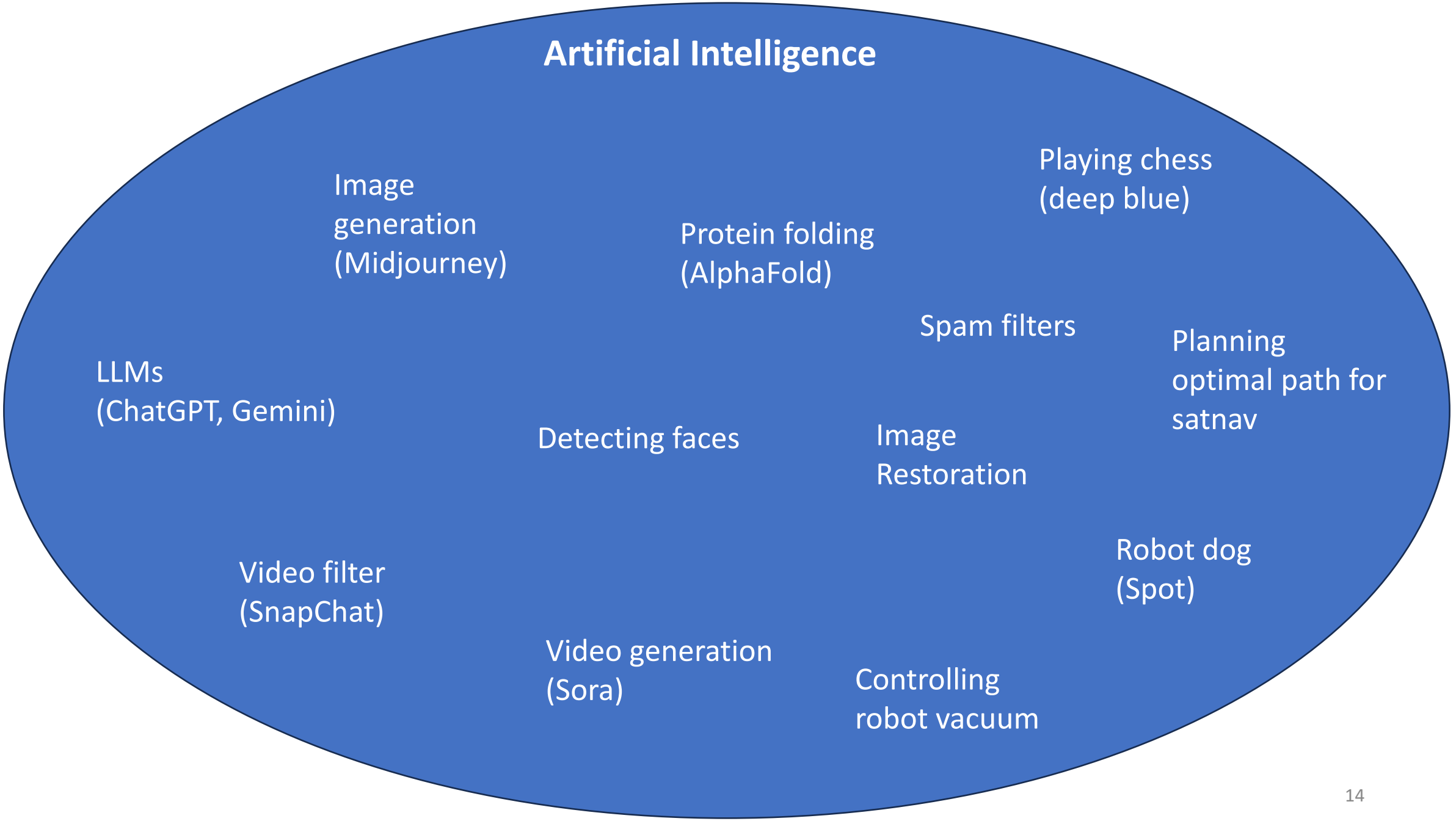
- Summer School
  - Deep Learning & Reinforcement Learning Summer School
    (http://videolectures.net/DLRLsummerschool2018_toronto/)
- Modules
  - DeepLearning.ai (Andrew Ng)
    (https://www.youtube.com/channel/UCcIXc5mJsHVYTZR1maL5l9w/playlists)
  - CS231n – Convolutional Neural Networks (Stanford)
    (https://www.youtube.com/playlist?list=PL3FW7Lu3i5JvHM8ljYj-zLfQRF3EO8sYv)
  - CS224d – Natural Language Processing (Stanford)
    (https://www.youtube.com/playlist?list=PL3FW7Lu3i5Jsnh1rnUwq_TcylNr7EkRe6)
  - Fast.ai
    (https://www.youtube.com/playlist?list=PLCdvEQLhYkYmKTKWTrH7bHtQ1CsKZaQBl)

Pre-requisite (knowledge)

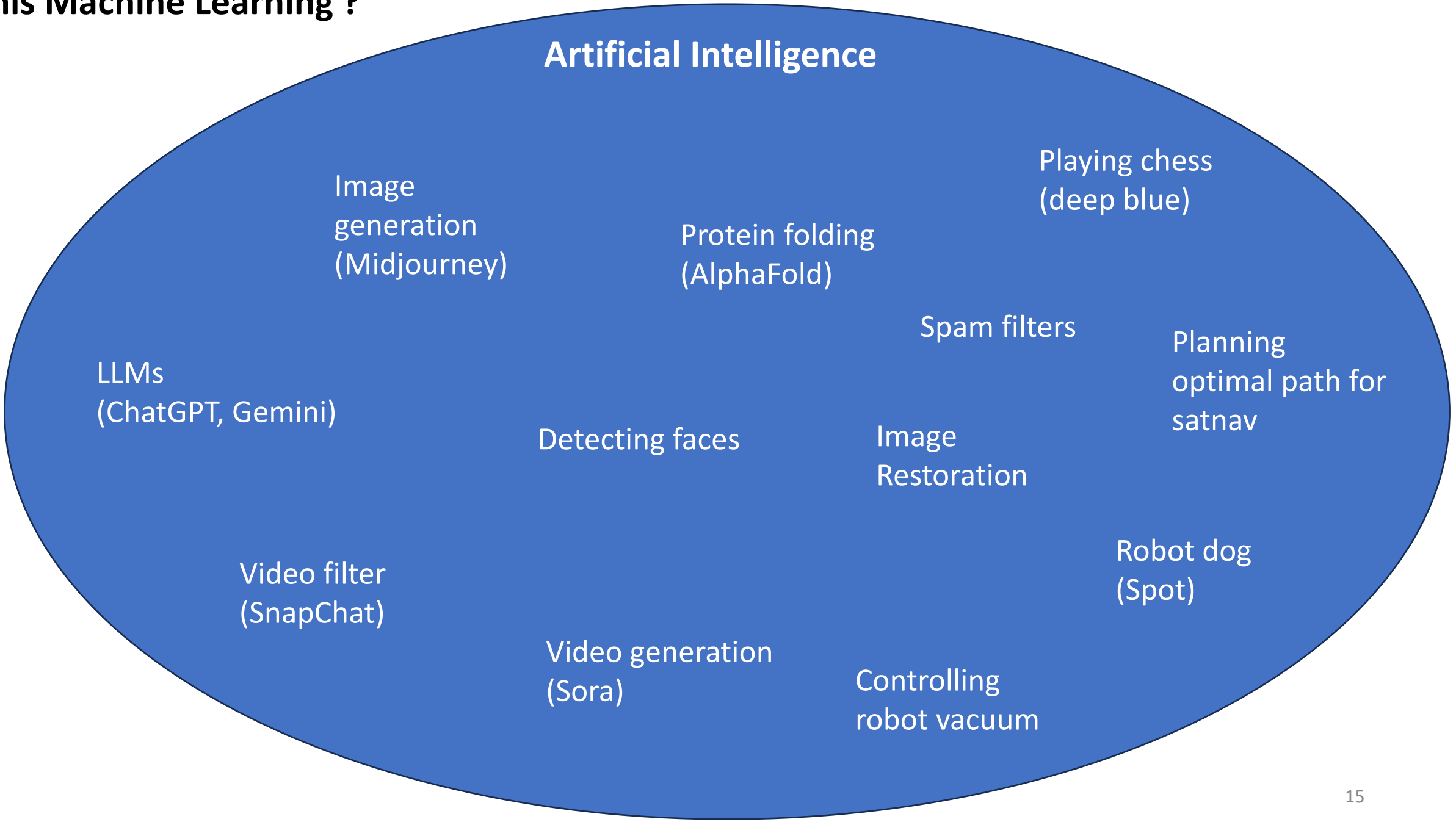
- This module requires a solid **Maths background**
  - Canvas self-learning module to refresh on Maths
    - https://canvas.bham.ac.uk/courses/60674
    - Email your lecturer to enrol, if you can't access it and need a refresher on Maths
  - Linear Algebra: vector/matrix manipulations
    - MIT Open Course Linear Algebra
  - Calculus: partial derivative, chain rule
    - MIT Open Course Multivariable Calculus
  - Probability
    - MIT Open Course Introduction to Probability
- This module expects familiarity in **Python programming**
  - and NumPy, SciPy, Matplotlib, scikit-learn, Pandas

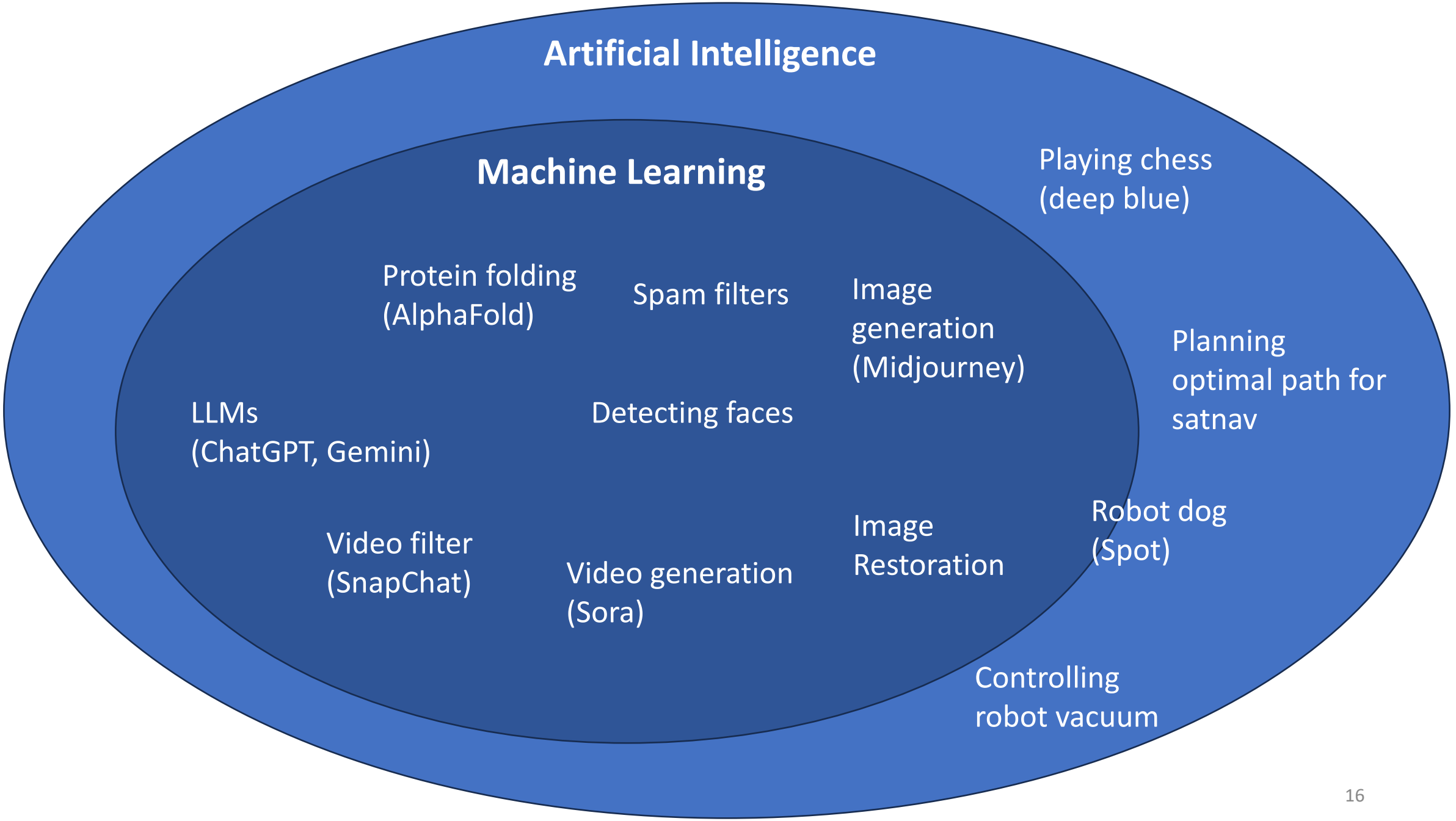
Module learning outcomes

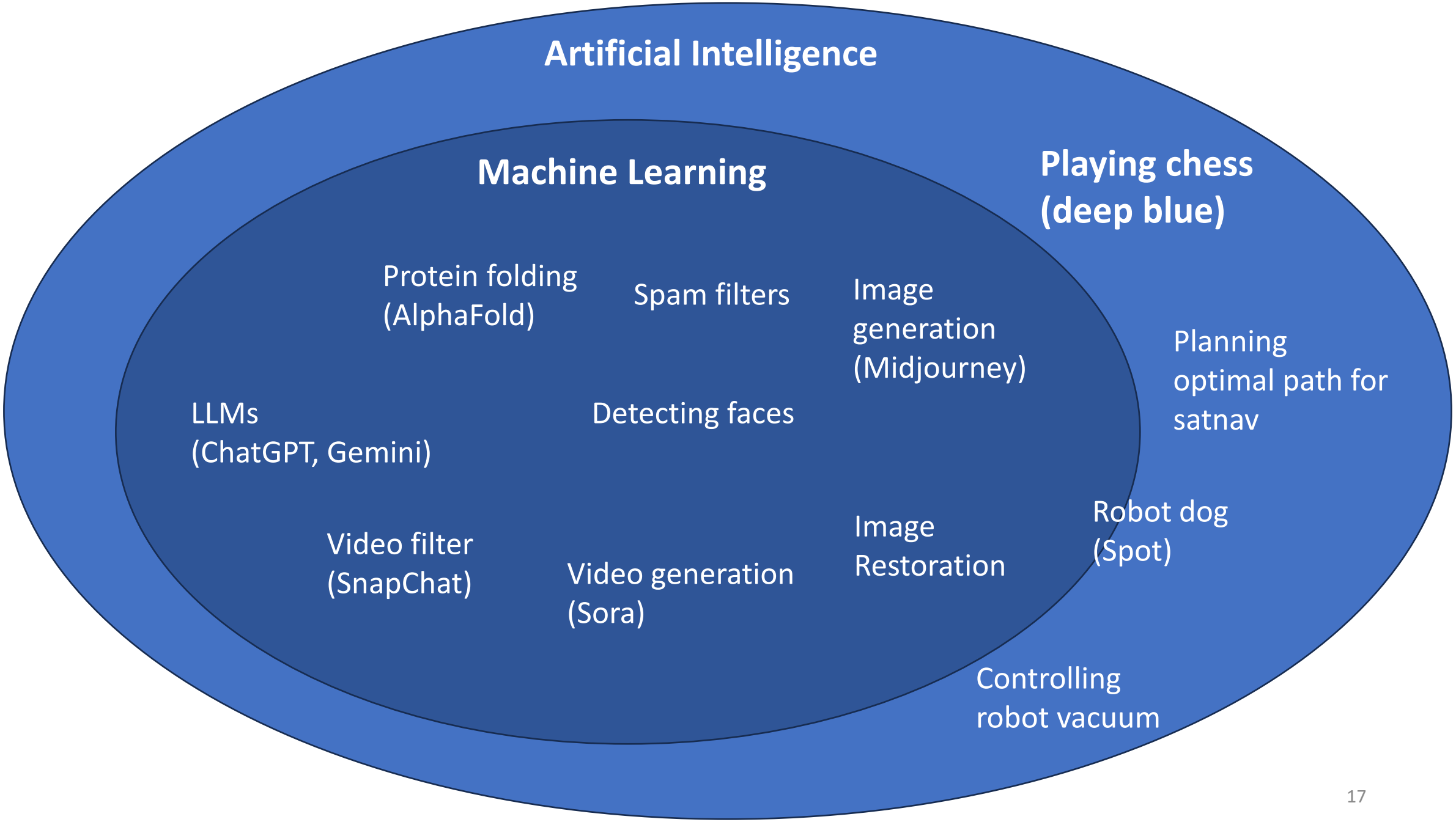
1. Understand the relationship between real brains and simple artificial neural network models
2. Describe and explain some of the principal architectures and learning algorithms of neural computation
3. Explain the learning and generalisation aspects of neural computation
4. Demonstrate an understanding of the benefits and limitations of neural-based learning techniques in context of other state-of-the-art methods of automated learning
5. Apply neural computation algorithms to specific technical and scientific problems


Artificial Intelligence
Image generation (Midjourney)
Protein folding (AlphaFold)
Playing chess (deep blue)
Spam filters
Planning optimal path for satnav
LLMs (ChatGPT, Gemini)
Detecting faces
Image Restoration
Robot dog (Spot)
Video filter (SnapChat)
Video generation (Sora)
Controlling robot vacuum

Is this Machine Learning ?



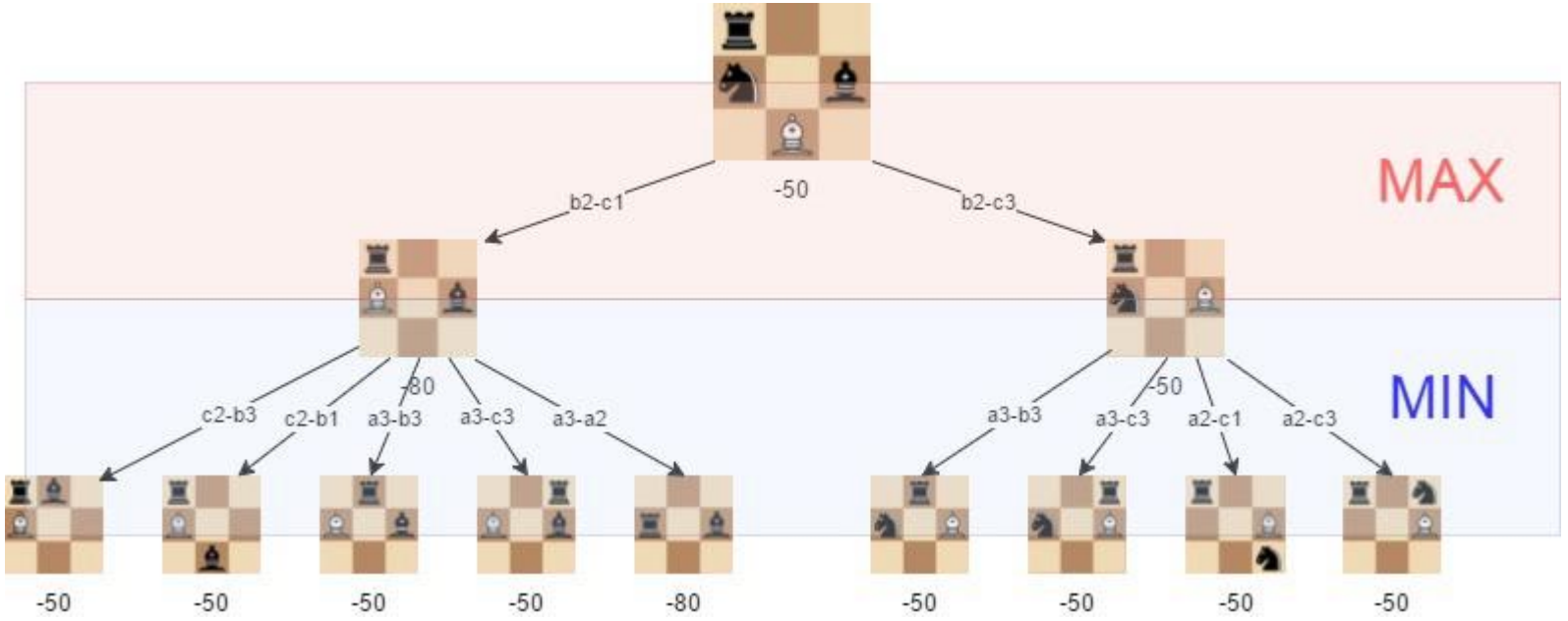

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Deep Blue – Chess

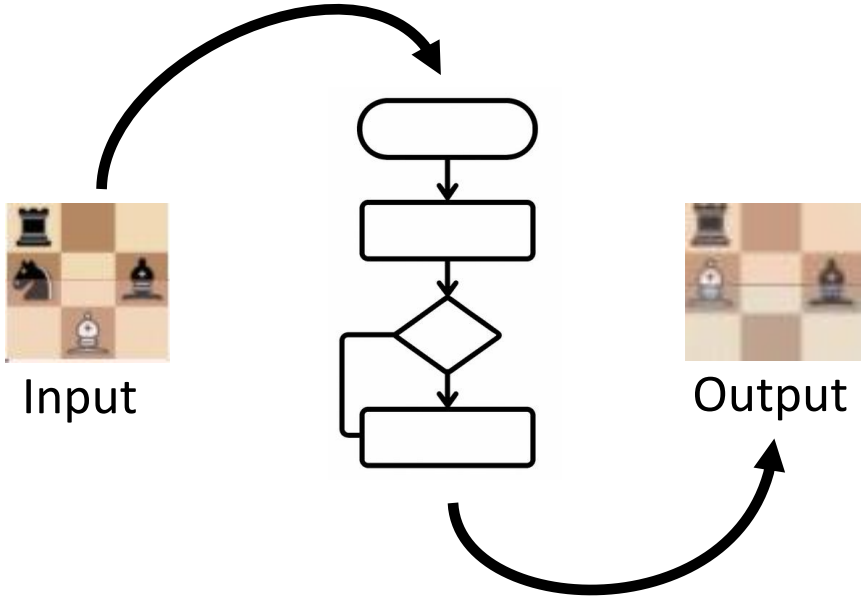
- MiniMax Algorithm
- Not Machine Learning!



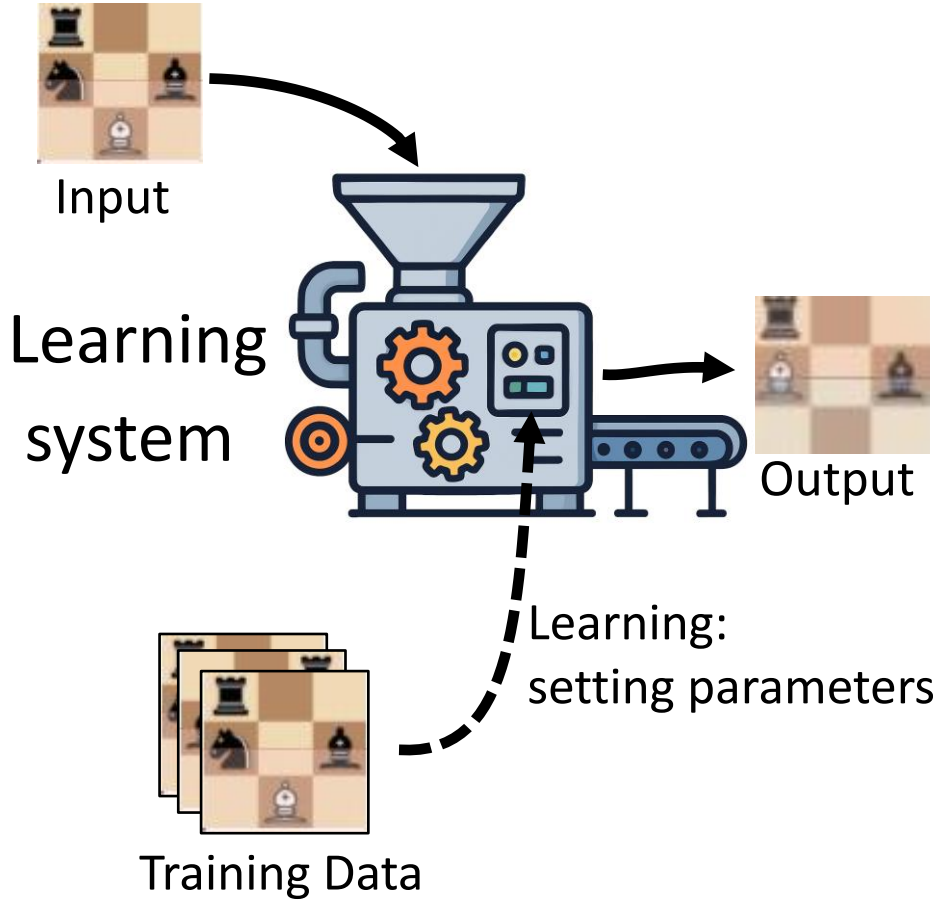


Engineered Algorithms vs Machine learning

Algorithm



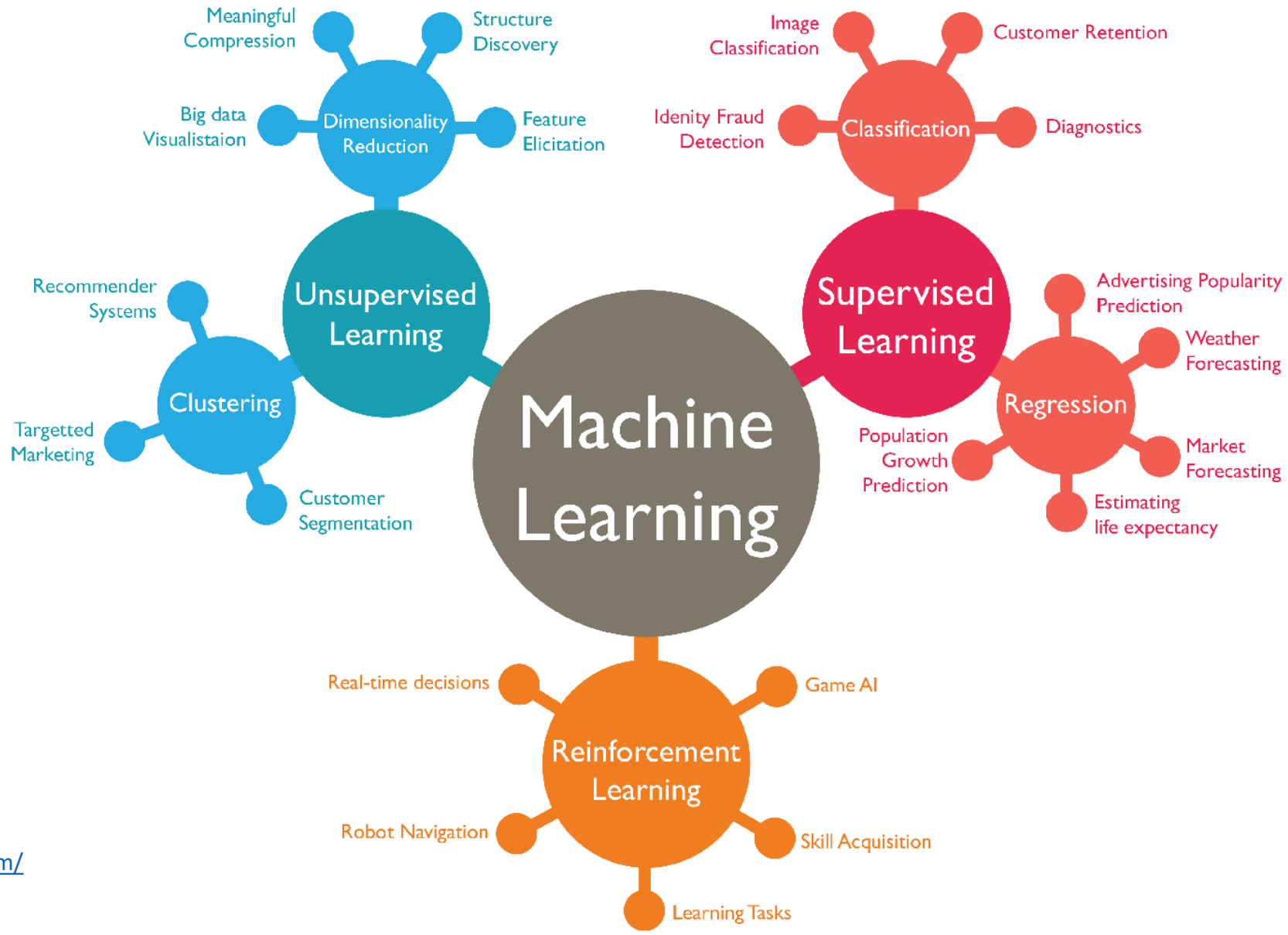
Machine Learning



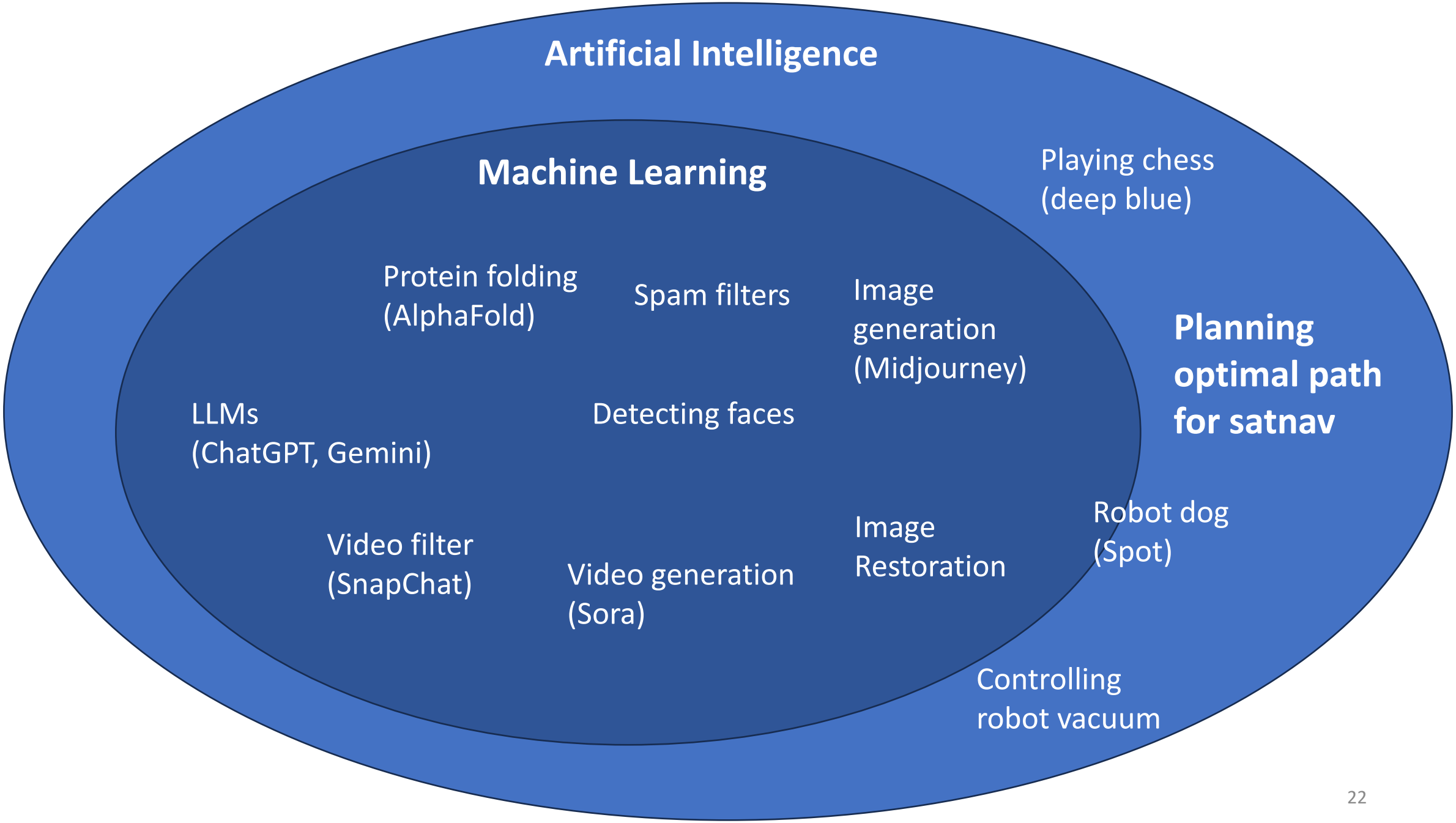
Learning is adjusting parameters from data



Types of machine learning

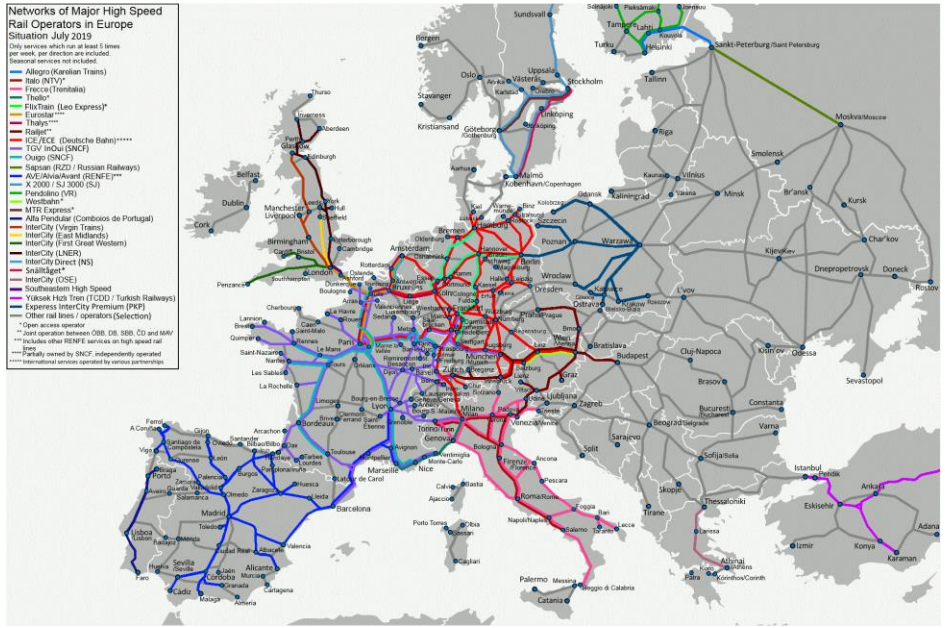


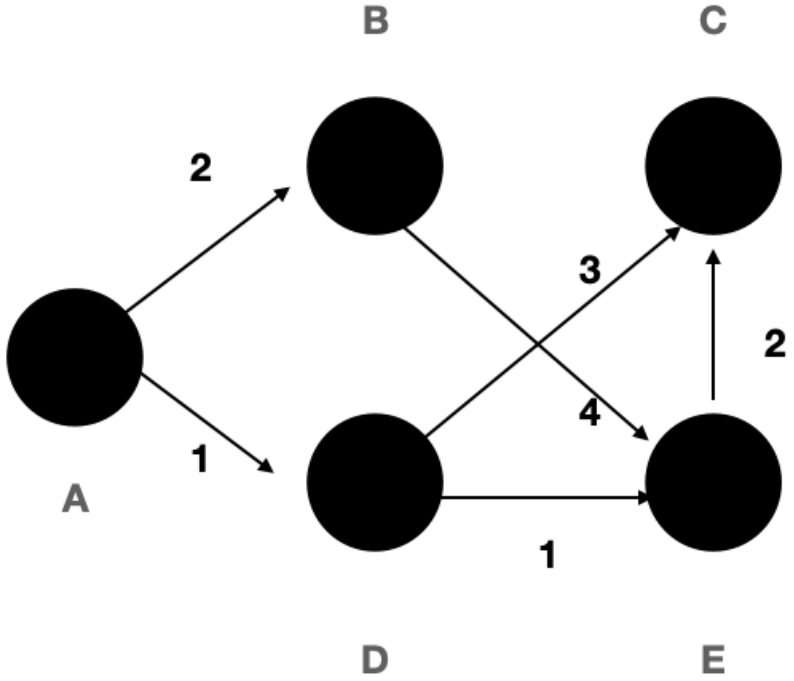
https://towardsdatascience.com/machine-learning-types-2-c1291d4f04b1


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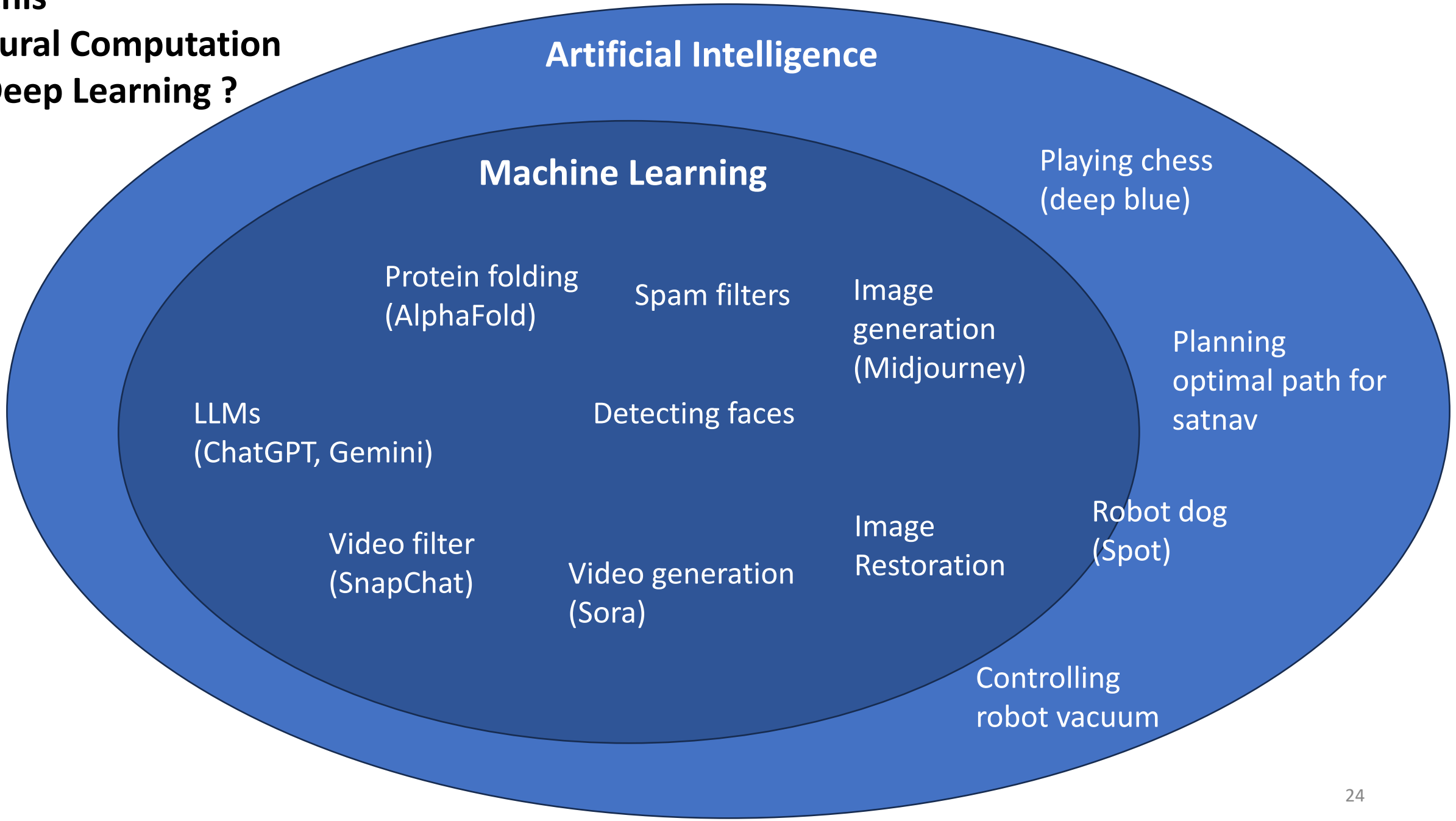
Navigation

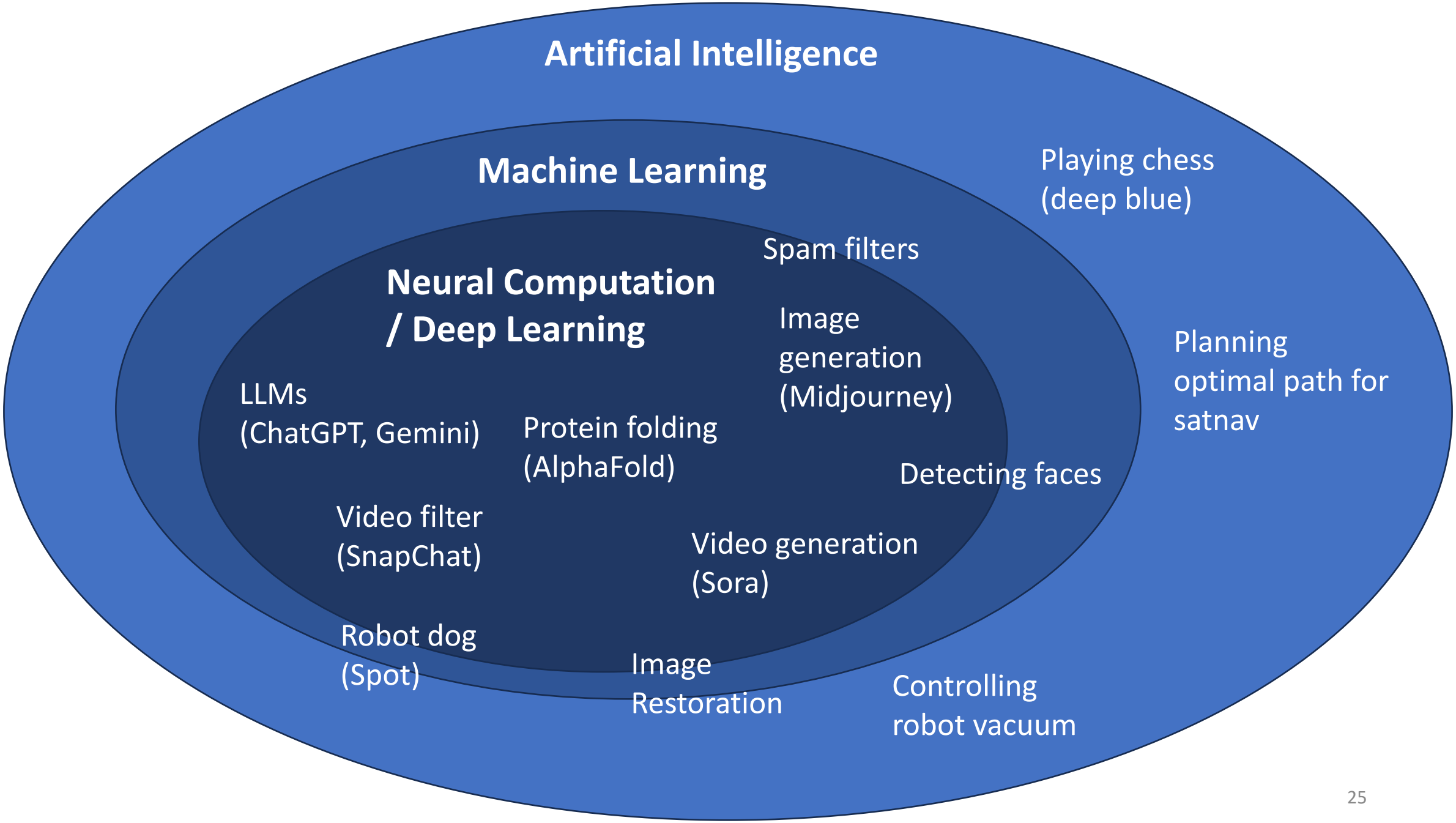
- Dijkstra or Astar Algorithm
- Not Machine Learning!



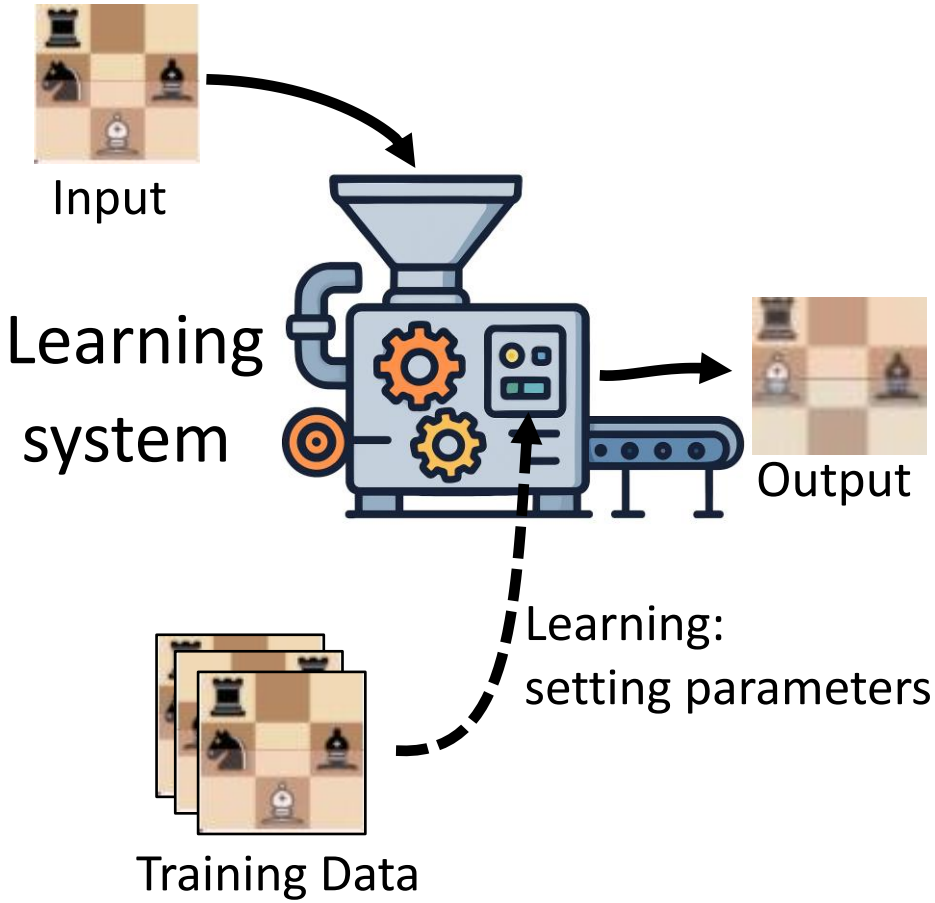


Is this Neural Computation / Deep Learning ?

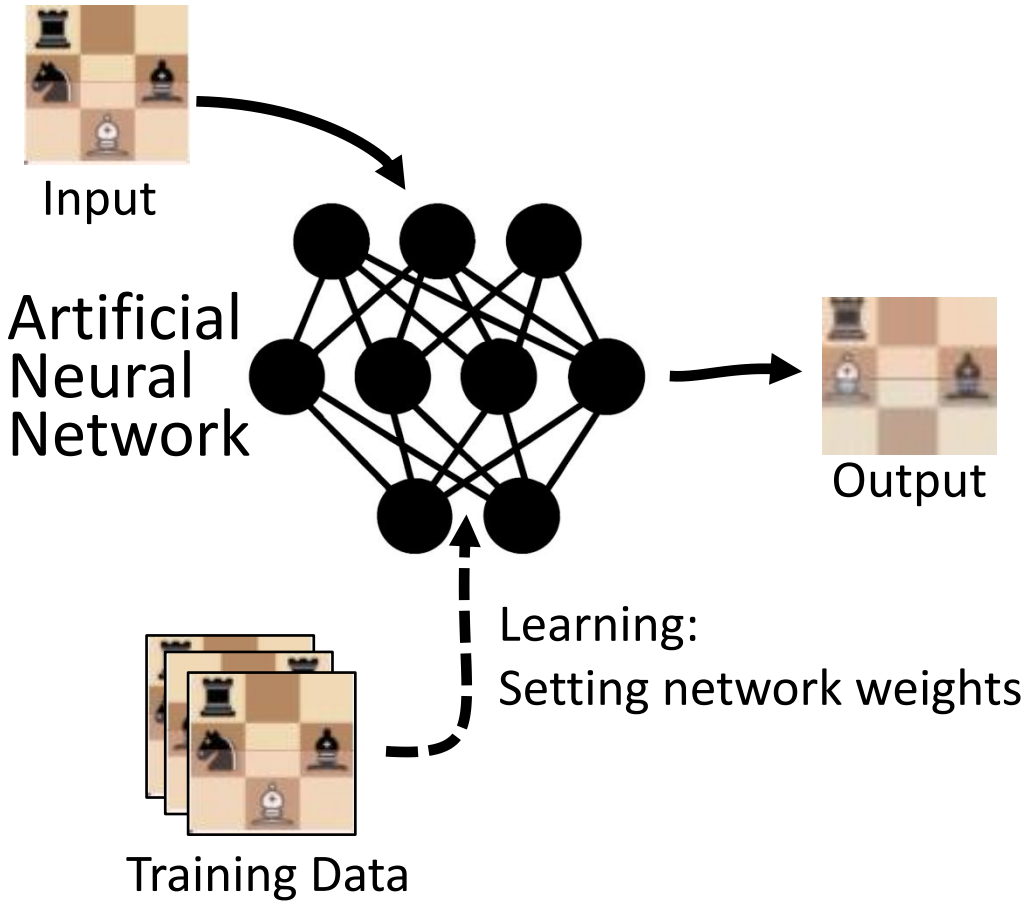


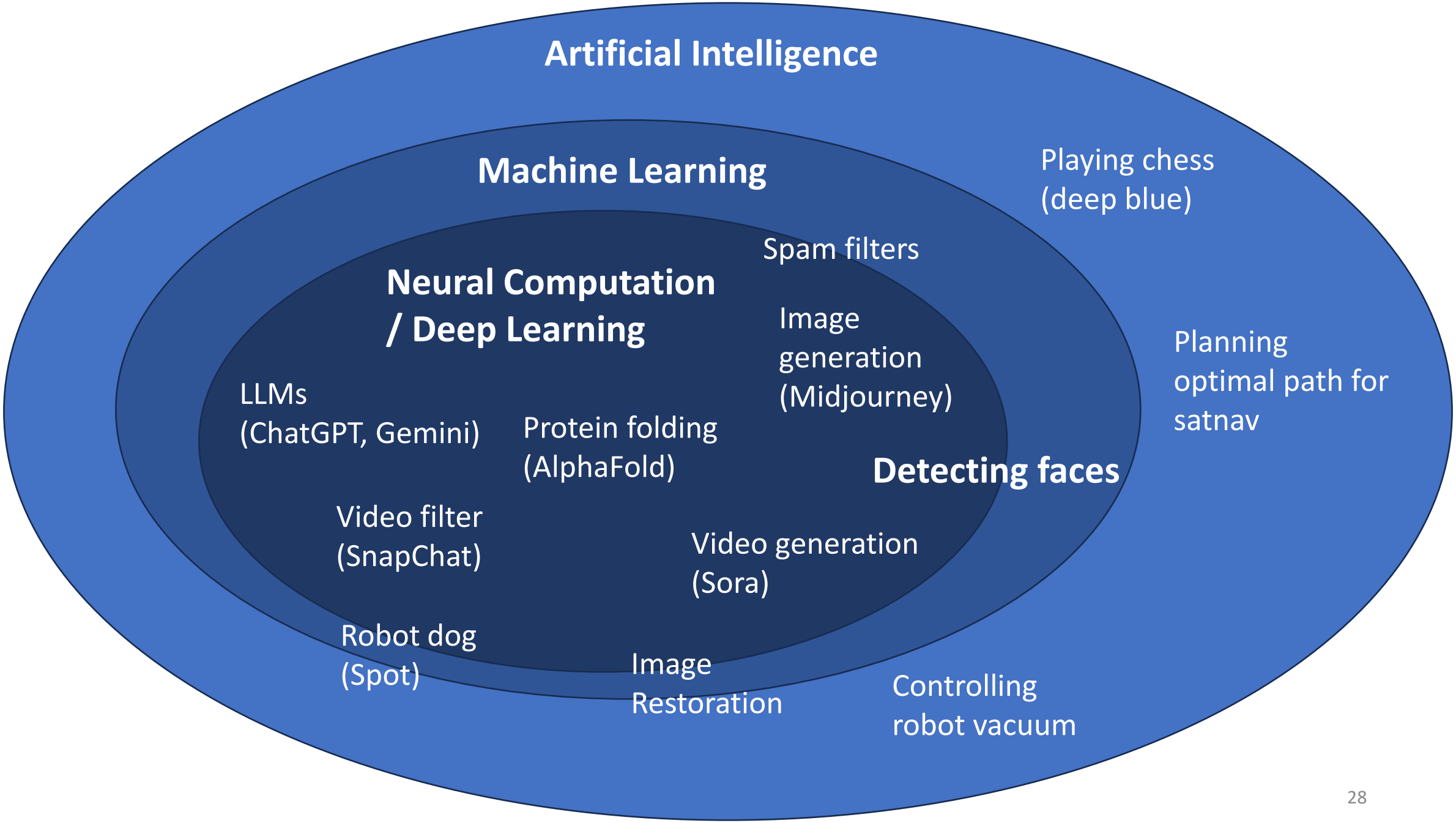

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Machine learning



Neural Computation / Deep Learning



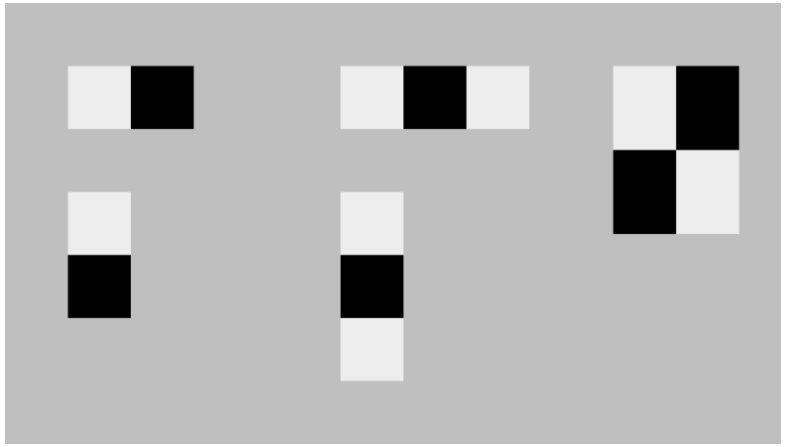

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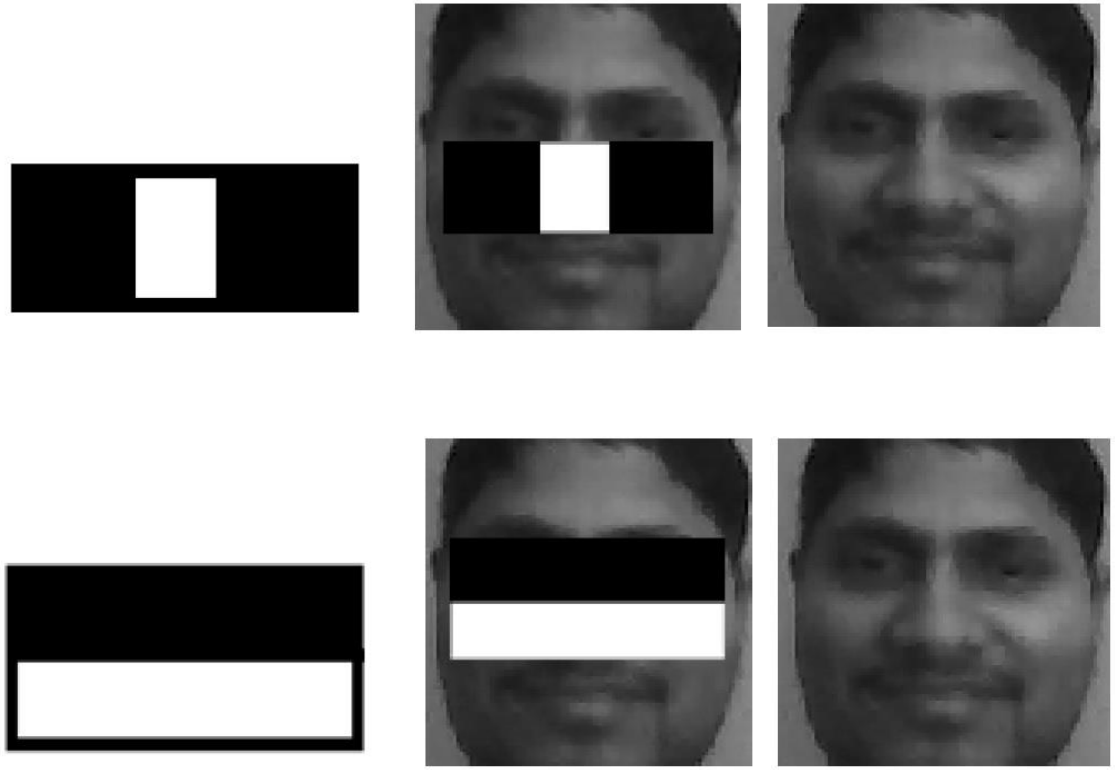
Face Detection Viola-Jones

- AdaBoost algorithm
- Learned classifiers
- Not Deep learning!



https://www.vrogue.co/post/face-detection-introduction-to-opencv-using-python-part-1-studytonight





Summary

Artificial Intelligence:

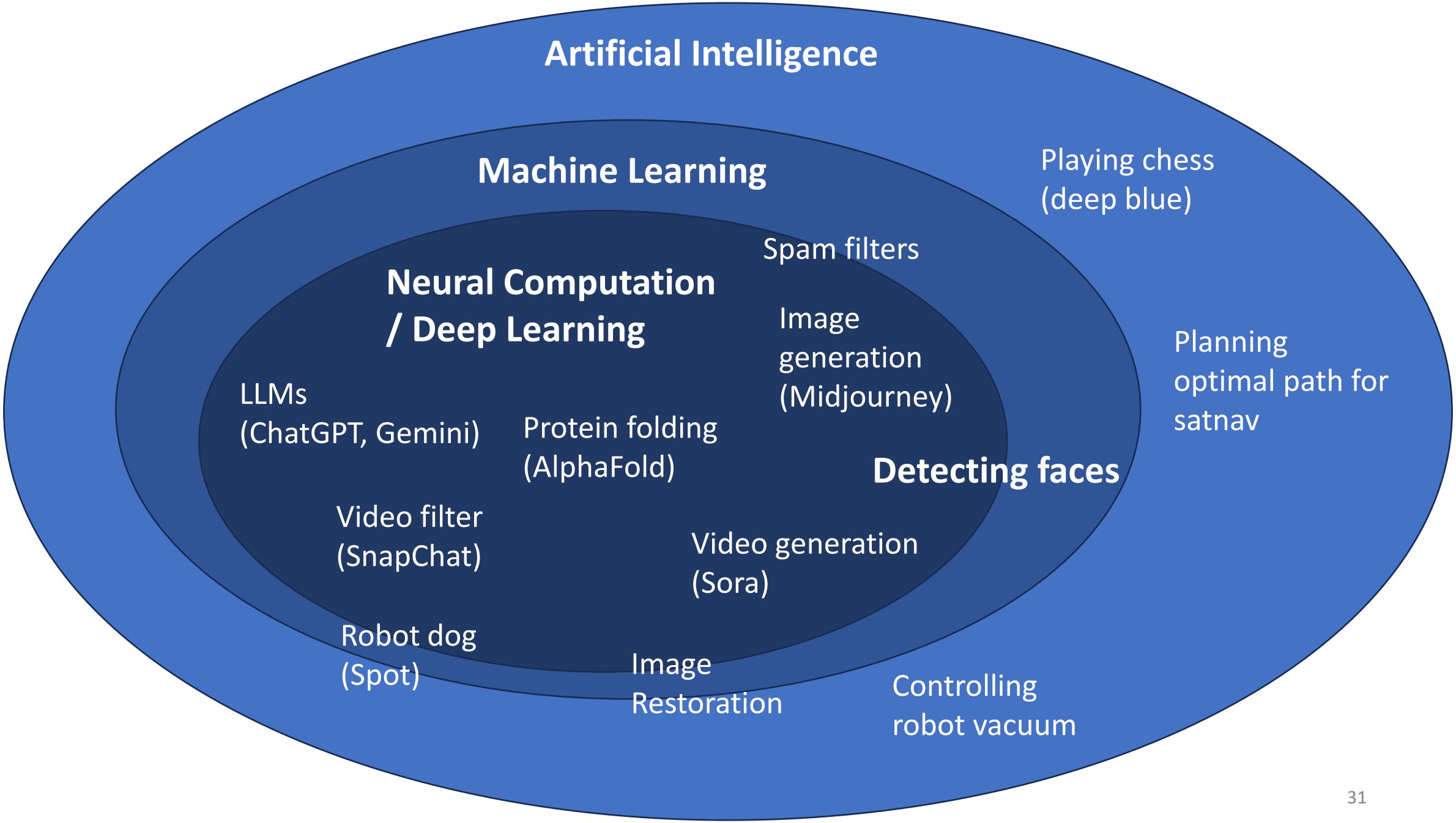
“capability of computational systems to perform tasks typically associated with human intelligence”

Machine Learning:

“statistical algorithms that can learn from data and generalise to unseen data and thus perform tasks without explicit instructions.”

Neural Computation / Deep Learning

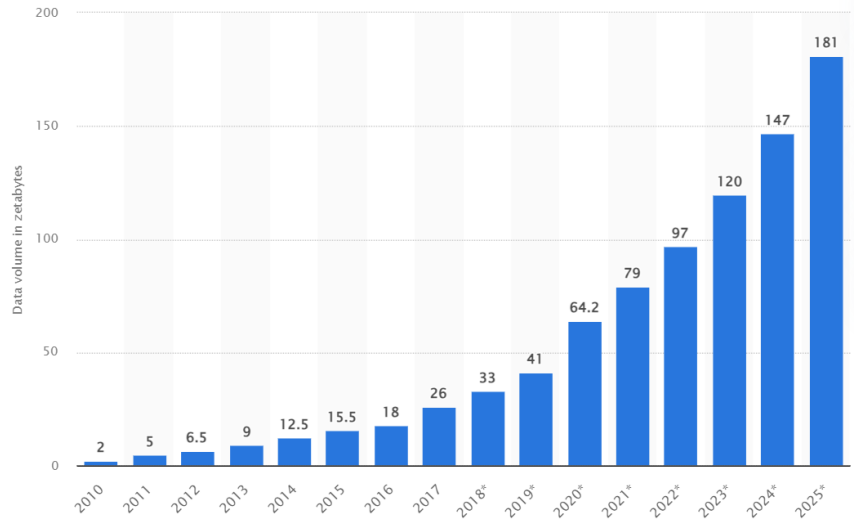
*“machine learning that utilises **multilayered neural networks**”*


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What has led to the surge in NNs?

1. Lots of digital data
   - e.g., web, pictures, videos

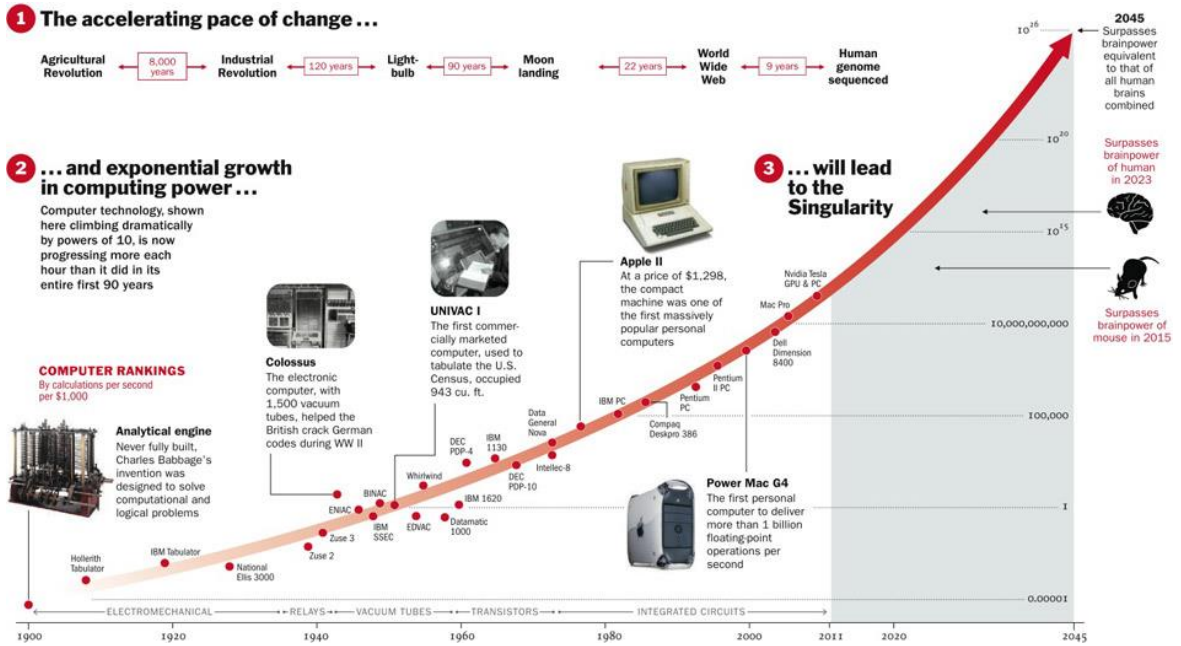
2. Lots of compute power
   - e.g., GPUs, TPUs



https://www.statista.com/statistics/871513/worldwide-data-created/





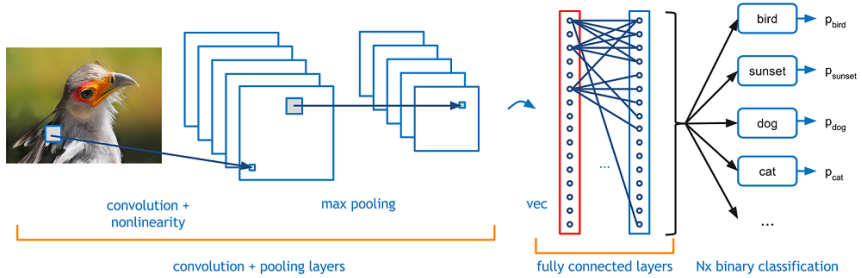


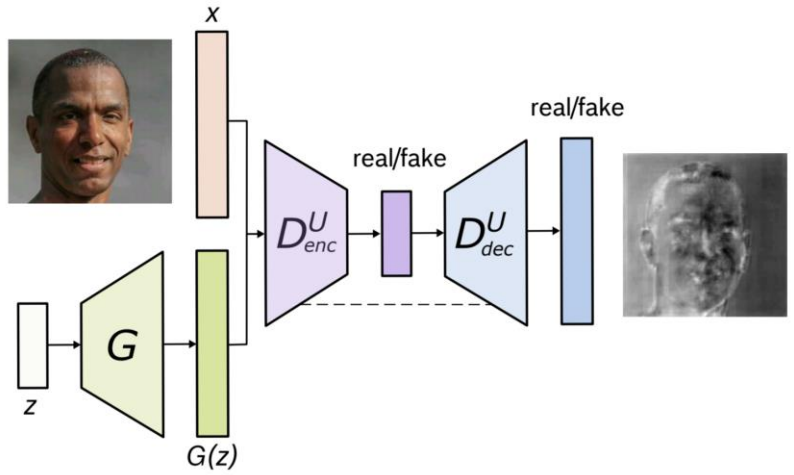
https://content.time.com/time/interactive/0%2C31813%2C2048601%2C00.html

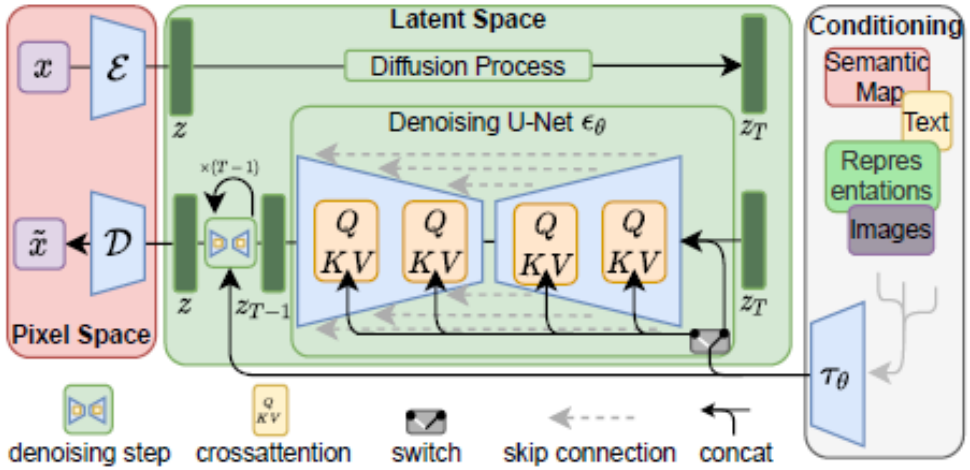
What has led to the surge in NNs?

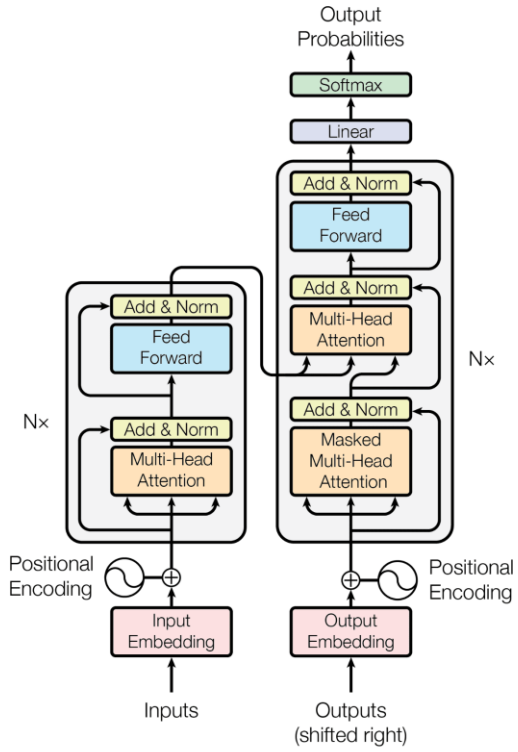
3. Lots of NN innovations and architectures
   - e.g., CNNs, RNNs, GANs, transformers, diffusion models



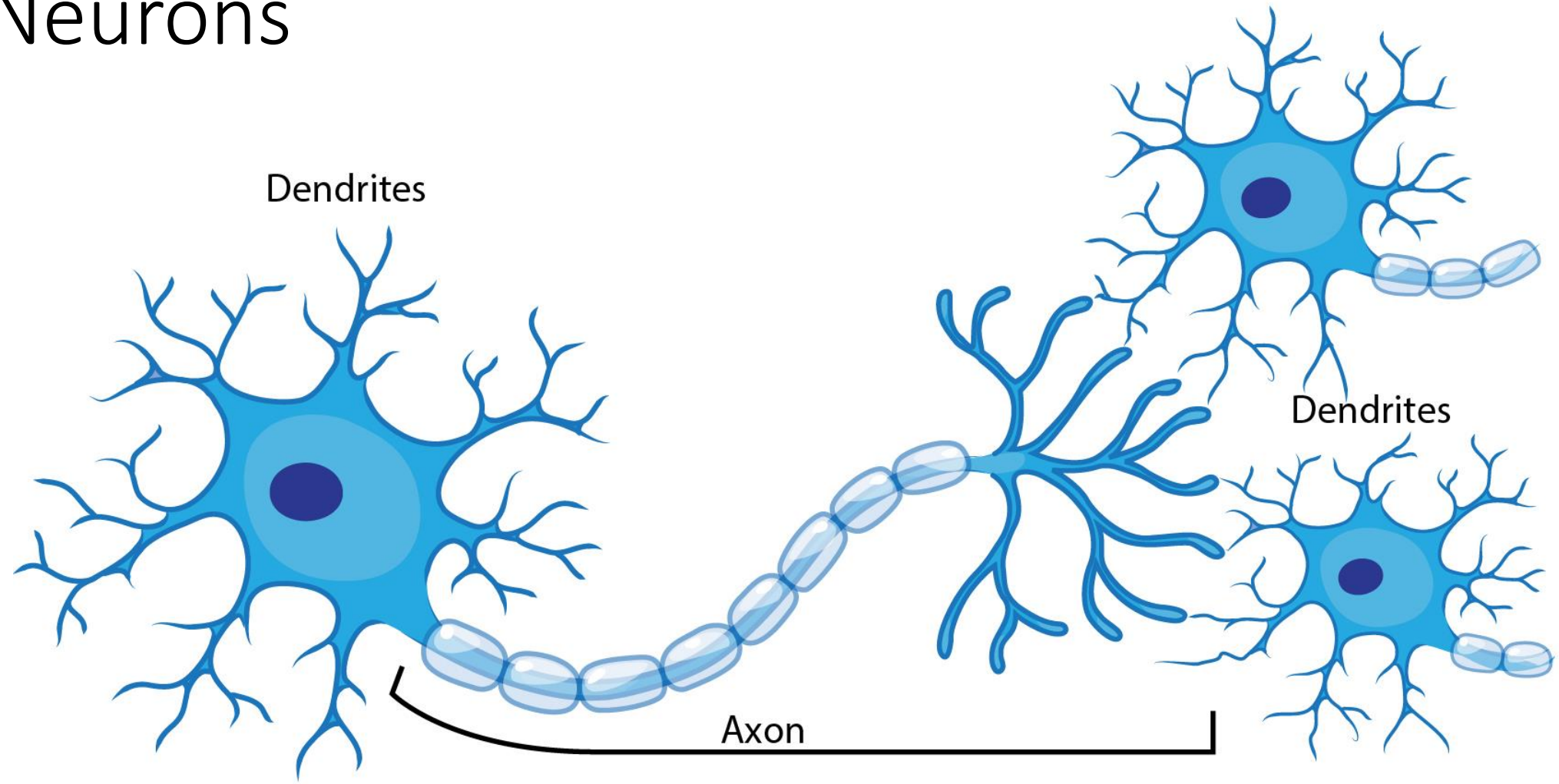






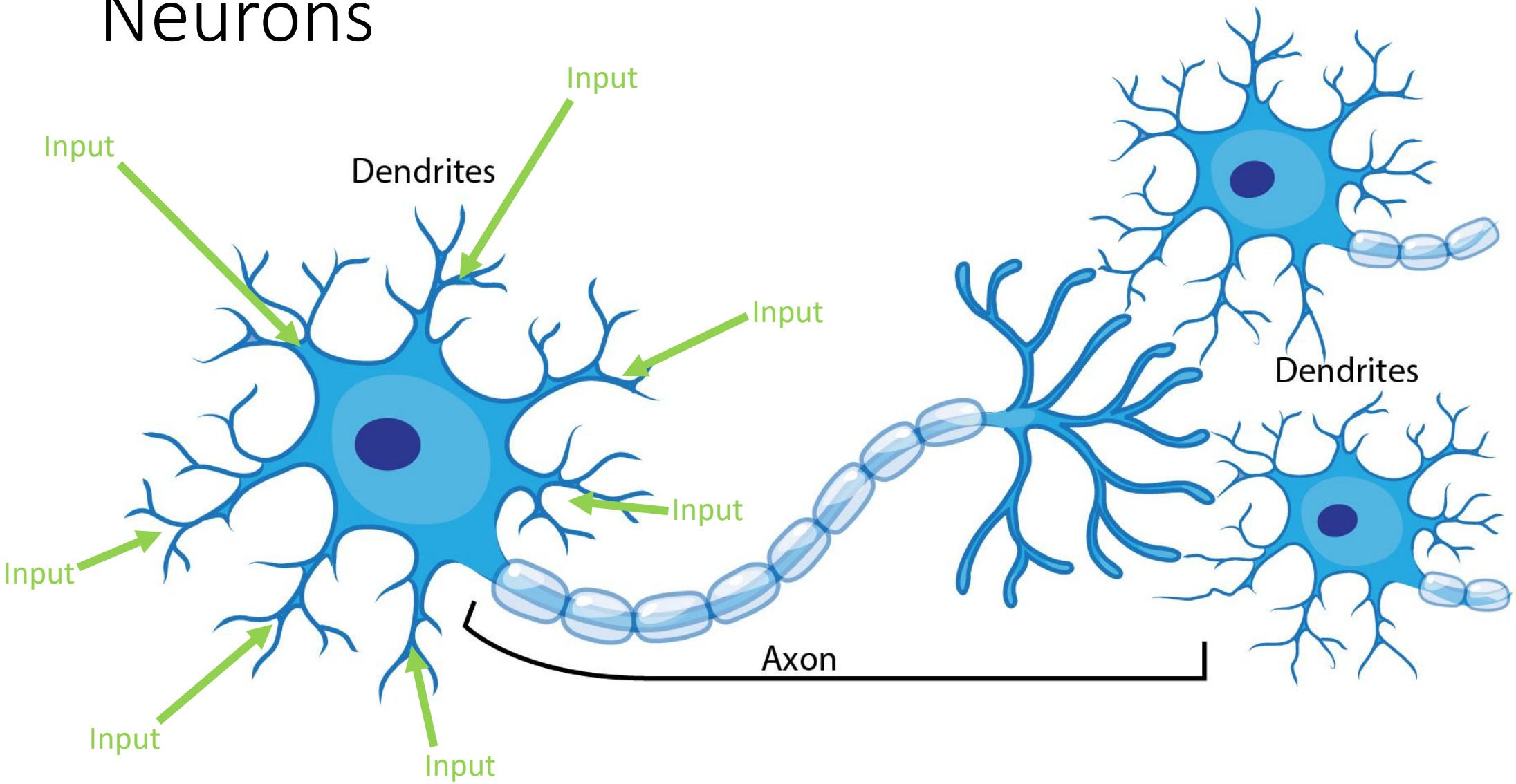


Neurons



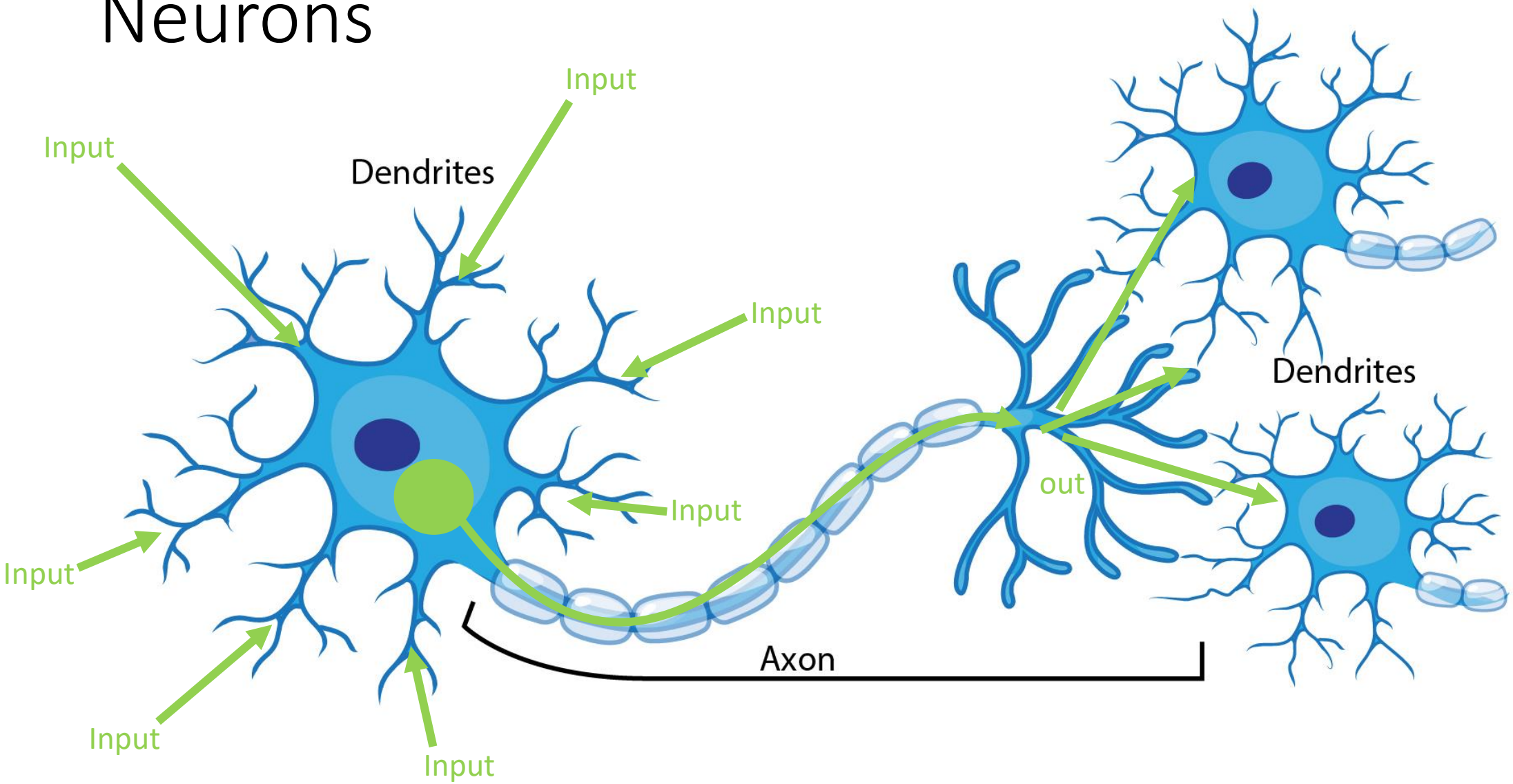
Devin K. Phillips. "Speed of the Human Brain". ASU - Ask A Biologist. 13 May, 2015. https://askabiologist.asu.edu/plosable/speed-human-brain

Neurons



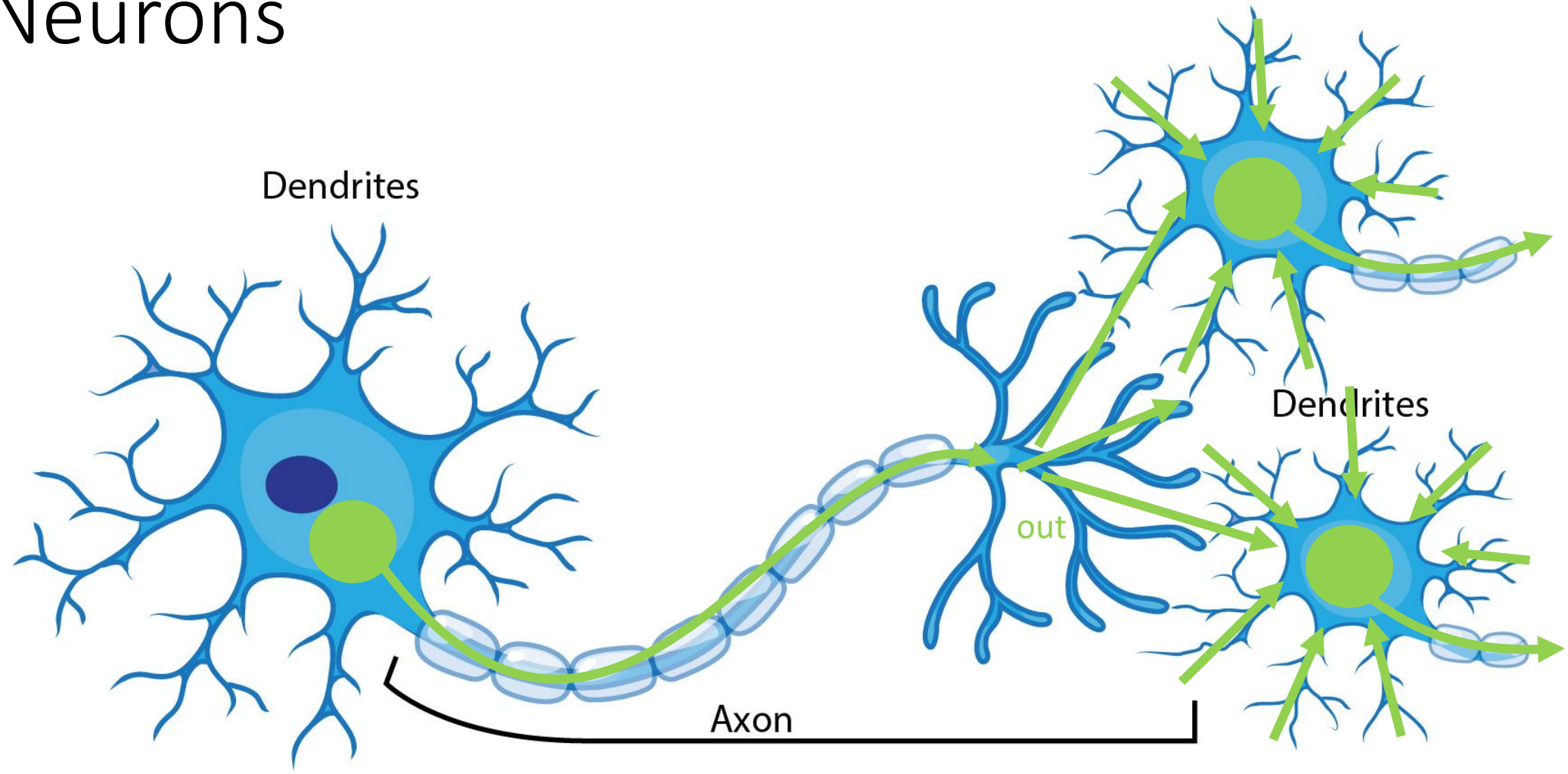
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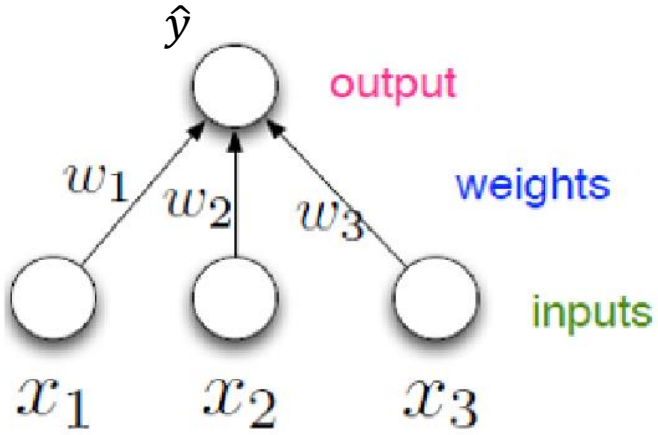
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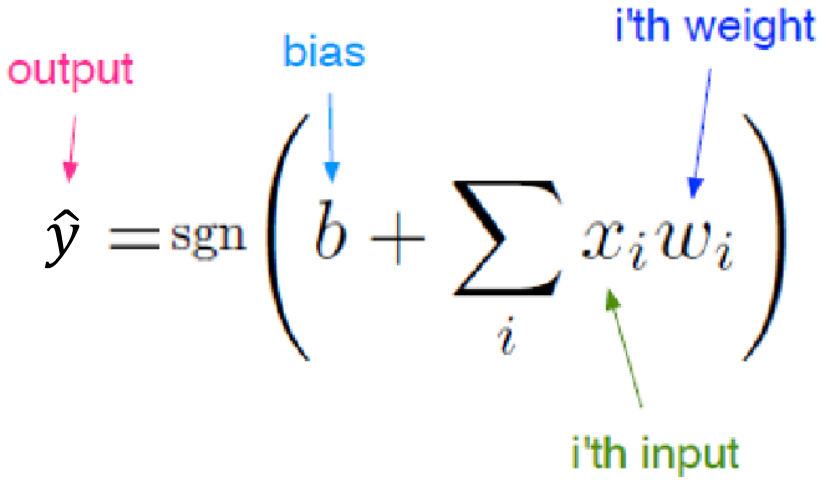
Neurons

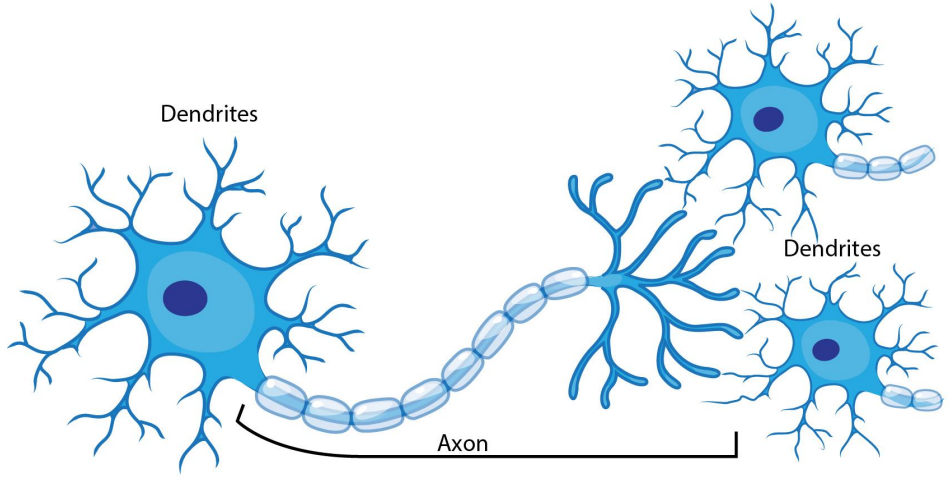


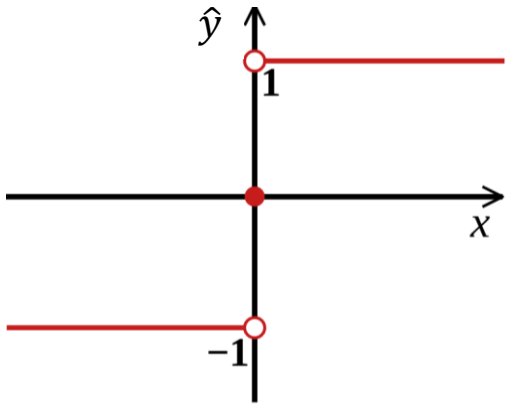
Devin K. Phillips. "Speed of the Human Brain". ASU - Ask A Biologist. 13 May, 2015. https://askabiologist.asu.edu/plosable/speed-human-brain

Perceptron aka McCulloch-Pitts Neuron



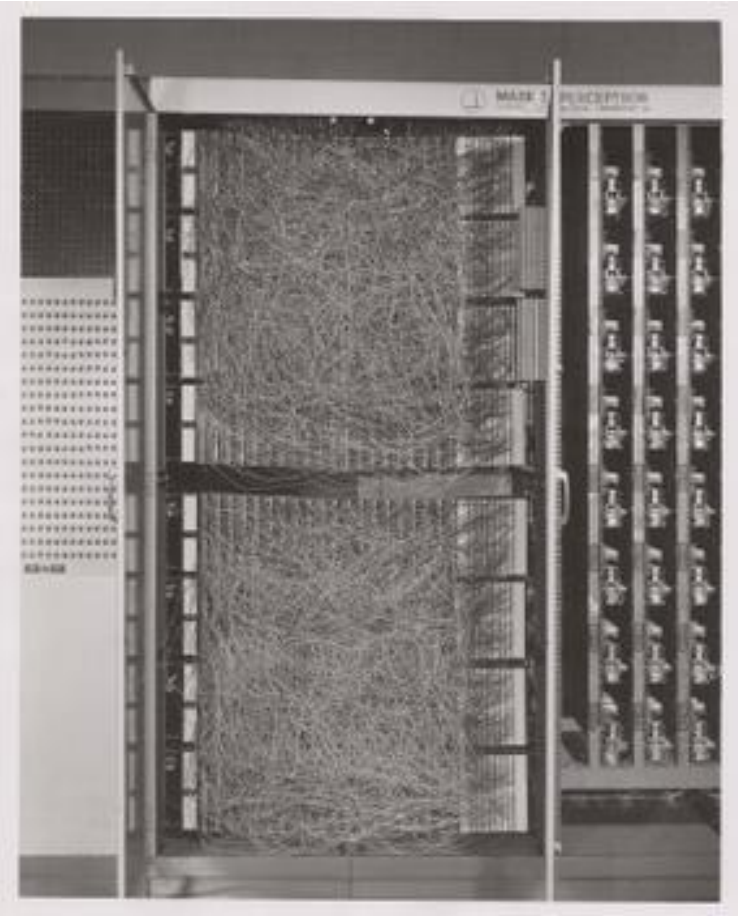


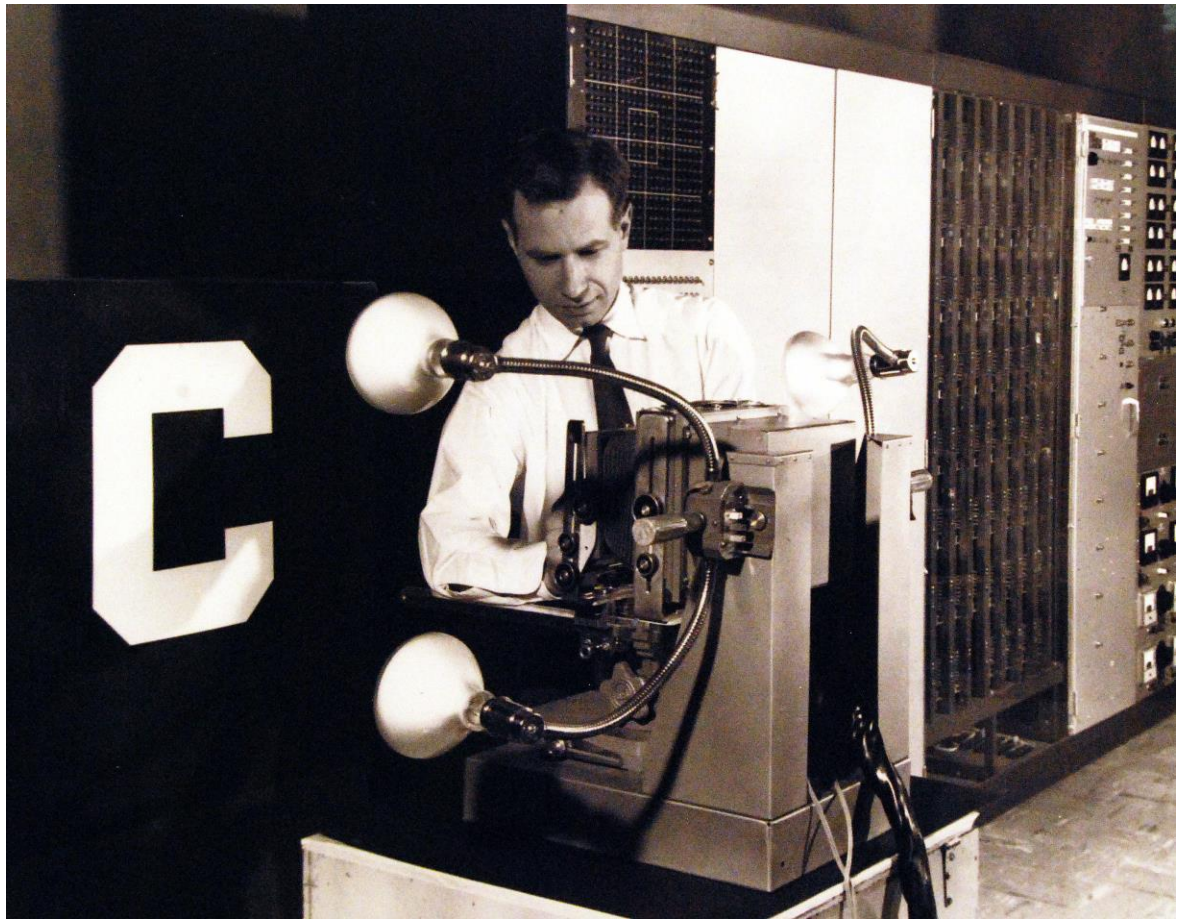




Sign function

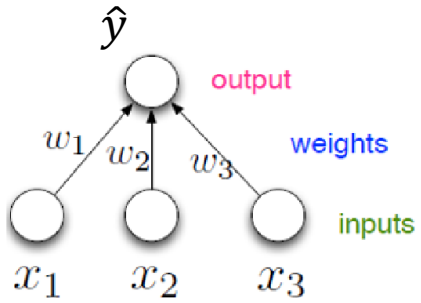
Frank Rosenblatt’s Perceptron





"the embryo of an electronic computer that [the Navy] expects will be able to walk, talk, see, write, reproduce itself and be conscious of its existence."- NY times 1958

Perceptron Classifier – Classifying Letters

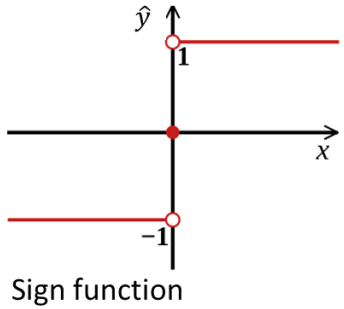


$$\text{output} \quad \hat{y} = \text{sgn}\left(b + \sum_i x_i w_i\right)$$

bias

i'th weight

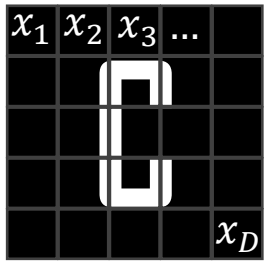
i'th input

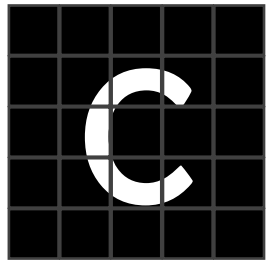


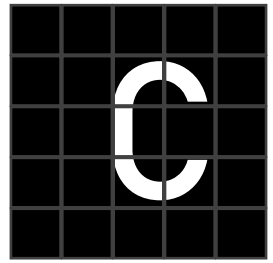
Sign function

Input $\boldsymbol{x} = (x_1, \ldots, x_n)$

Label $y = -1$

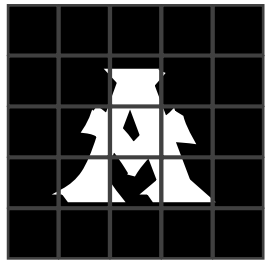


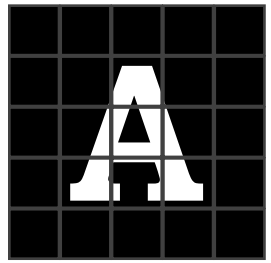


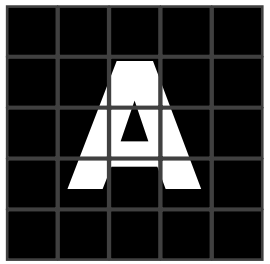


...

Label $y = 1$



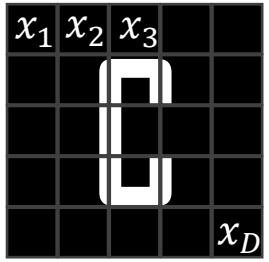




...

Perceptron Classifier

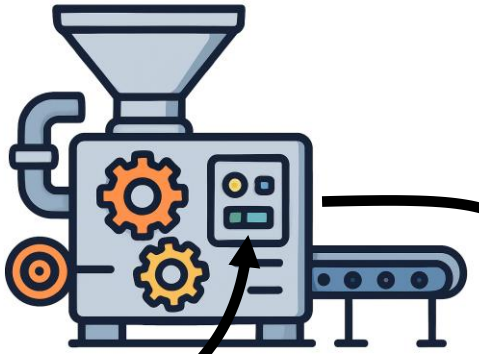
Input $\boldsymbol{x} = (x_1, \ldots, x_D)$



output bias i'th weight

$$\hat{y} = \text{sgn}\left(b + \sum_i x_i w_i\right)$$

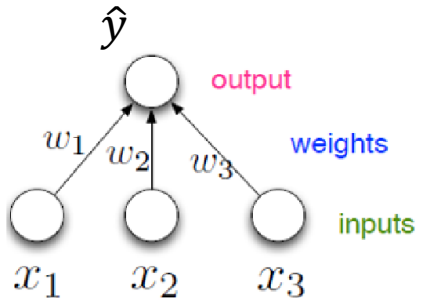
i'th input

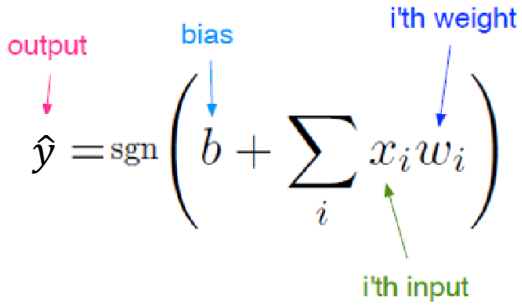


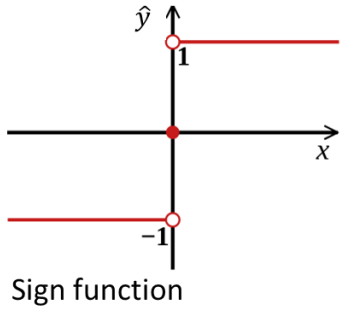
$\hat{y} = -1$

Parameters $\boldsymbol{w} = (w_1, \ldots, w_D), b$

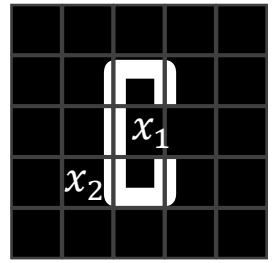
Perceptron Classifier

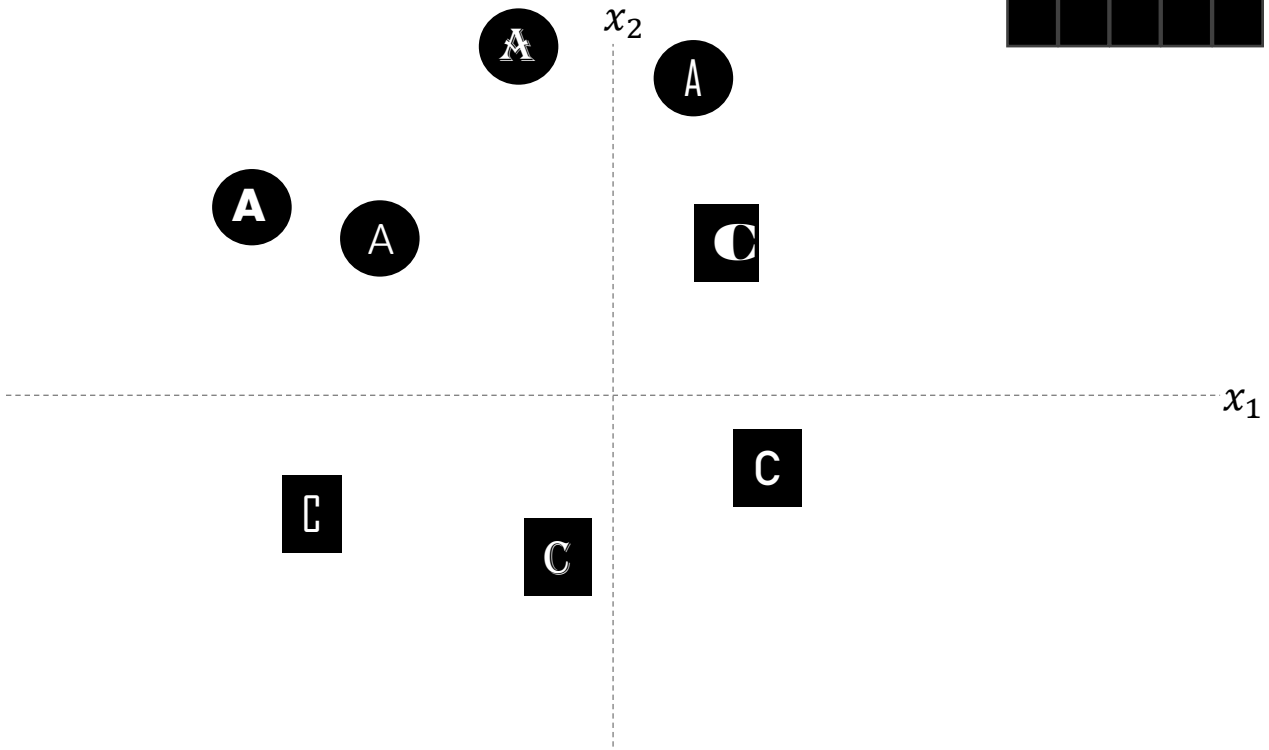




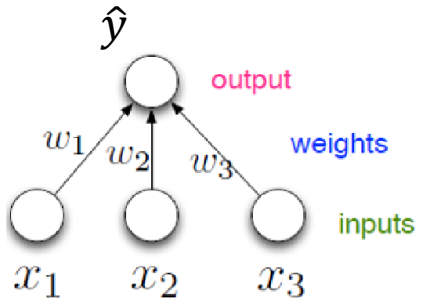


Sign function





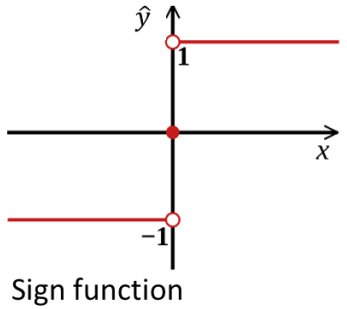
Perceptron Classifier



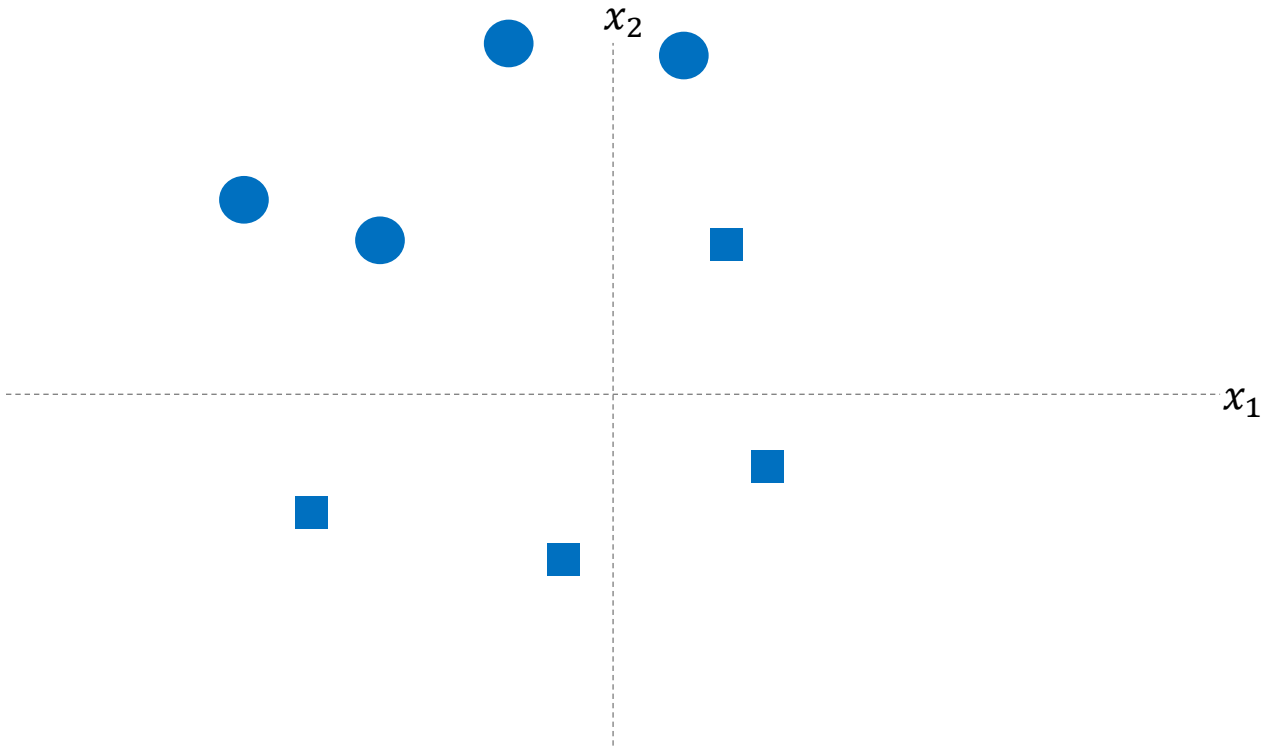
output bias i'th weight

$$\hat{y} = \text{sgn}\left(b + \sum_i x_i w_i\right)$$

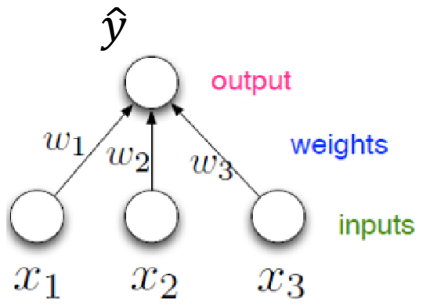
i'th input

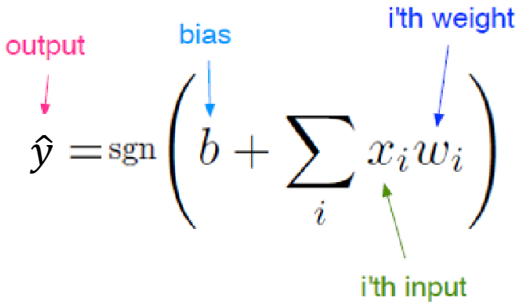


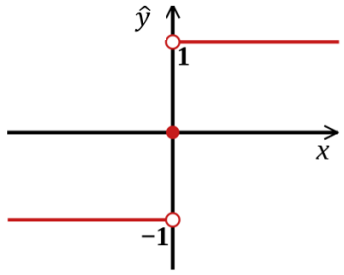
Sign function



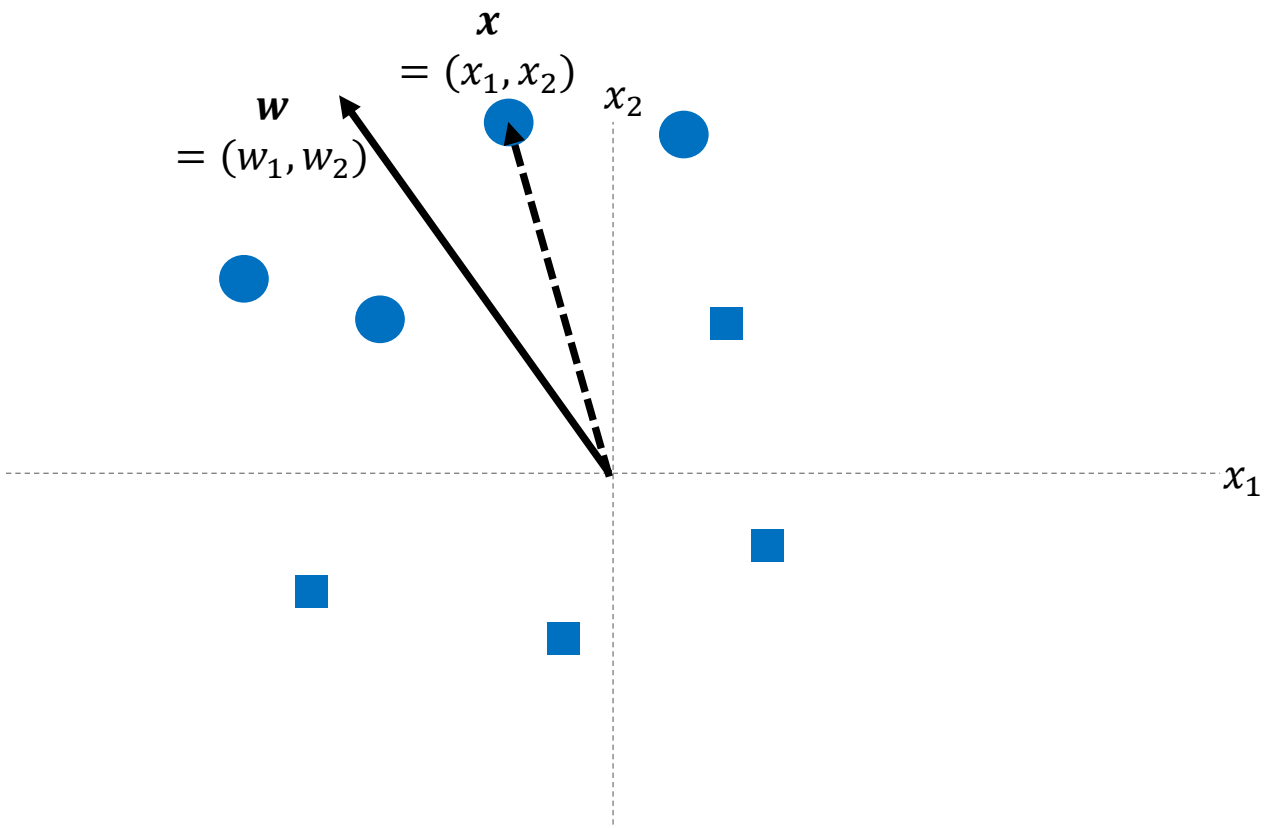
Perceptron Classifier



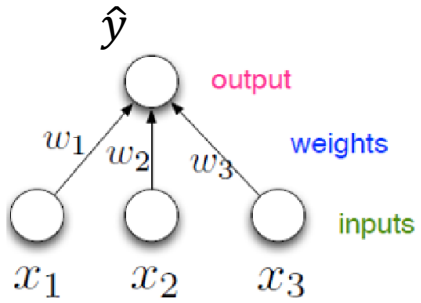


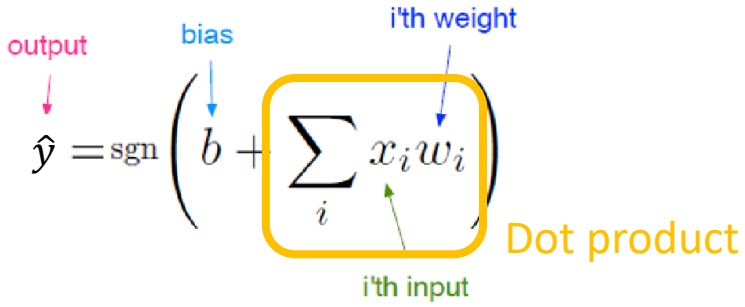


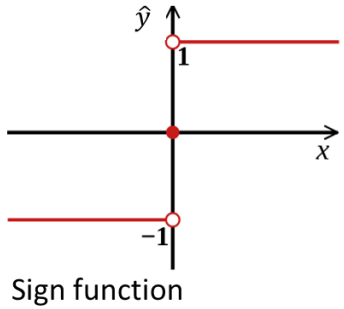
Sign function



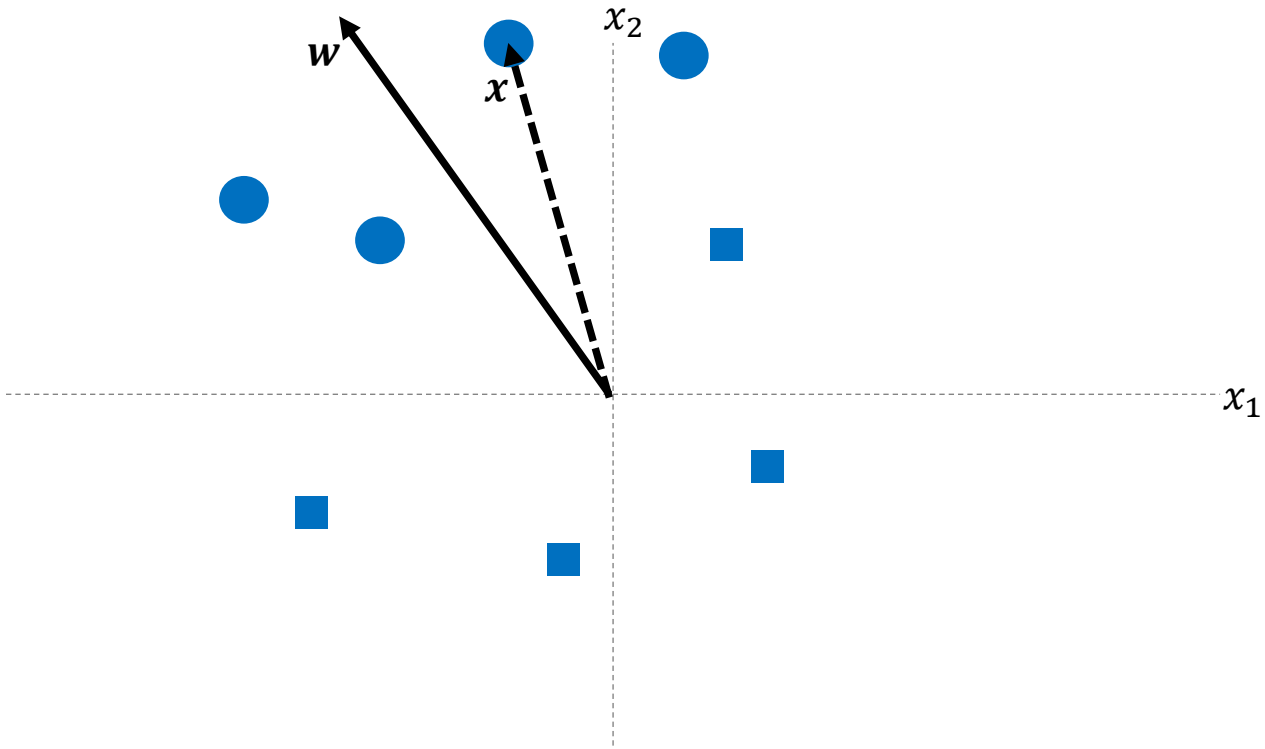
Perceptron Classifier







Sign function



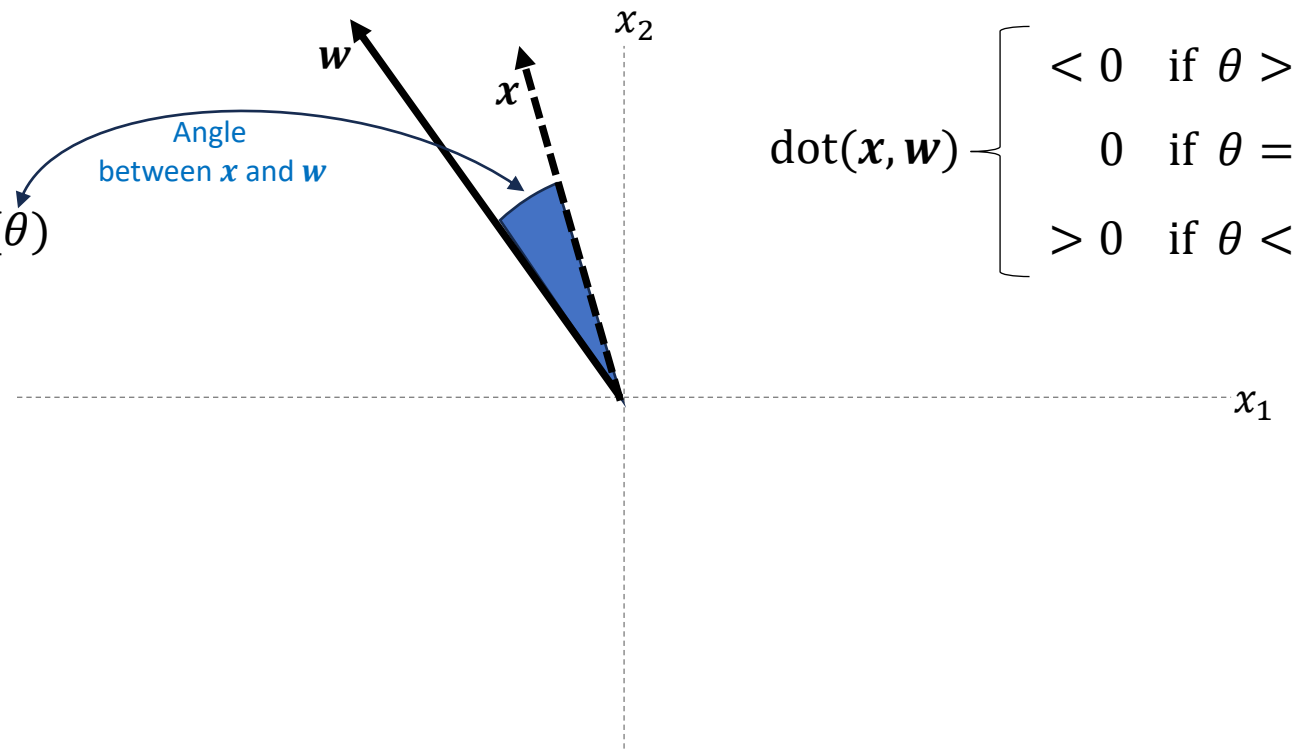
Dot Products – Geometric Interpretation

$$\text{dot}(\boldsymbol{x}, \boldsymbol{w}) = \sum_{i=1}^{D} x_i \cdot w_i$$

$$= \big|\big|\boldsymbol{x}\big|\big| \cdot \big|\big|\boldsymbol{w}\big|\big| \cdot \cos(\theta)$$

Euclidian length of $\boldsymbol{x}$

Euclidian length of $\boldsymbol{w}$



$$\text{dot}(\boldsymbol{x}, \boldsymbol{w}) \begin{cases} < 0 & \text{if } \theta > 90\text{deg} \\ 0 & \text{if } \theta = 90\text{deg} \\ > 0 & \text{if } \theta < 90\text{deg} \end{cases}$$

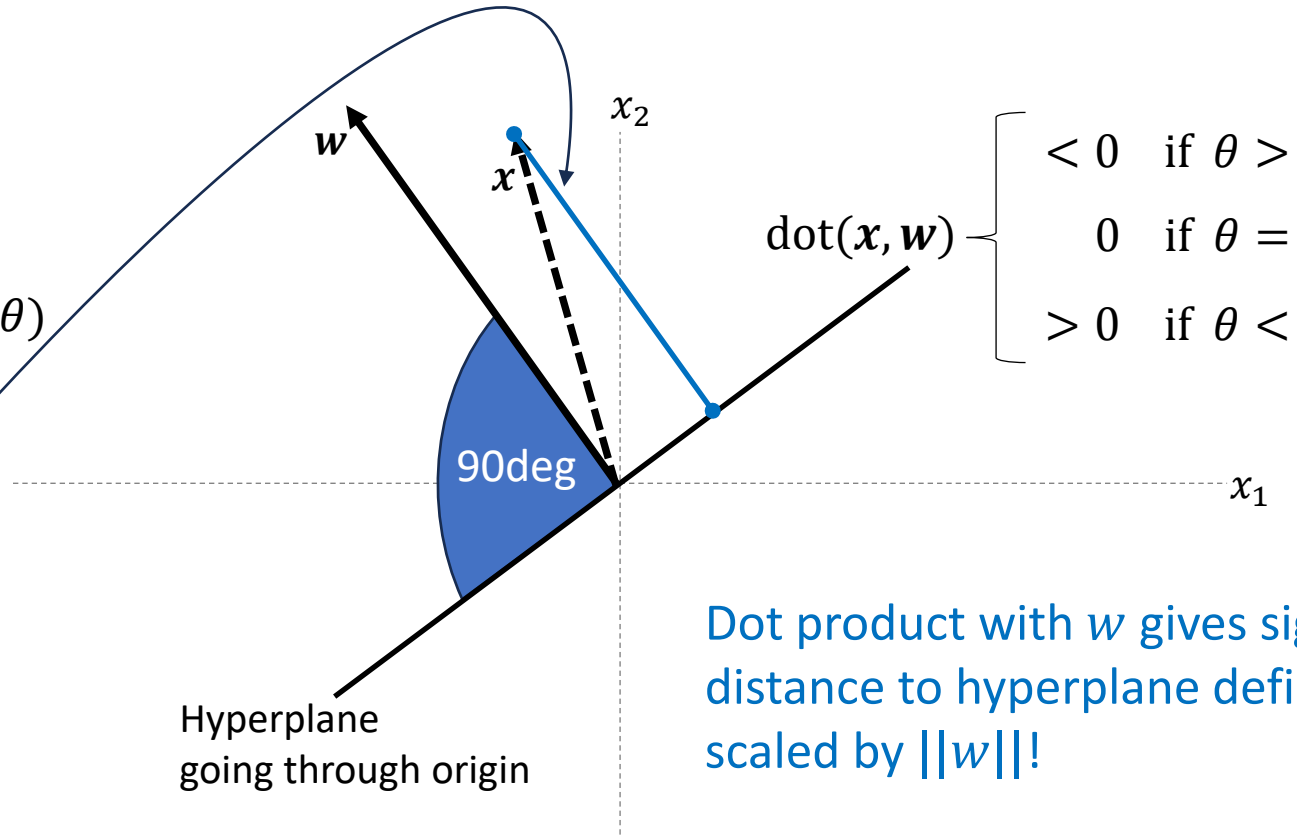
Dot Products – Geometric Interpretation

$$\text{dot}(\boldsymbol{x}, \boldsymbol{w}) = \sum_{i=1}^{n} x_i \cdot w_i$$

$$= ||\boldsymbol{x}|| \cdot ||\boldsymbol{w}|| \cdot \cos(\theta)$$

$$= ||\boldsymbol{w}|| \cdot d$$

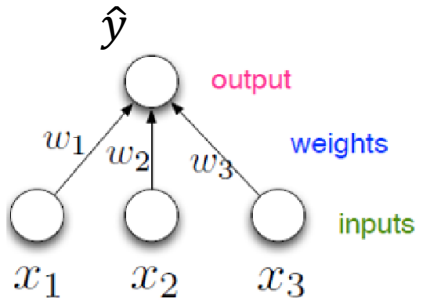
Signed distance to hyperplane

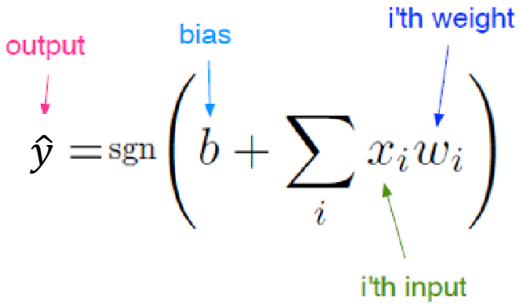


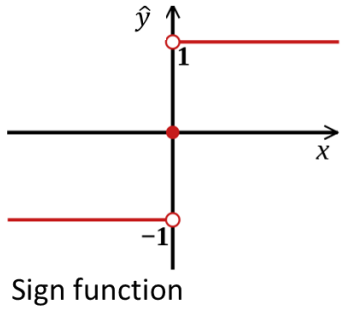
$$\text{dot}(\boldsymbol{x}, \boldsymbol{w}) \begin{cases} < 0 & \text{if } \theta > 90\text{deg} \\ 0 & \text{if } \theta = 90\text{deg} \\ > 0 & \text{if } \theta < 90\text{deg} \end{cases}$$

Dot product with w gives signed distance to hyperplane defined by w scaled by $||w||$!

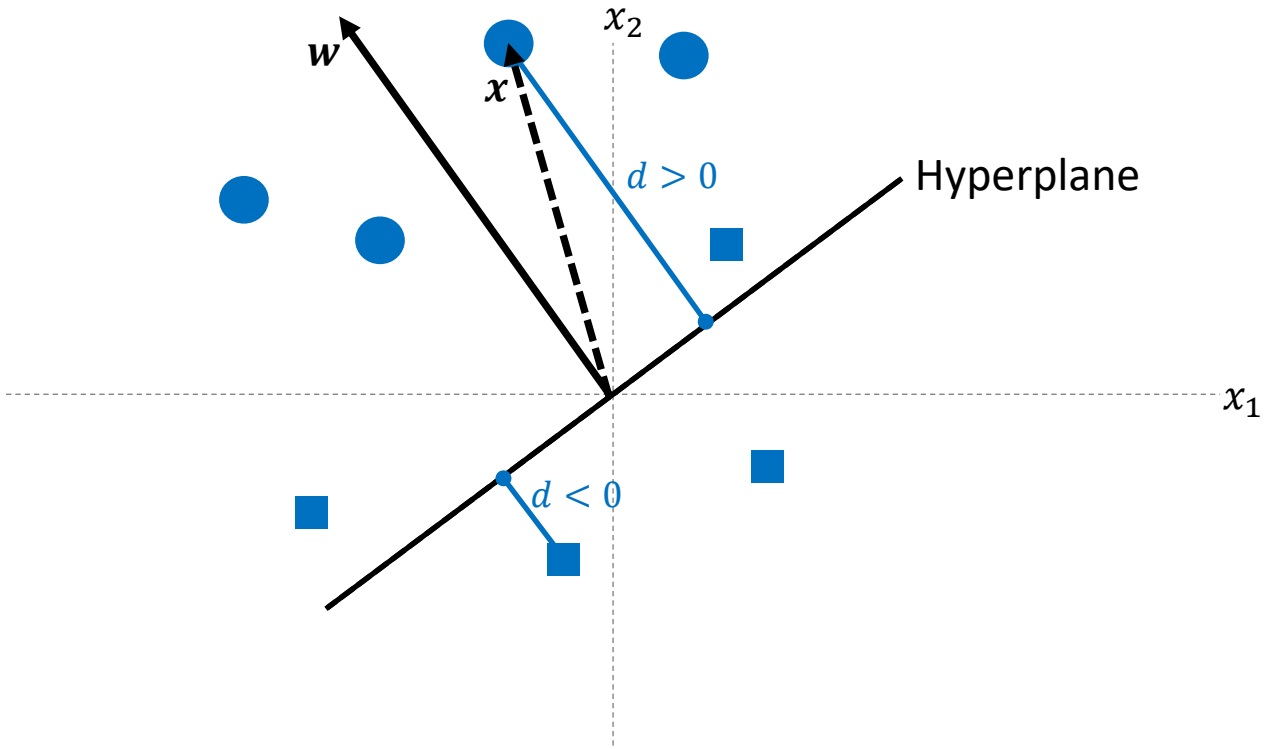
Perceptron Classifier



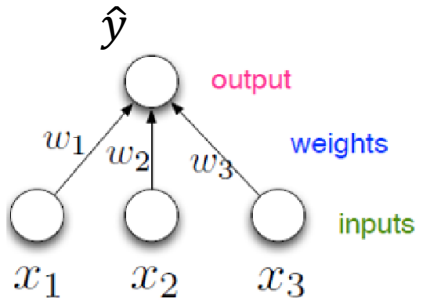


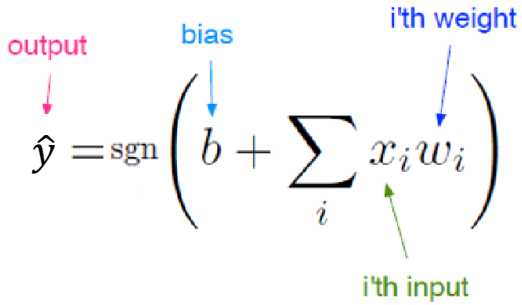


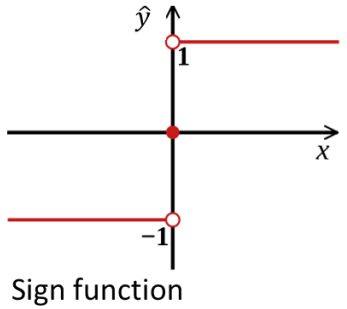
Sign function



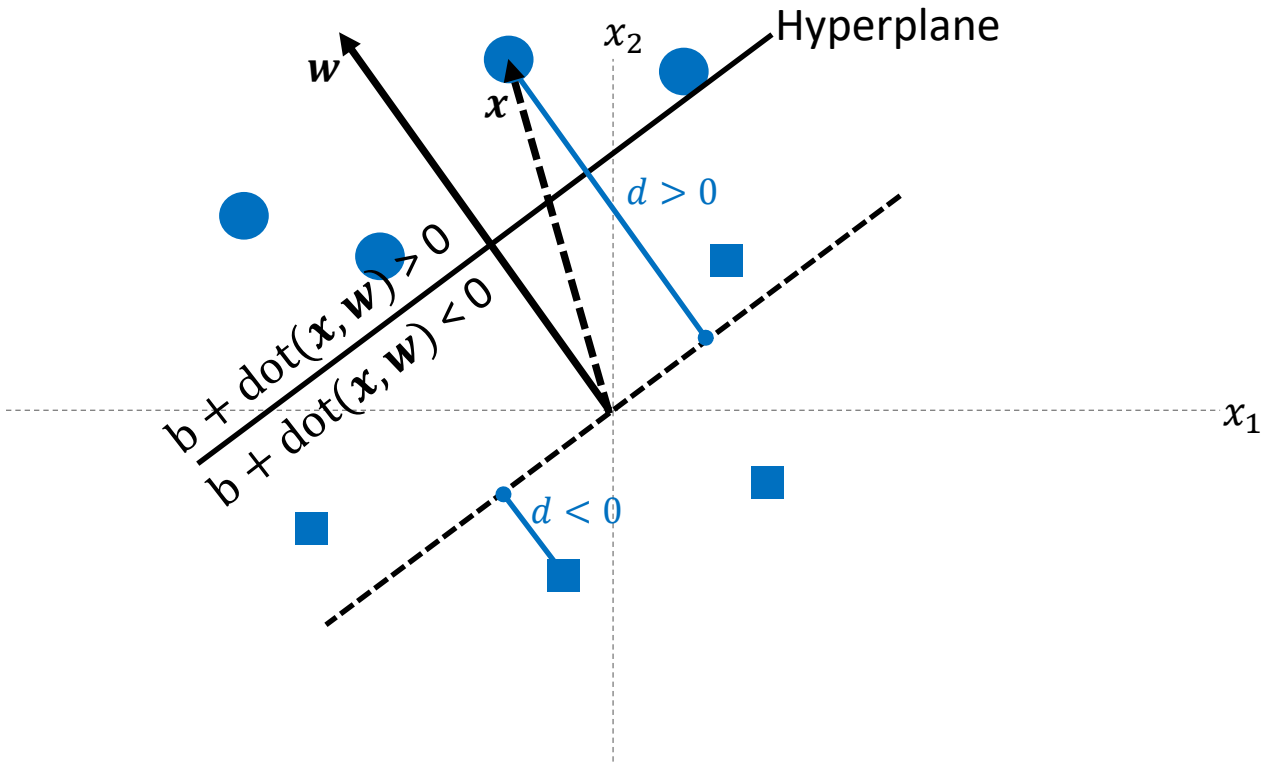
Perceptron Classifier



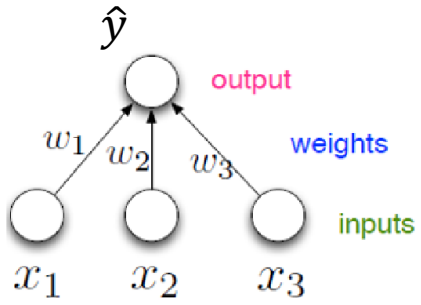


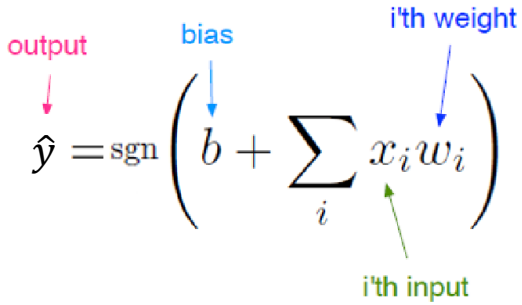


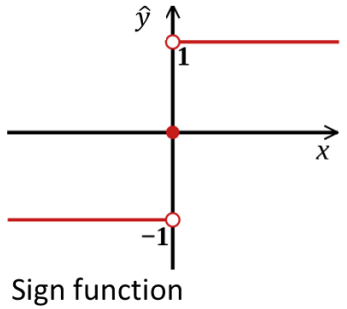
Sign function

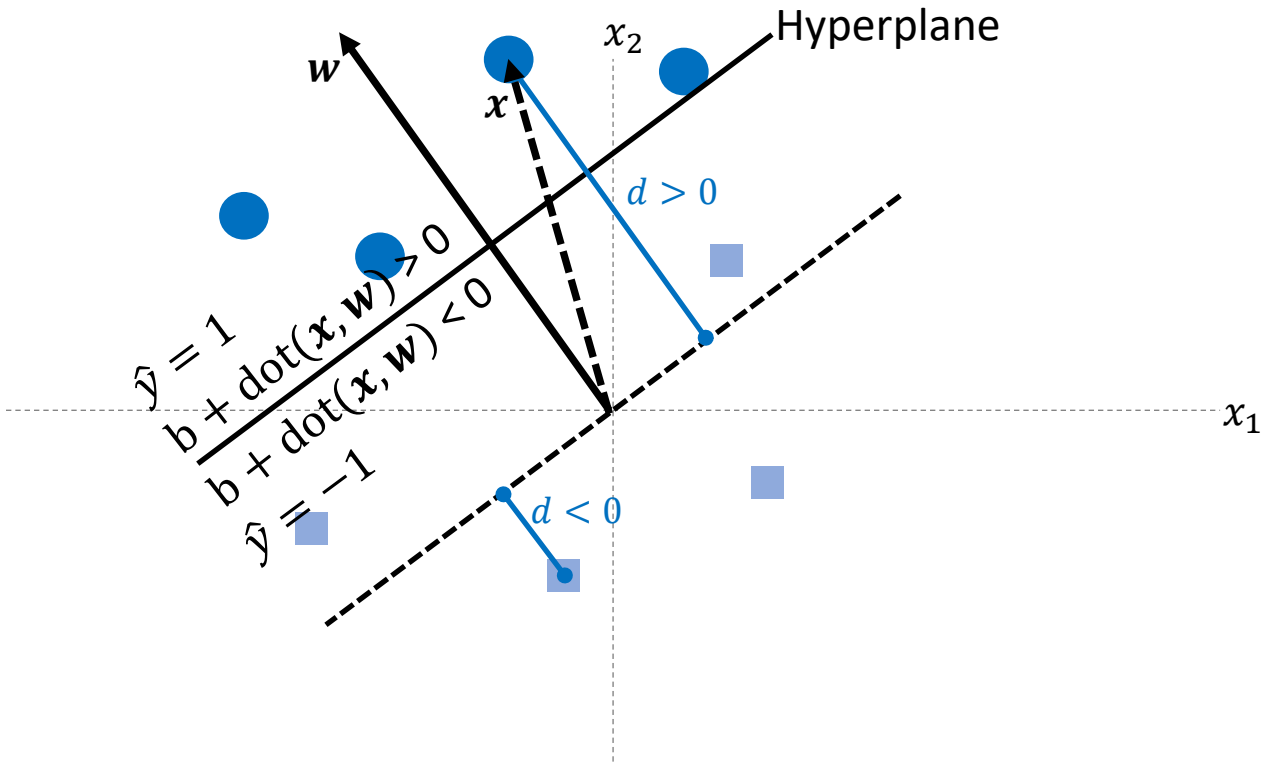


Perceptron Classifier

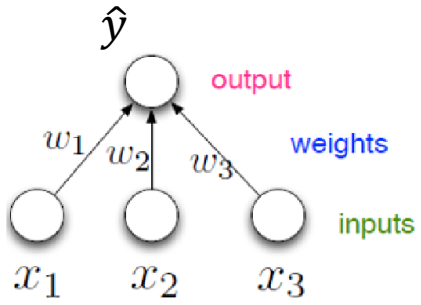


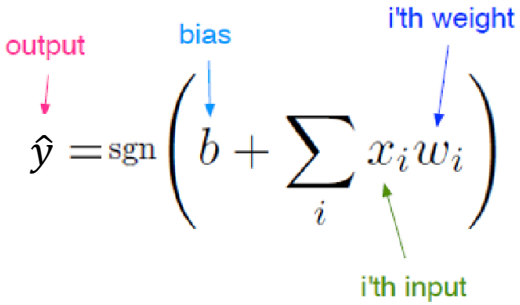


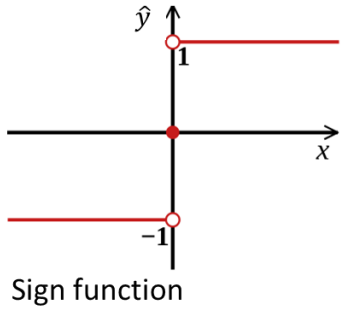




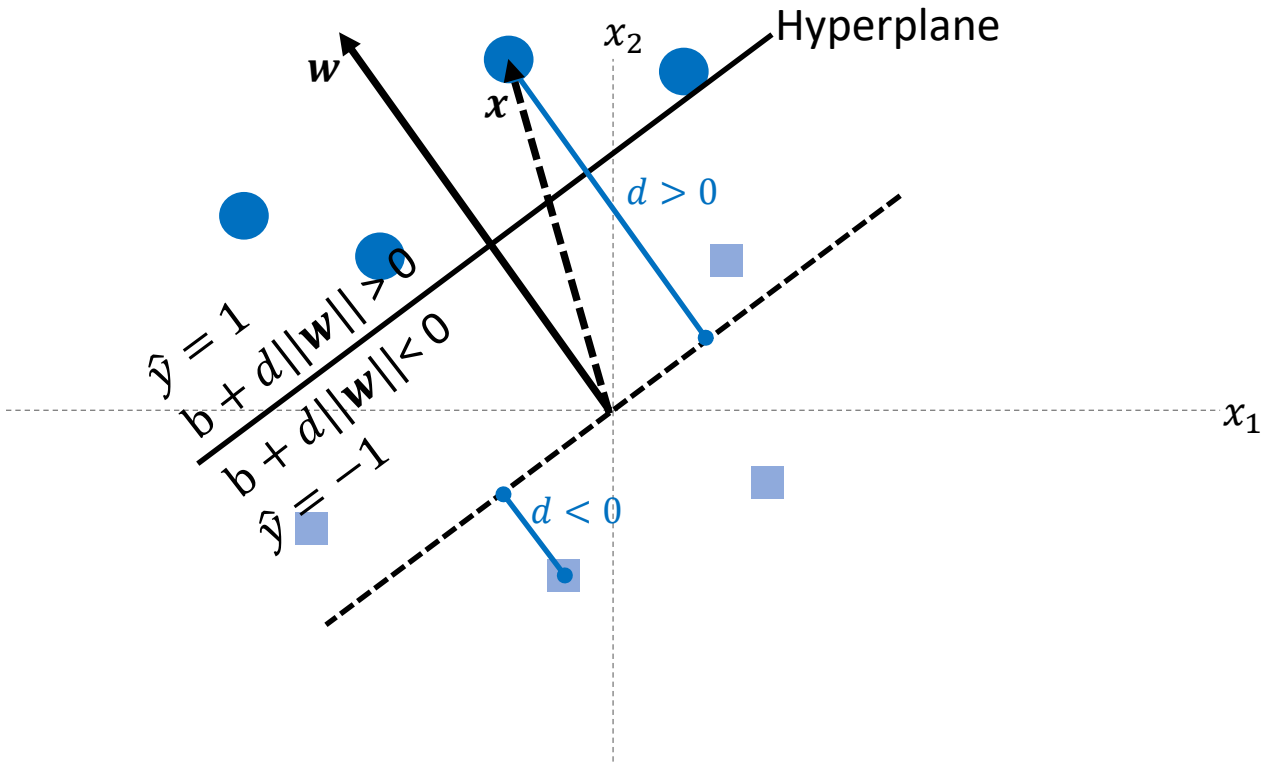
Perceptron Classifier



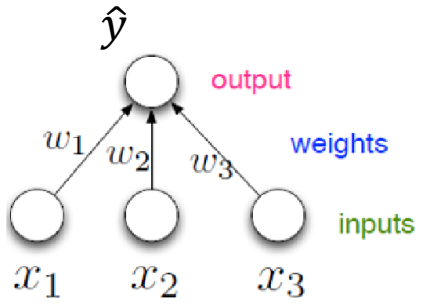


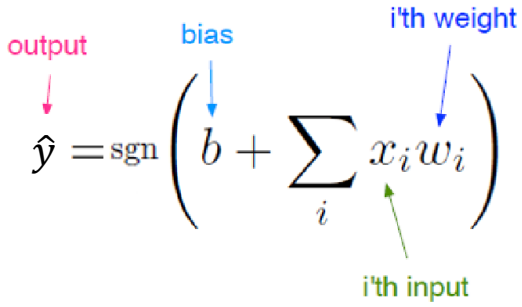


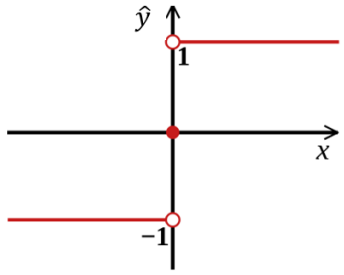
Sign function



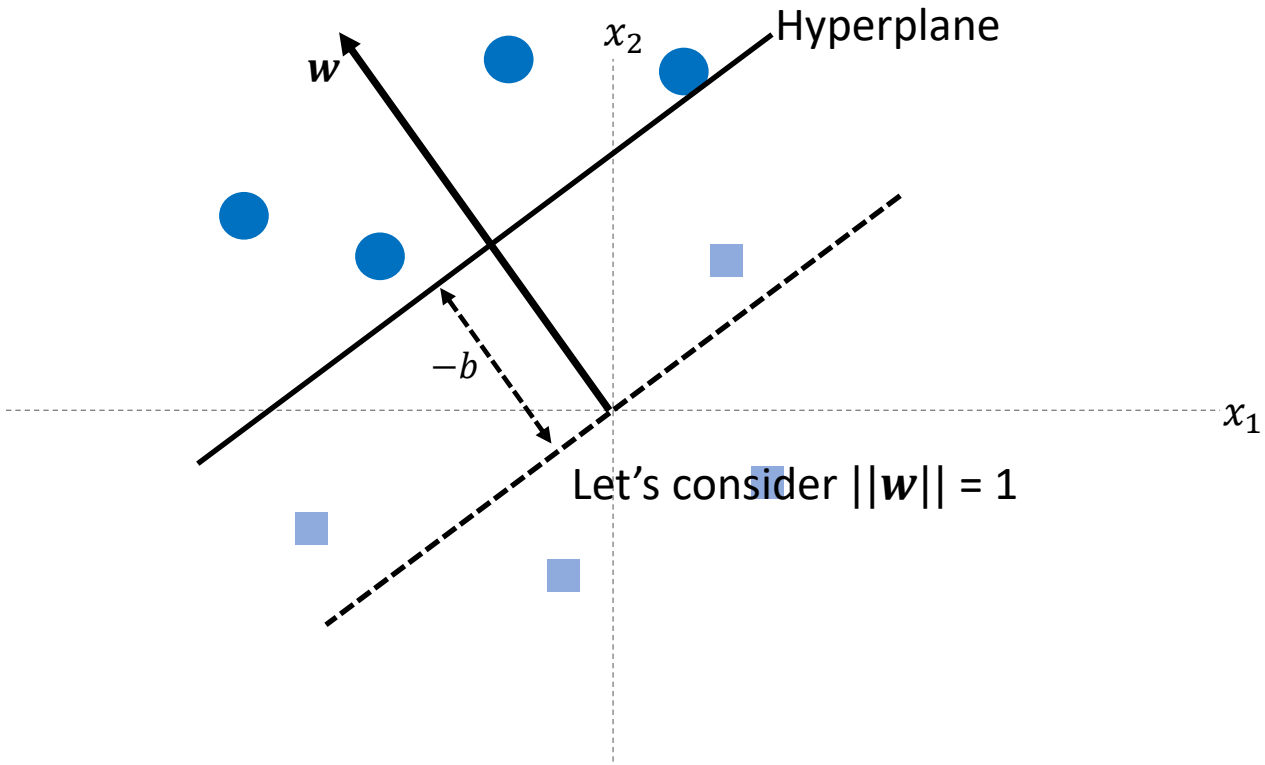
Perceptron Classifier



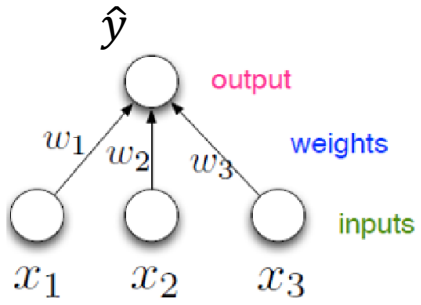




Sign function



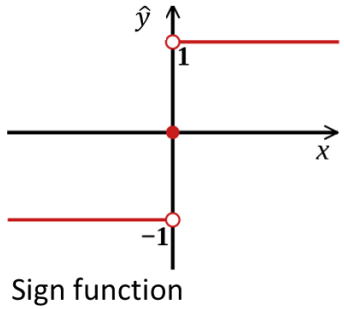
Perceptron Classifier



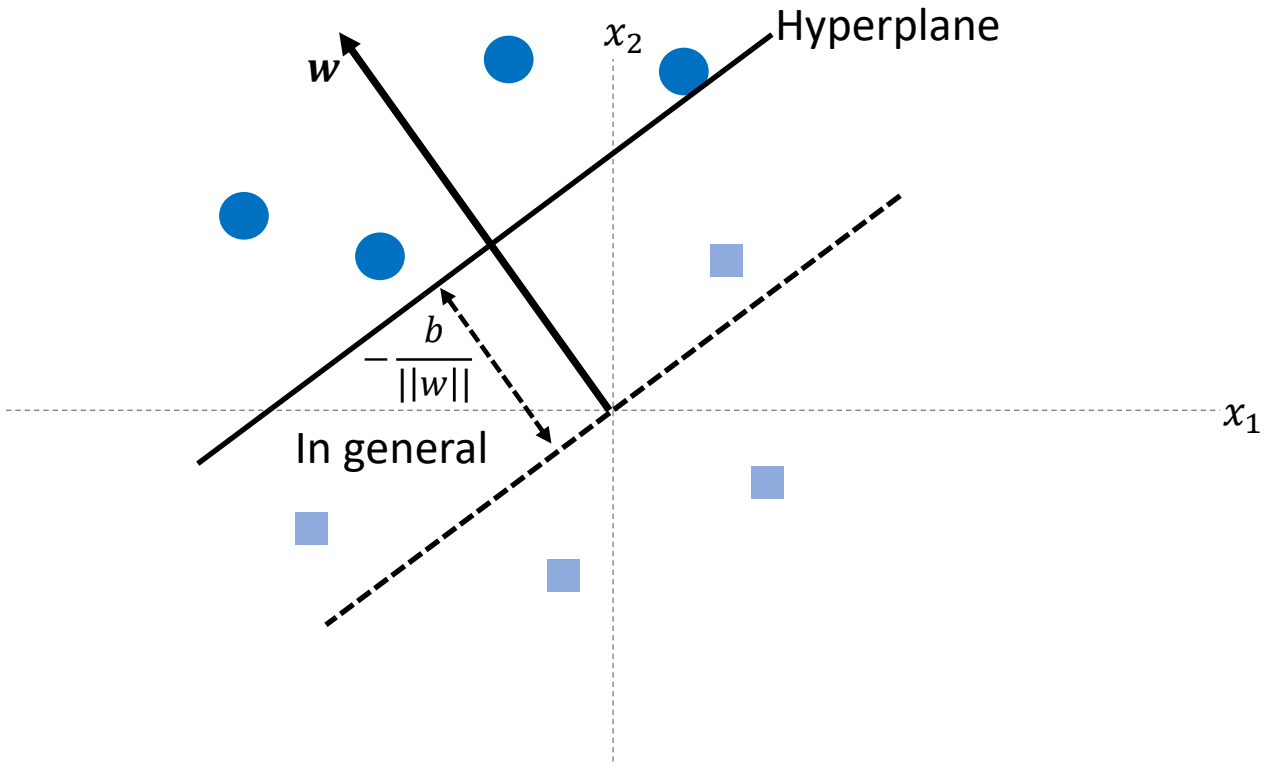
output bias i'th weight

$$\hat{y} = \text{sgn}\left(b + \sum_i x_i w_i\right)$$

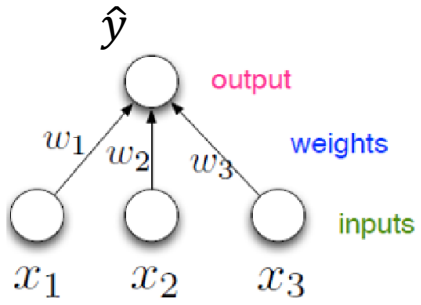
i'th input

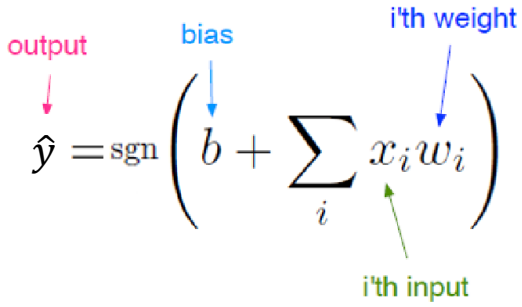


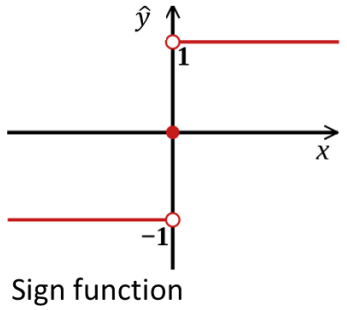
Sign function



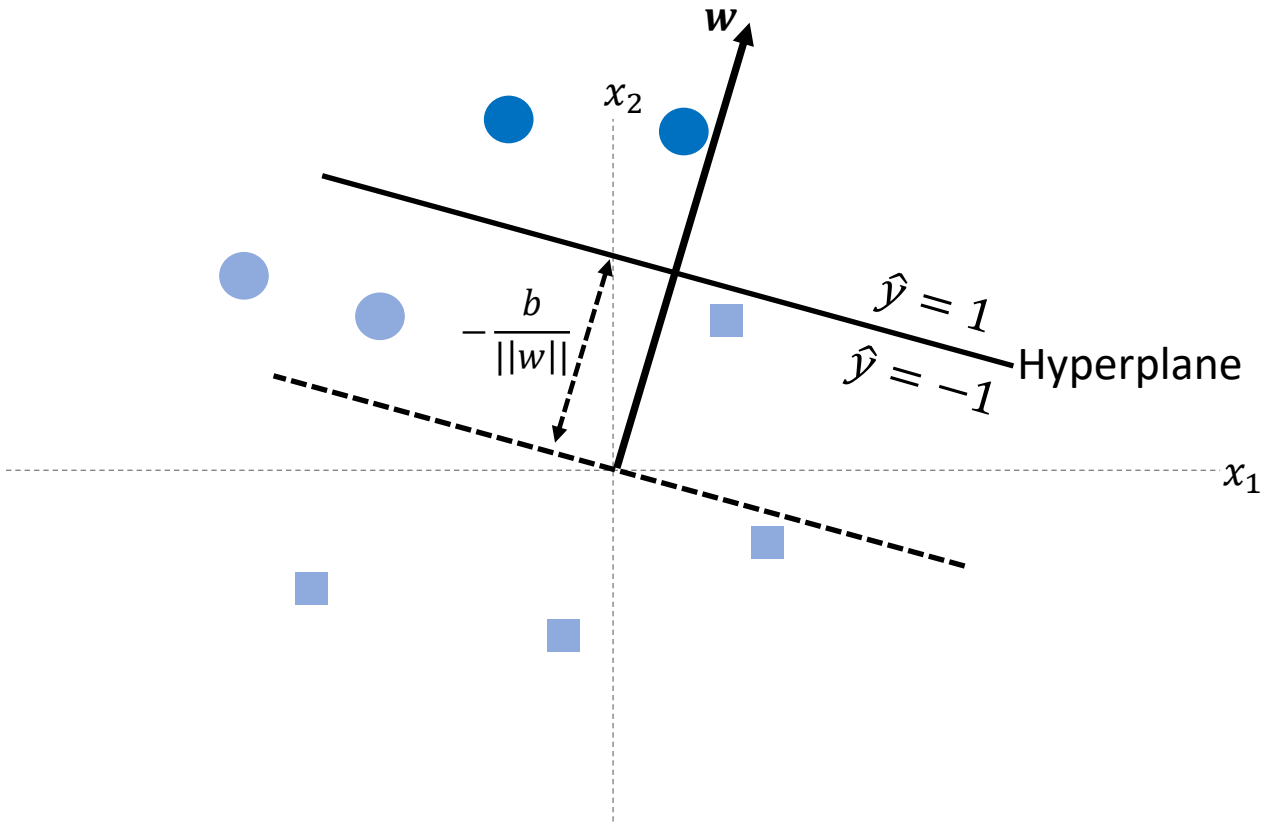
Perceptron Classifier





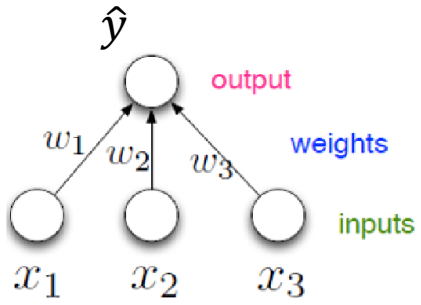


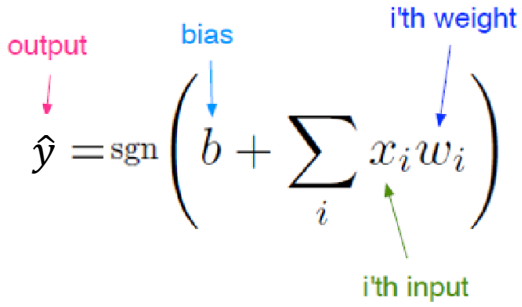
Sign function

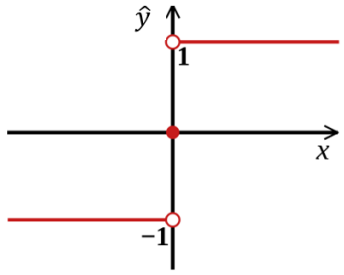


Perceptron Classifier

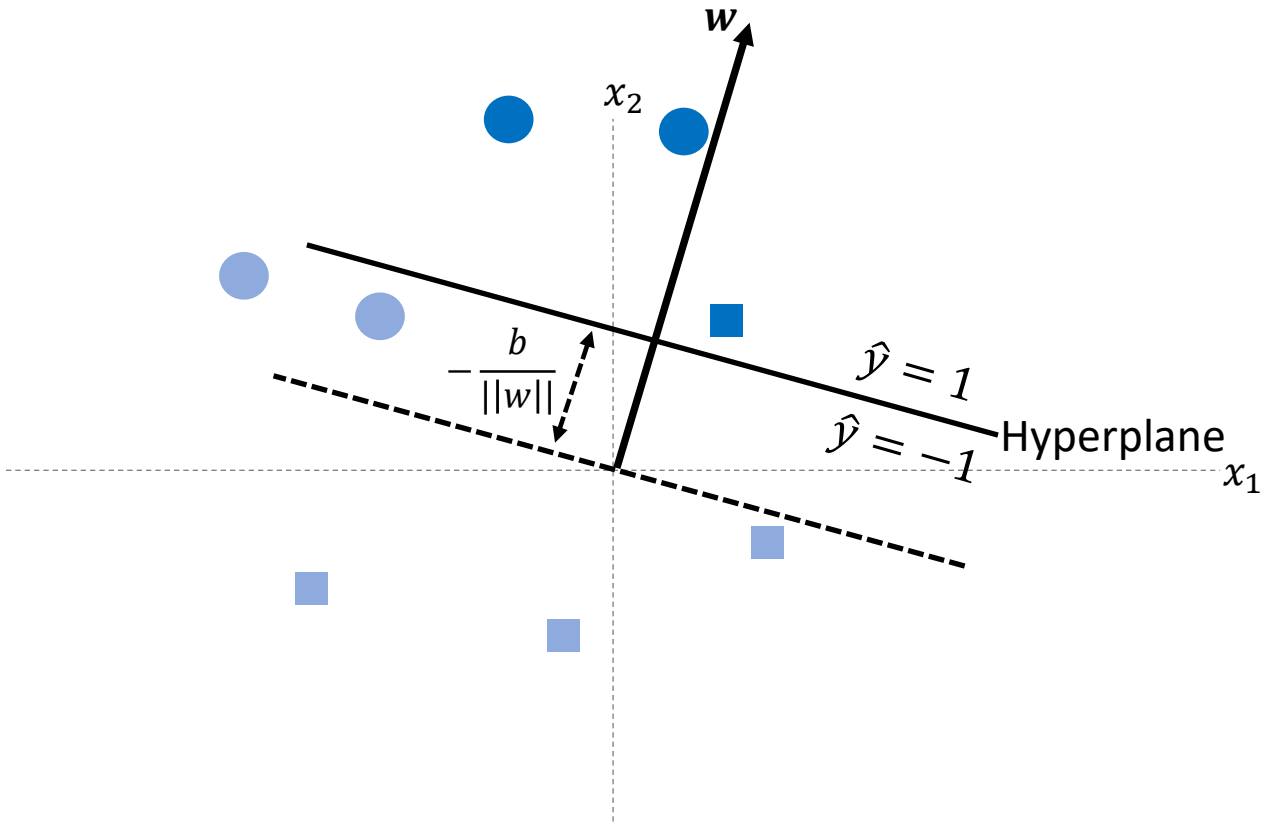
Question:
What if $b = 0$?



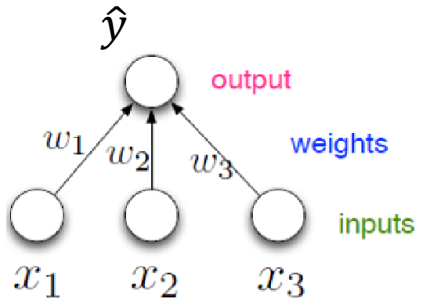


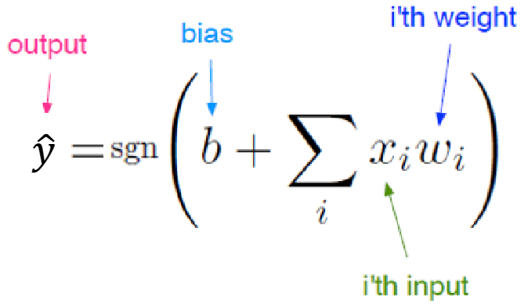


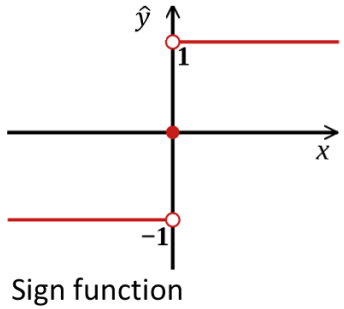
Sign function



Perceptron Classifier







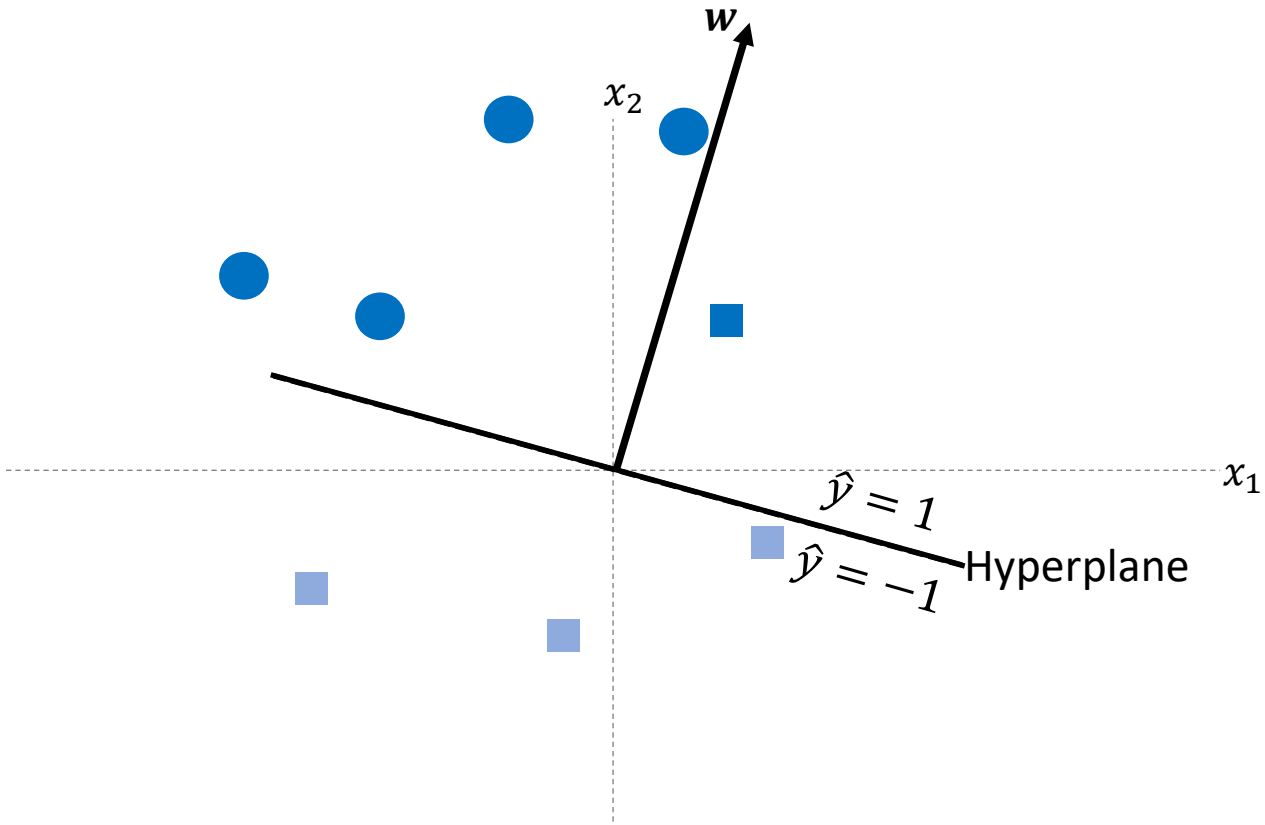
Sign function

Question:

What if $b = 0$?

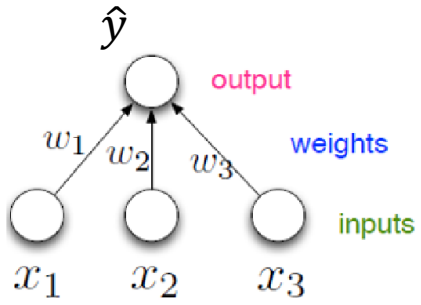
Answer:

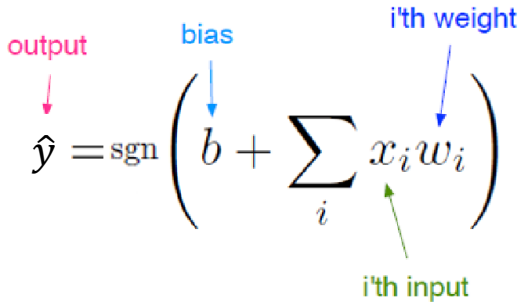
Hyperplane at origin

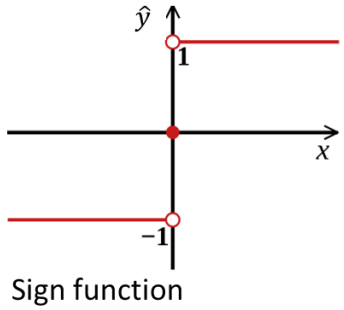


Perceptron Classifier

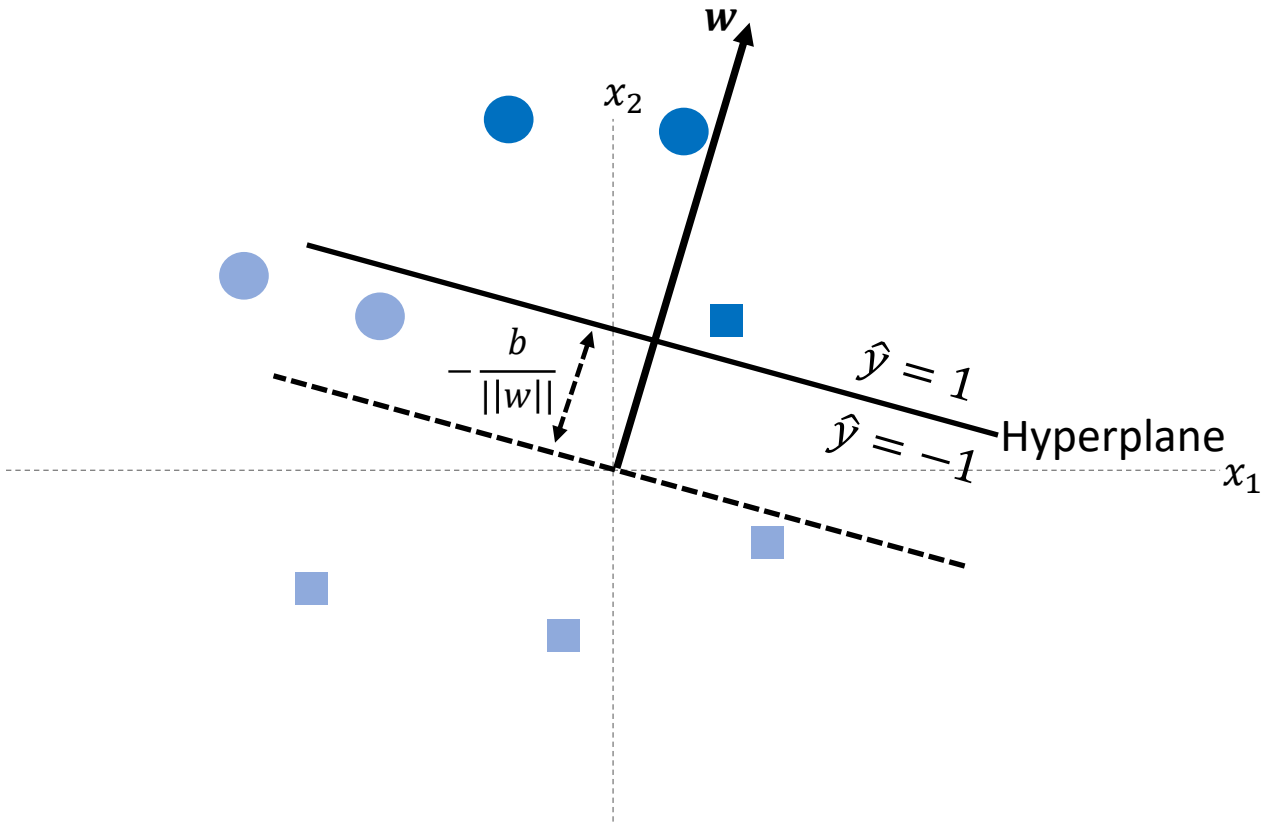
Question:
Is $b < 0$ or is $b > 0$?



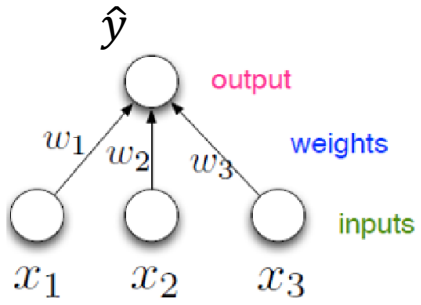


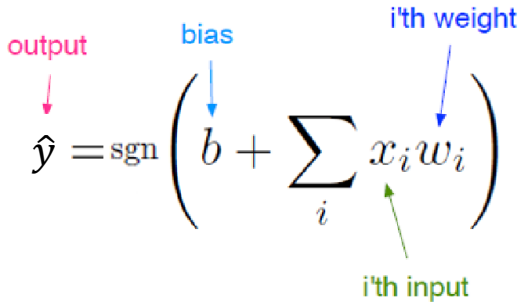


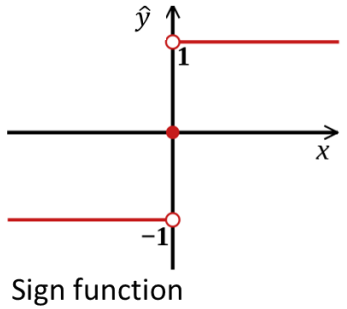
Sign function



Perceptron Classifier







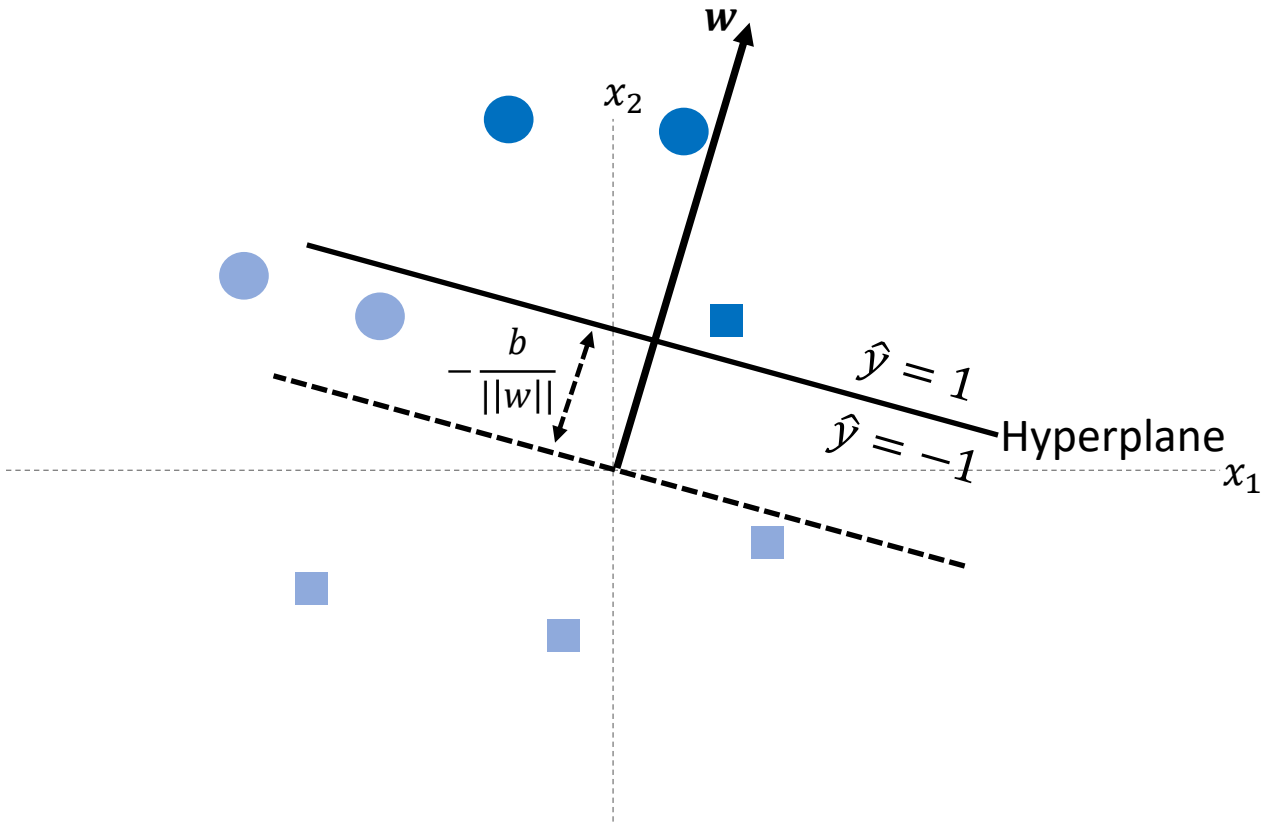
Sign function

Question:

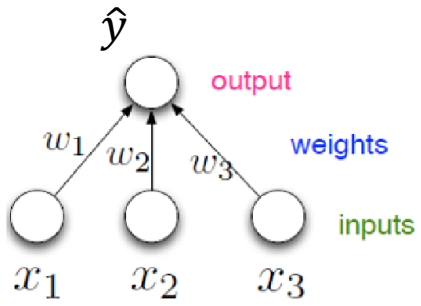
Is $b < 0$ or is $b > 0$?

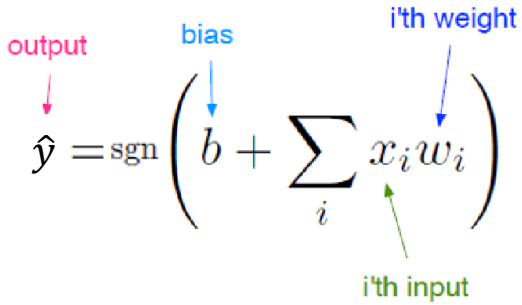
Answer:

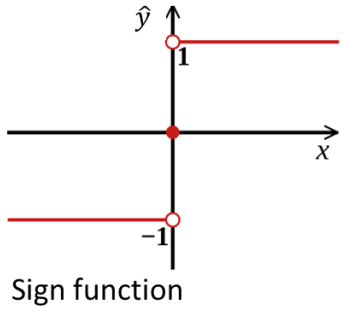
It is negative: $b < 0$.



Perceptron Classifier







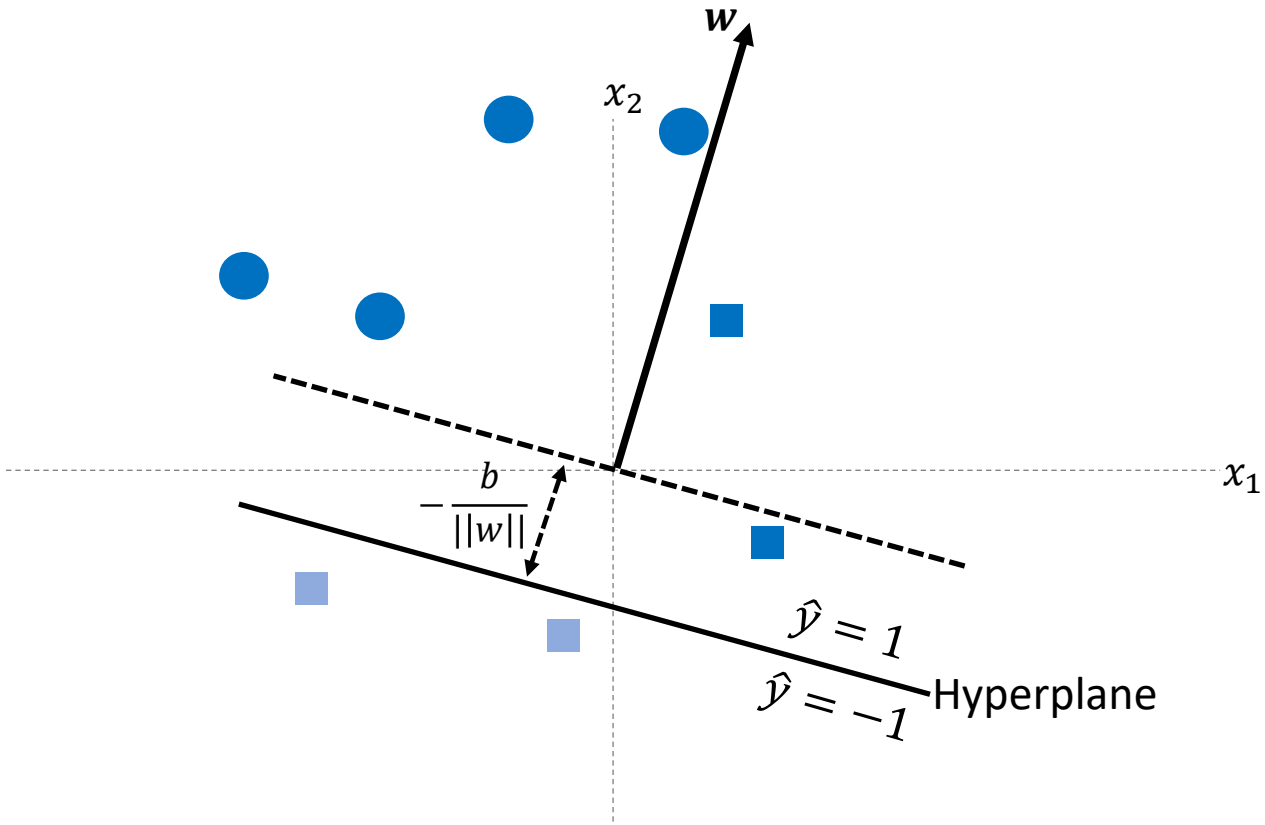
Sign function

Question:

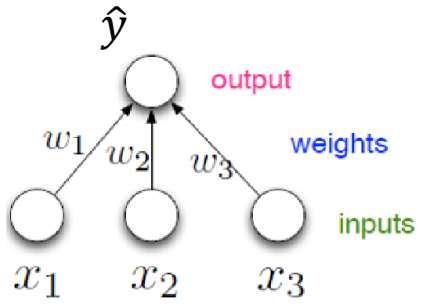
Is $b < 0$ or is $b > 0$?

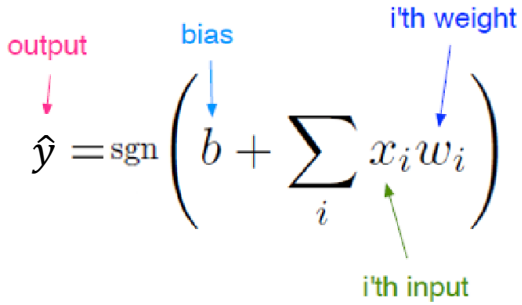
Answer:

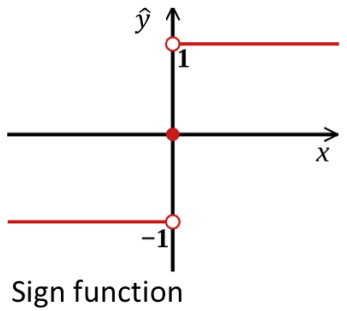
It is positive: $b > 0$.



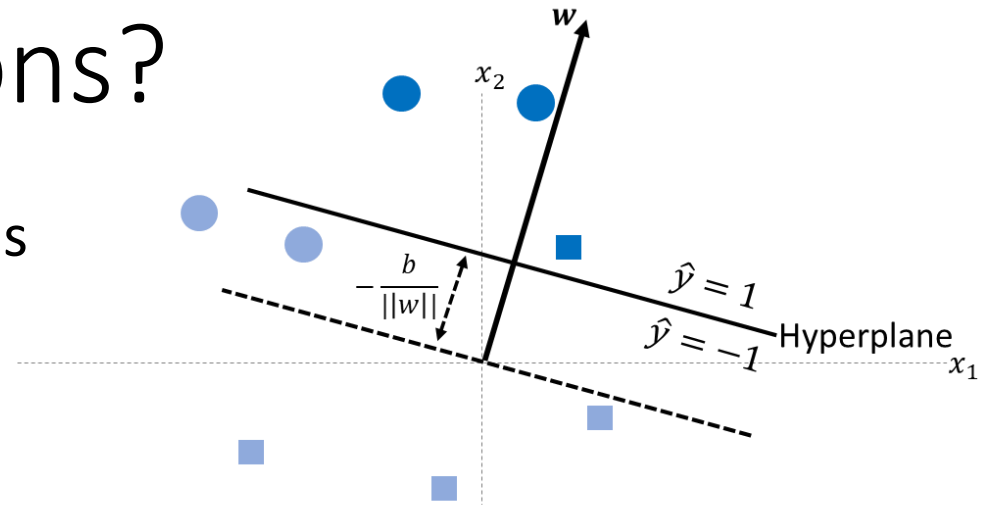
What about higher dimensions?

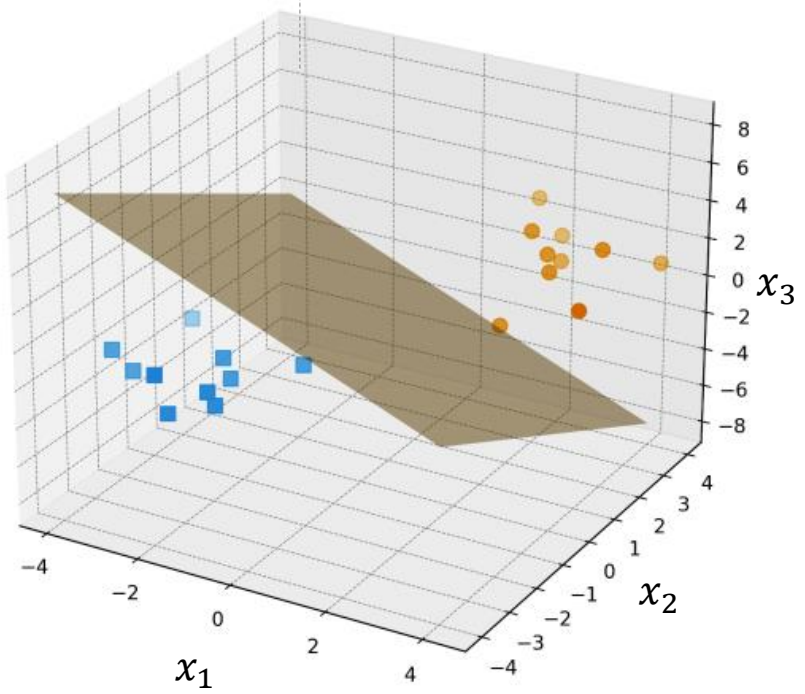




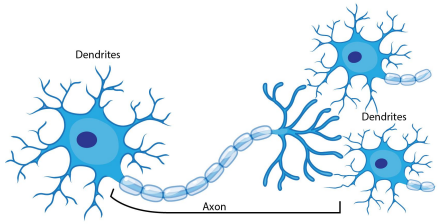


- Everything works in higher dims as well ☺.
- In D-dimensional space, Hyperplane has D-1 dimensions.
- No simple intuition for higher dims ☹.
- Tipp: try to get intuition for 2D/3D and trust the process for higher dims.

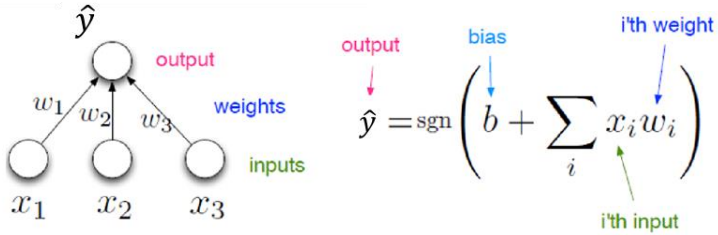




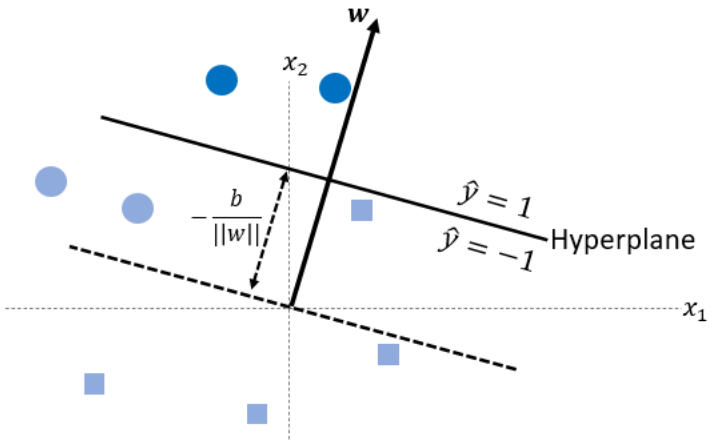
Perceptron Classifier Summary



- Simple model of biological Neuron



- Hyperplane separating points
  - w and b define position and orientation
    - Weight vector w determines orientation
    - Negative b shifts it in direction of w
    - Positive b shifts it against direction of w

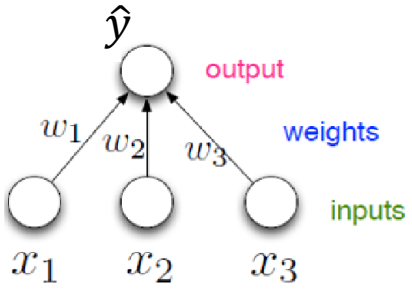


‘Perceptron’ Regressor



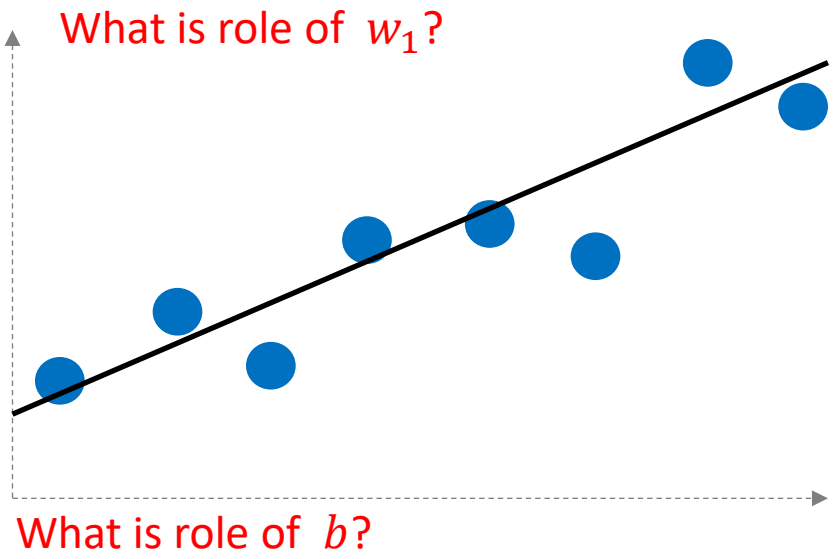
- Predict commute time to University
  - Input variables (features):
    - Distance to University (km)
    - Day of the week (1 for weekday, 0 for weekend)
  - Output / target?
    - Commute time (min)
  - Solution:
    - Perceptron without sign-function
    - Equal to standard linear regression

Dist. (km) x_1	Day x_2	Commute (min) y
2.7	1	25
4.1	1	33
1.0	0	15
5.2	1	45
2.8	0	22

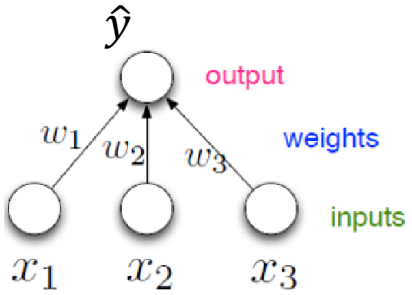


$$\hat{y} = b + \sum_{i}^{D} w_i x_i$$

‘Perceptron’ Regressor - 1D input



- Predict commute time to University
  - Input variables (features):
    - Distance to University (km)
    - Day of the week (1 for weekday, 0 for weekend)
  - Output / target?
    - Commute time (min)
  - Solution:
    - Perceptron without sign-function
    - Equal to standard linear regression

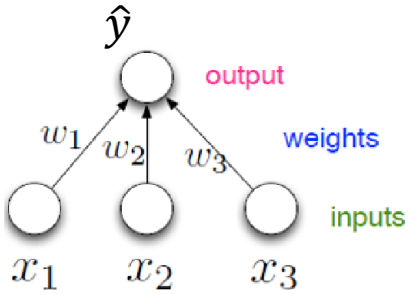


$$\hat{y} = b + \sum_{i}^{D} w_i x_i$$

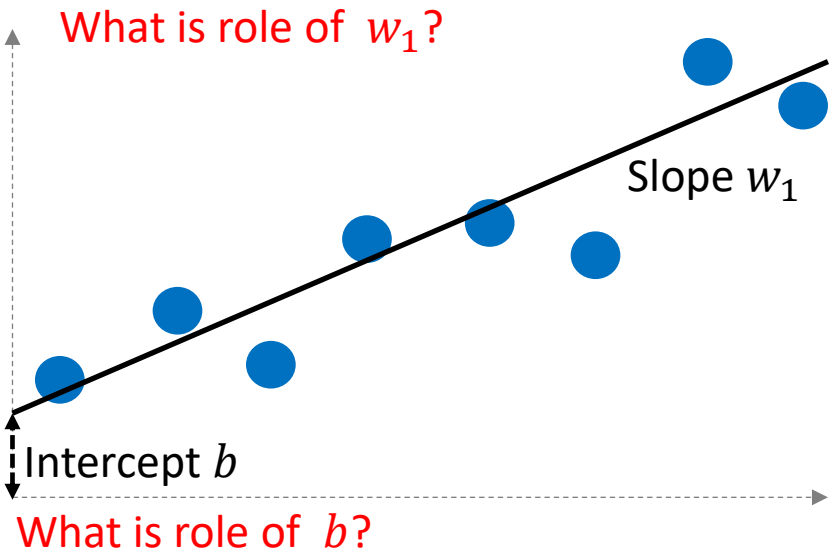
Dist. (km) x_1	Commute (min) y
2.7	25
4.1	33
1.0	15
5.2	45
2.8	22

‘Perceptron’ Regressor - 1D input

- Predict commute time to University
  - Input variables (features):
    - Distance to University (km)
    - Day of the week (1 for weekday, 0 for weekend)
  - Output / target?
    - Commute time (min)
  - Solution:
    - Perceptron without sign-function
    - Equal to standard linear regression



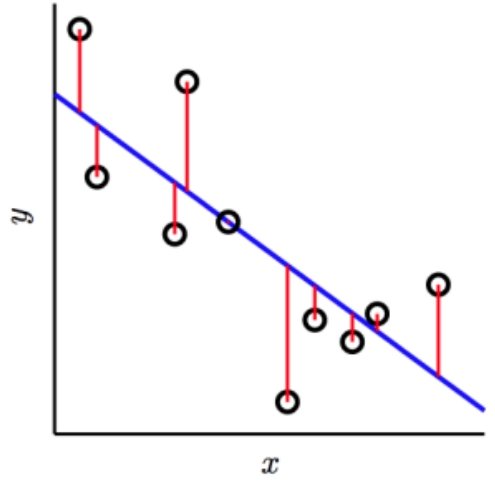
$$\hat{y} = b + \sum_{i}^{D} w_i x_i = b + w_1 x_1$$



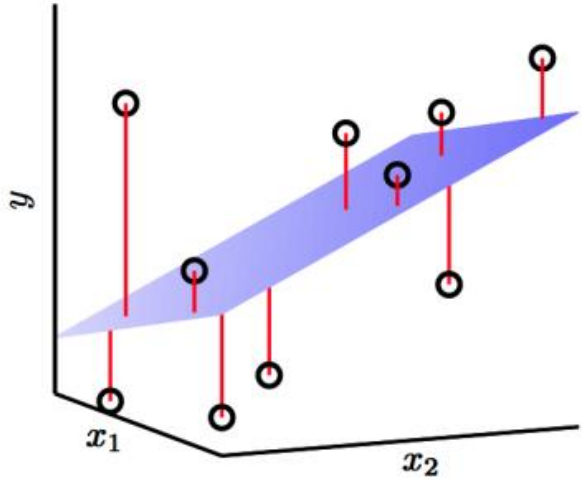
Dist. (km) x_1	Commute (min) y
2.7	25
4.1	33
1.0	15
5.2	45
2.8	22

Perceptron Regressor Summary

- Perceptron without sign function is linear regressor

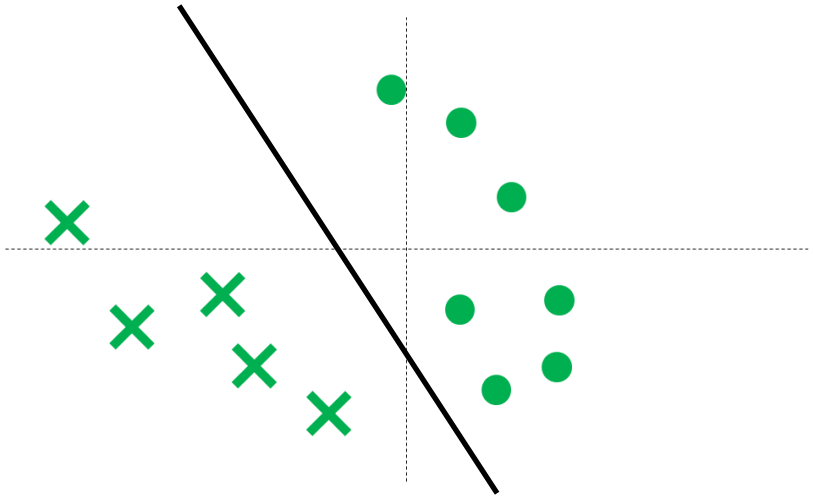


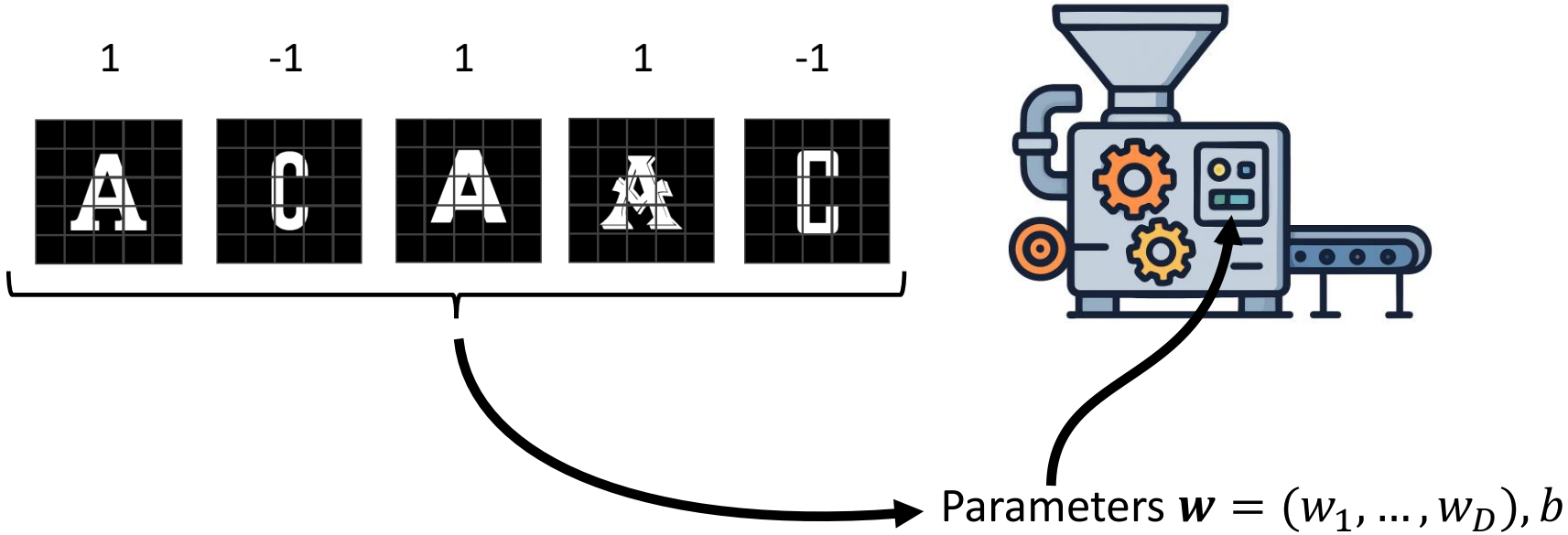
- Output is linear function of inputs
  - w and b define function
    - Weight vector w determines slope in each direction
    - Bias b shifts function up and down
  - Don't mix up with hyperplane in classifier!



Outlook

- How do we find w and b?
  - Adjusting parameters from data to solve task
  - Called supervised learning/training (week 2)
  - For regression and classification

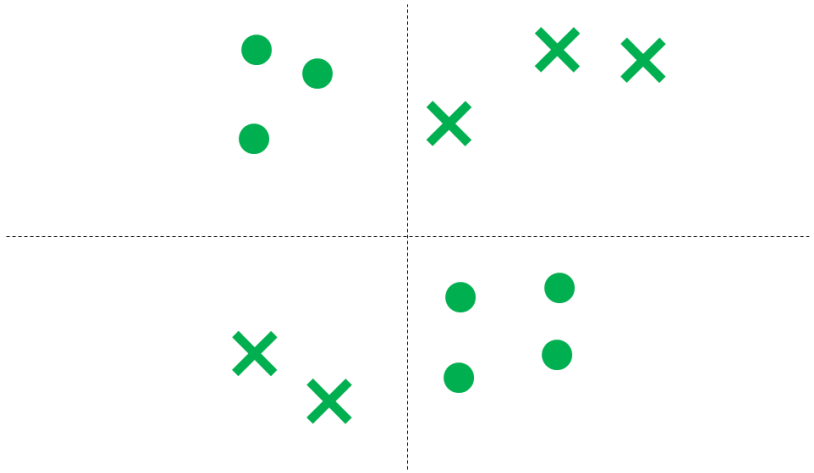




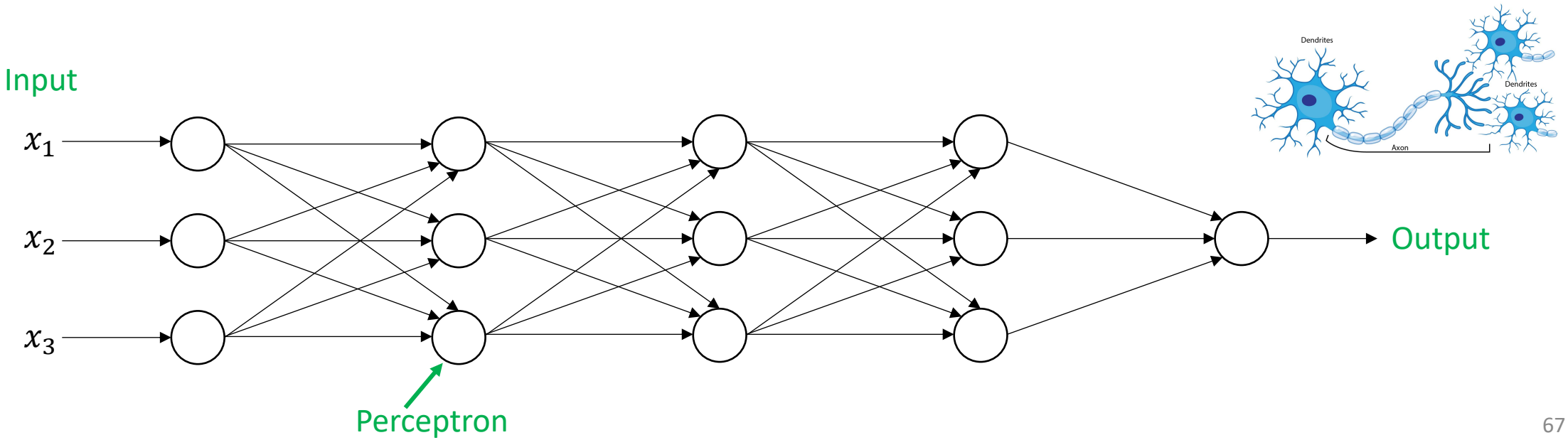
- But first revisiting …
  - Optimisation problems
  - Maximum likelihood

(Rest of week 1)

Outlook

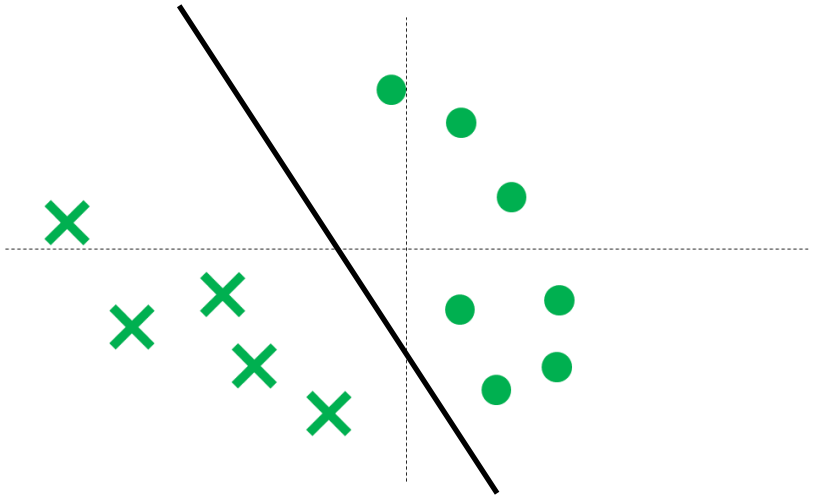


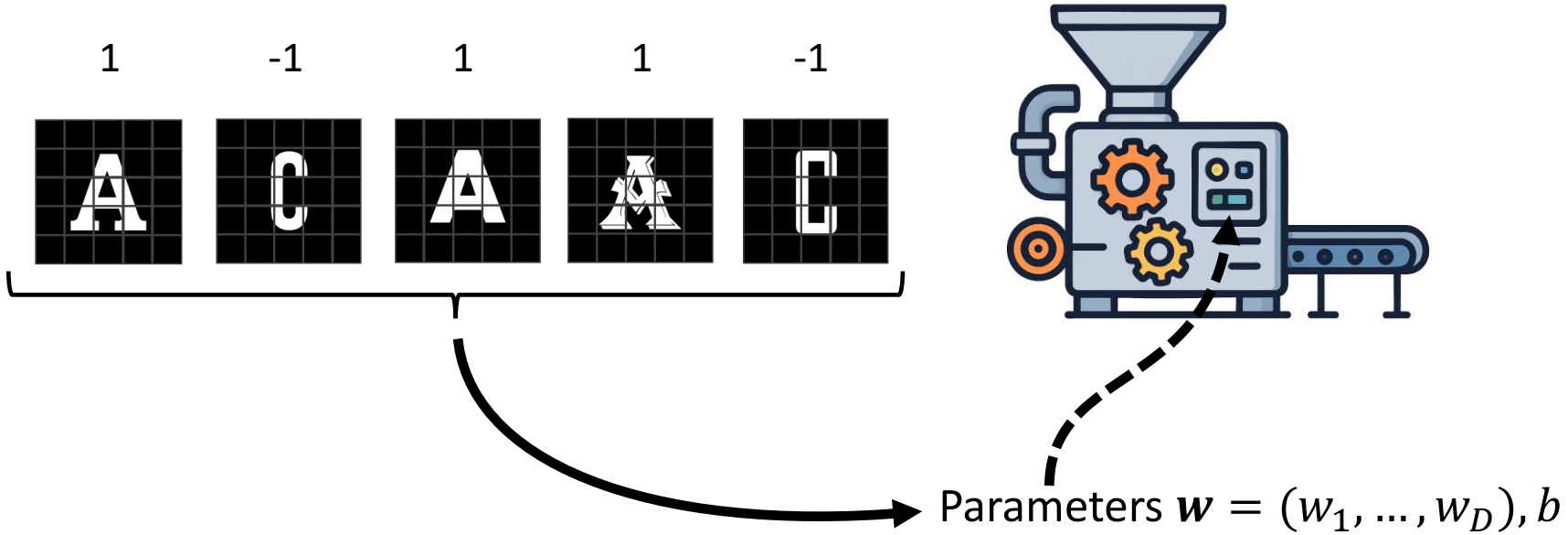
- What about more difficult data?
  - Data can be not-linearly-separable
  - Combine multiple neurons/perceptrons
  - Train them together (week 3)



Outlook - Learning

- How do we find w and b?
  - Adjusting parameters from data to solve task
  - Called supervised learning/training (week 2)
  - For regression and classification





- But first revisiting …
  - Optimisation problems
  - Maximum likelihood

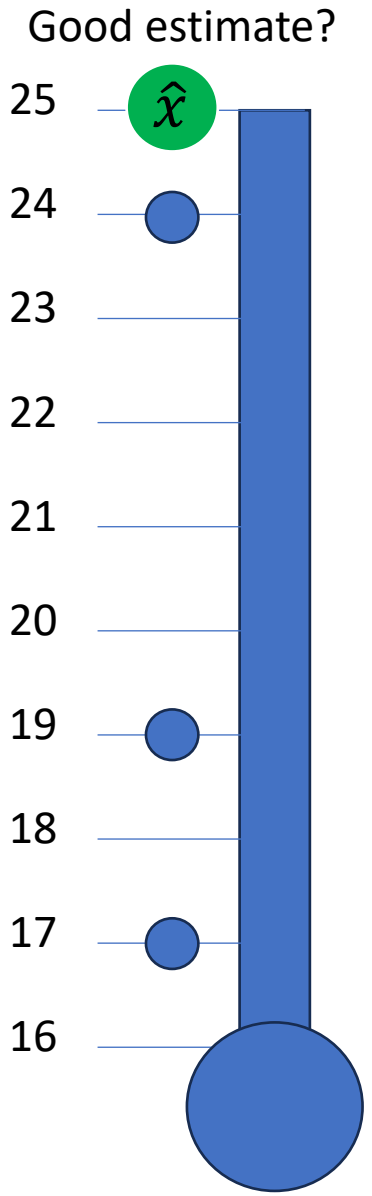
  (This is a tool we need)

Optimisation Problems

- Find estimate of true temperature
- Measurements:

$$x_1 = 19, x_2 = 17, x_3 = 24$$

$$\boldsymbol{x} = (x_1, x_2, x_3)$$



Optimisation Problems – Absolute Error

- Find estimate of true temperature
- Measurements:

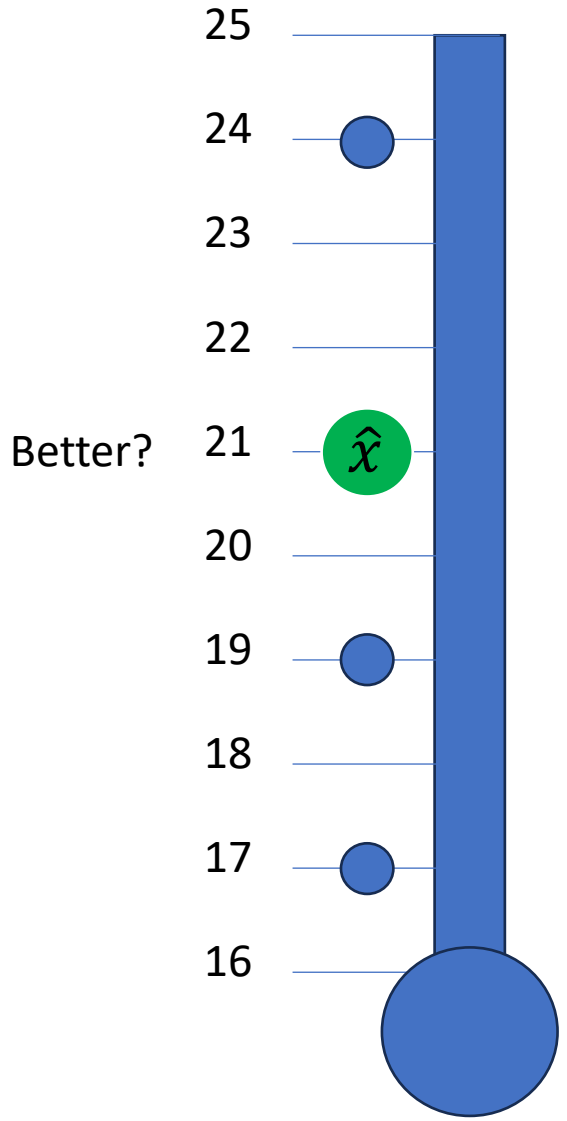
$$x_1 = 19, x_2 = 17, x_3 = 24$$

$$\boldsymbol{x} = (x_1, x_2, x_3)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x})$$

$$L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$



Optimisation Problems – Absolute Error

- Find estimate of true temperature
- Measurements:

$$x_1 = 19, x_2 = 17, x_3 = 24$$

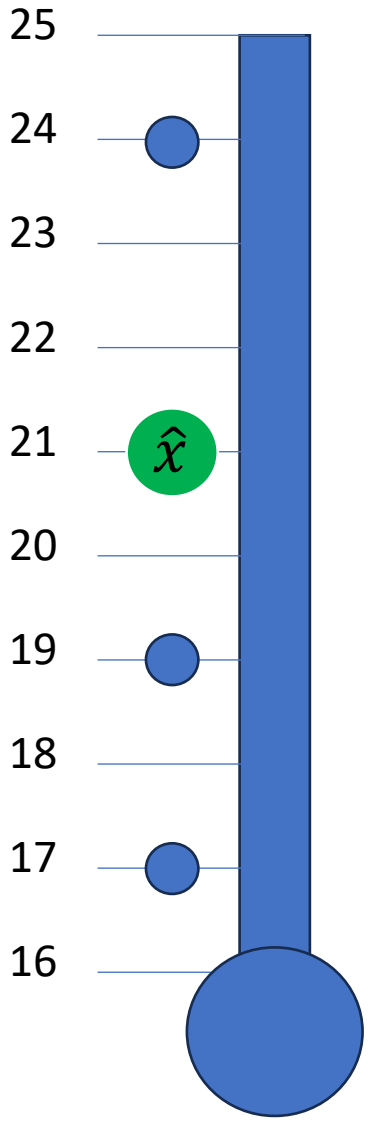
$$\boldsymbol{x} = (x_1, x_2, x_3)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x})$$

$$L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$

$$L(\hat{x} = 21, \boldsymbol{x}) =$$



Optimisation Problems – Absolute Error

- Find estimate of true temperature
- Measurements:

$$x_1 = 19, x_2 = 17, x_3 = 24$$

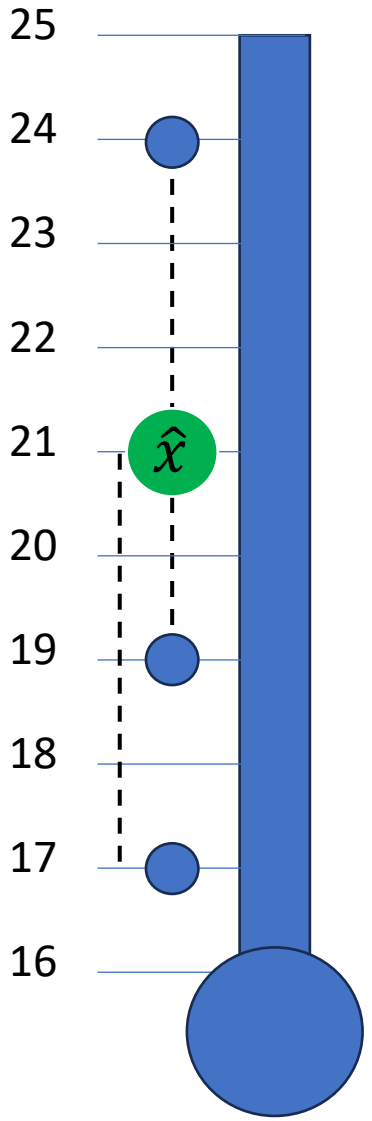
$$\boldsymbol{x} = (x_1, x_2, x_3)$$

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Optimisation Problems – Absolute Error

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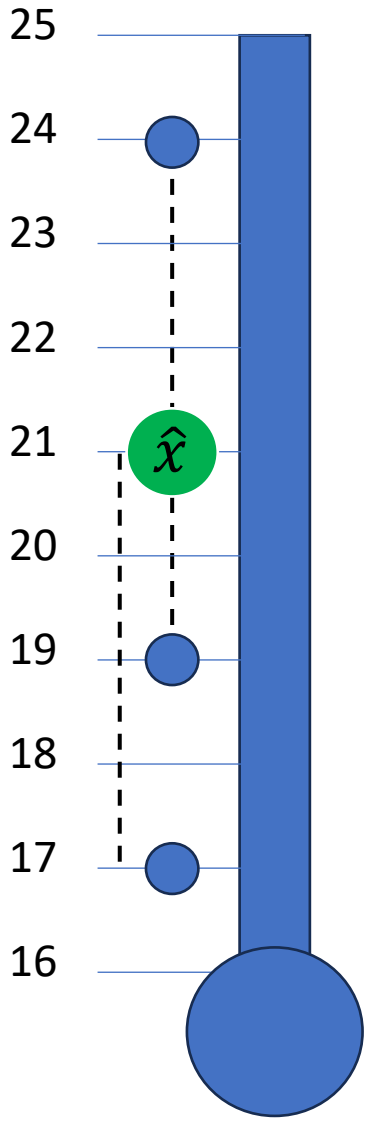
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$$L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$

$L(\hat{x} = 21, \boldsymbol{x}) = 3+2+4=9$



Optimisation Problems – Absolute Error

- Find estimate of true temperature
- Measurements:

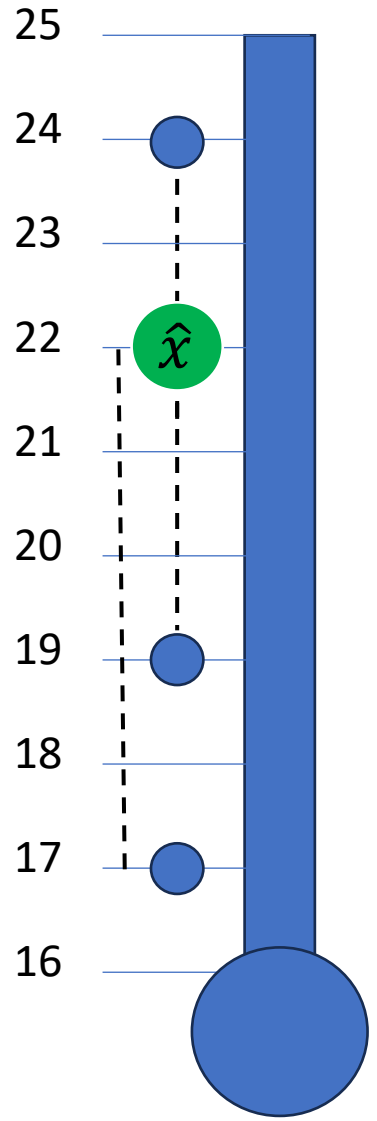
$$x_1 = 19, x_2 = 17, x_3 = 24$$

$$\boldsymbol{x} = (x_1, x_2, x_3)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x}) \qquad L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$

$L(\hat{x} = 22, \boldsymbol{x}) = 2+3+5=10$



Optimisation Problems – Absolute Error

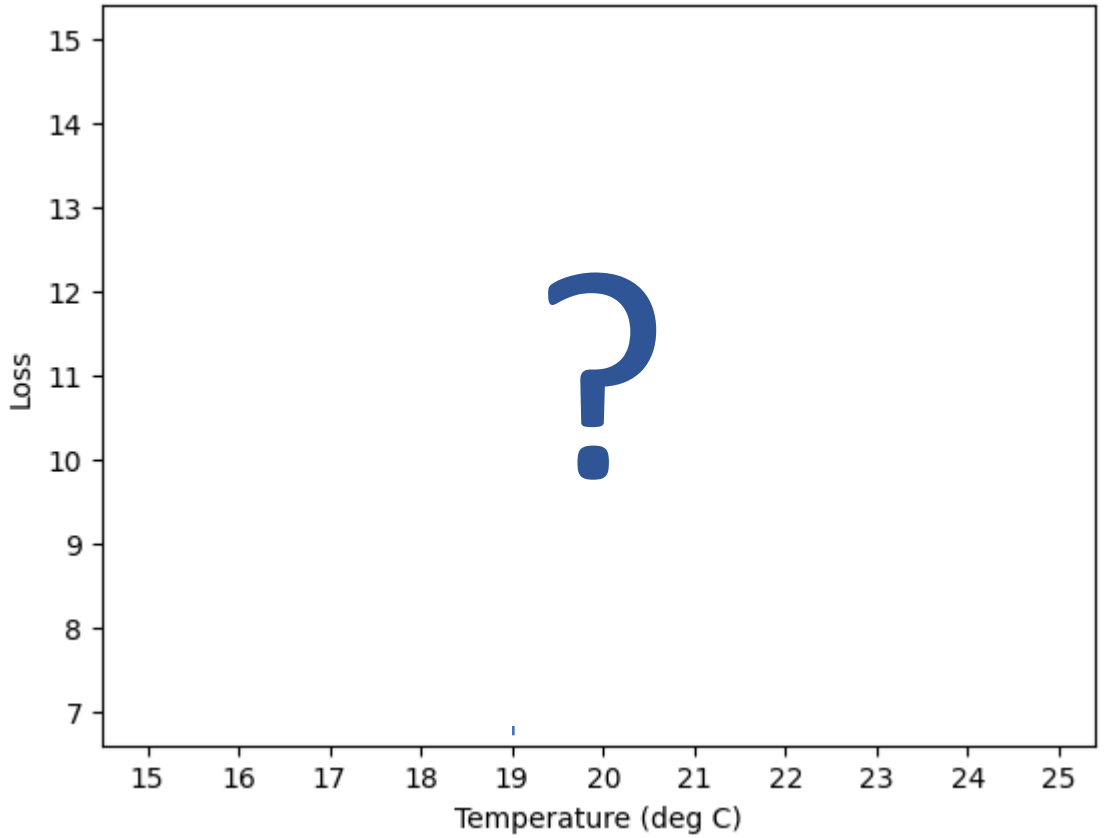
- Find estimate of true temperature
- Measurements:

$$x_1 = 19, x_2 = 17, x_3 = 24$$

$$\boldsymbol{x} = (x_1, x_2, x_3)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x}) \qquad L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$



Optimisation Problems – Absolute Error

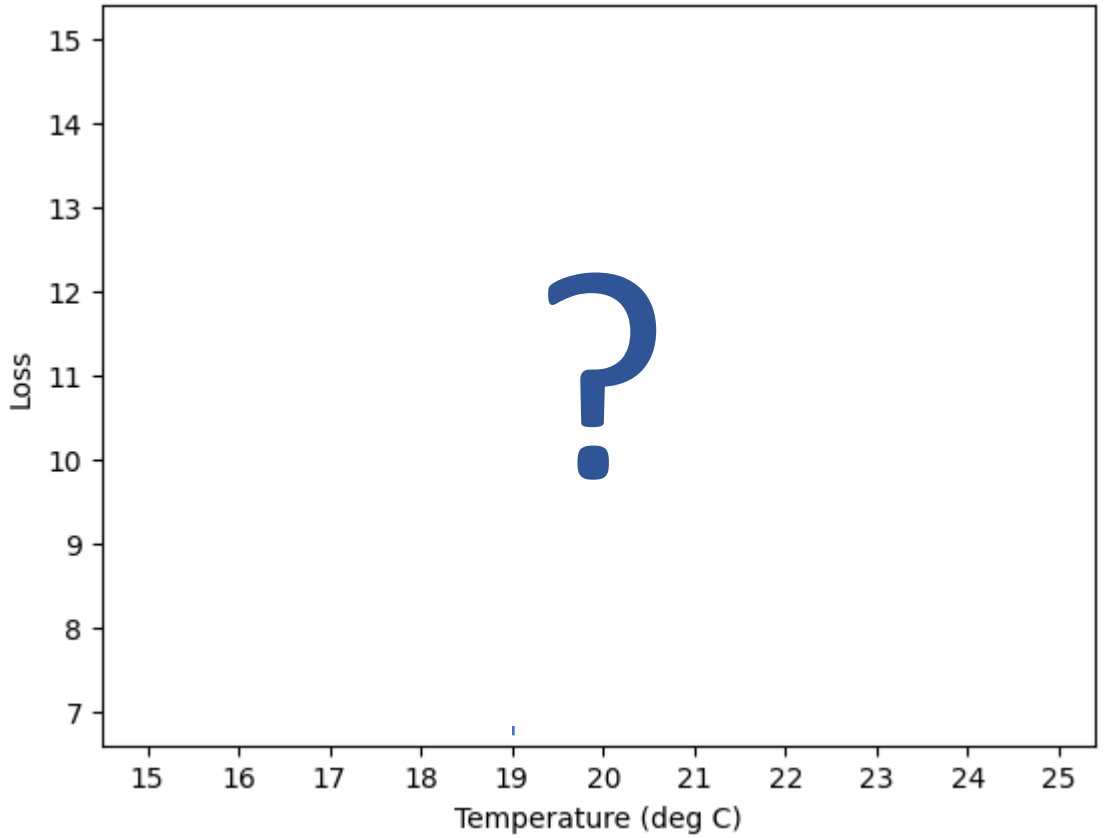
- Find estimate of true temperature
- Measurements:

$$x_1 = 19$$

$$\boldsymbol{x} = (x_1)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x}) \qquad L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$



Optimisation Problems – Absolute Error

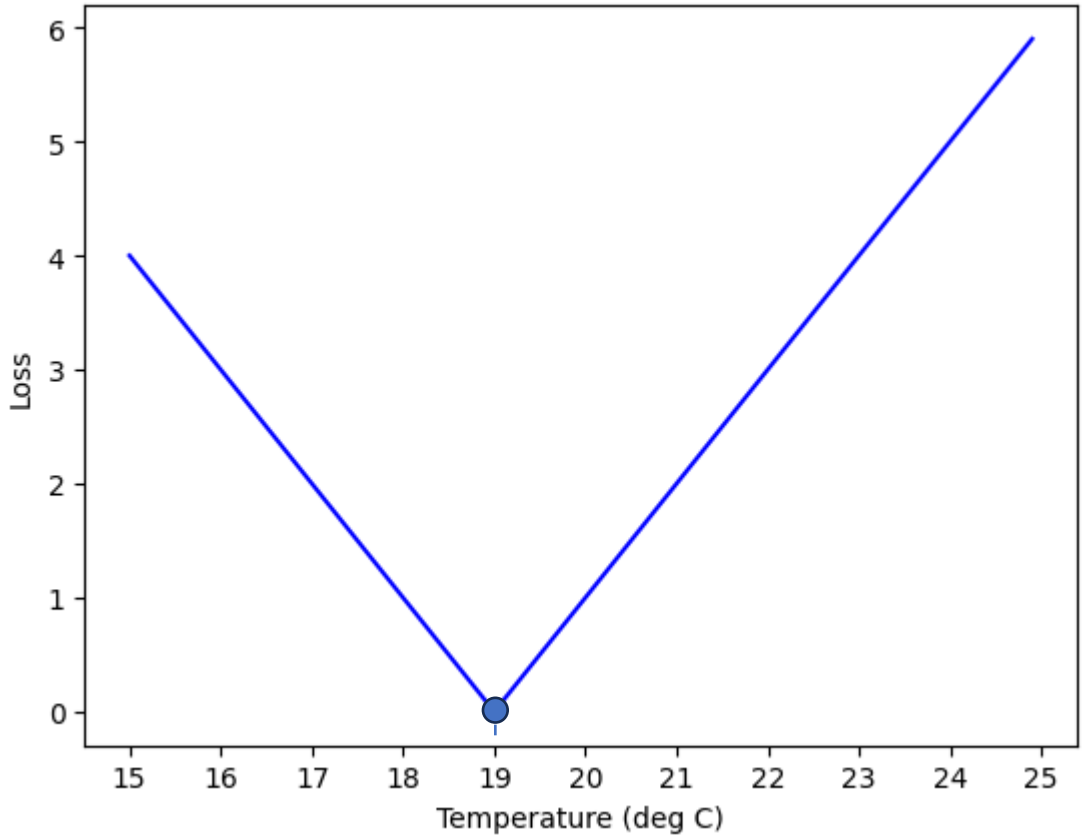
- Find estimate of true temperature
- Measurements:

$$x_1 = 19$$

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Optimisation Problems – Absolute Error

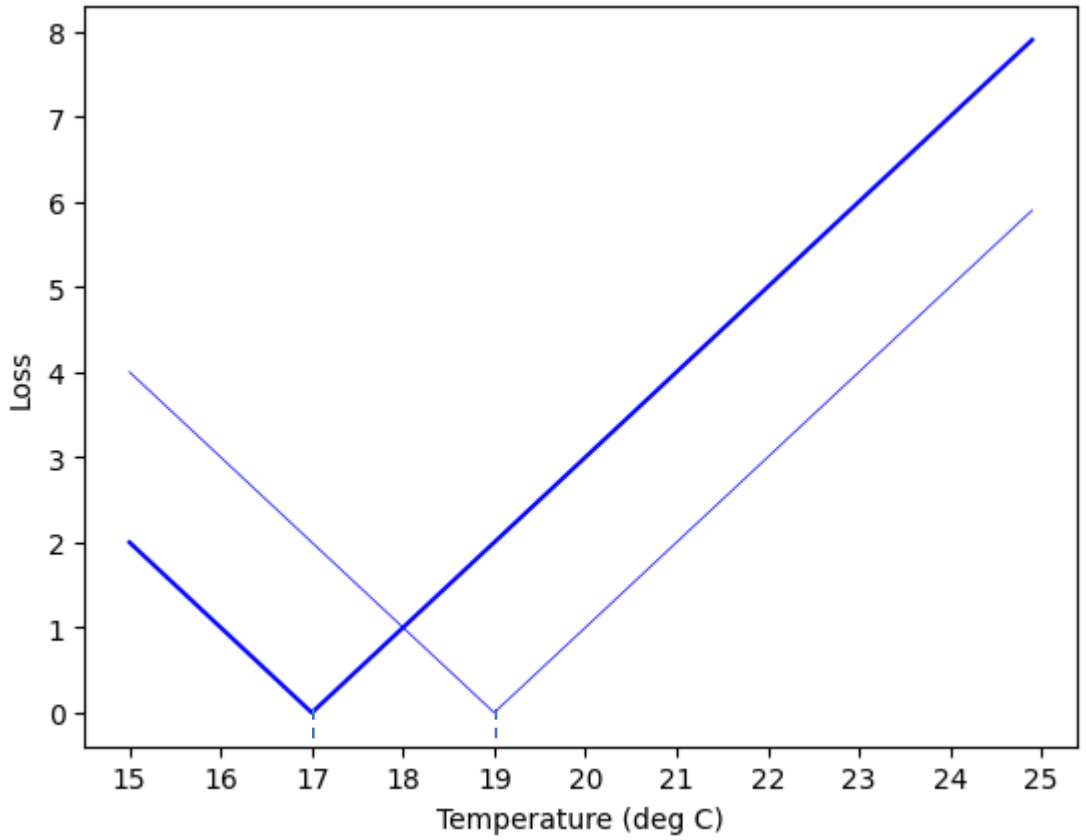
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- Measurements:

$$x_1 = 19, x_2 = 17$$

$$\boldsymbol{x} = (x_1, x_2)$$

- Strategy: find estimate that minimises loss.

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Optimisation Problems – Absolute Error

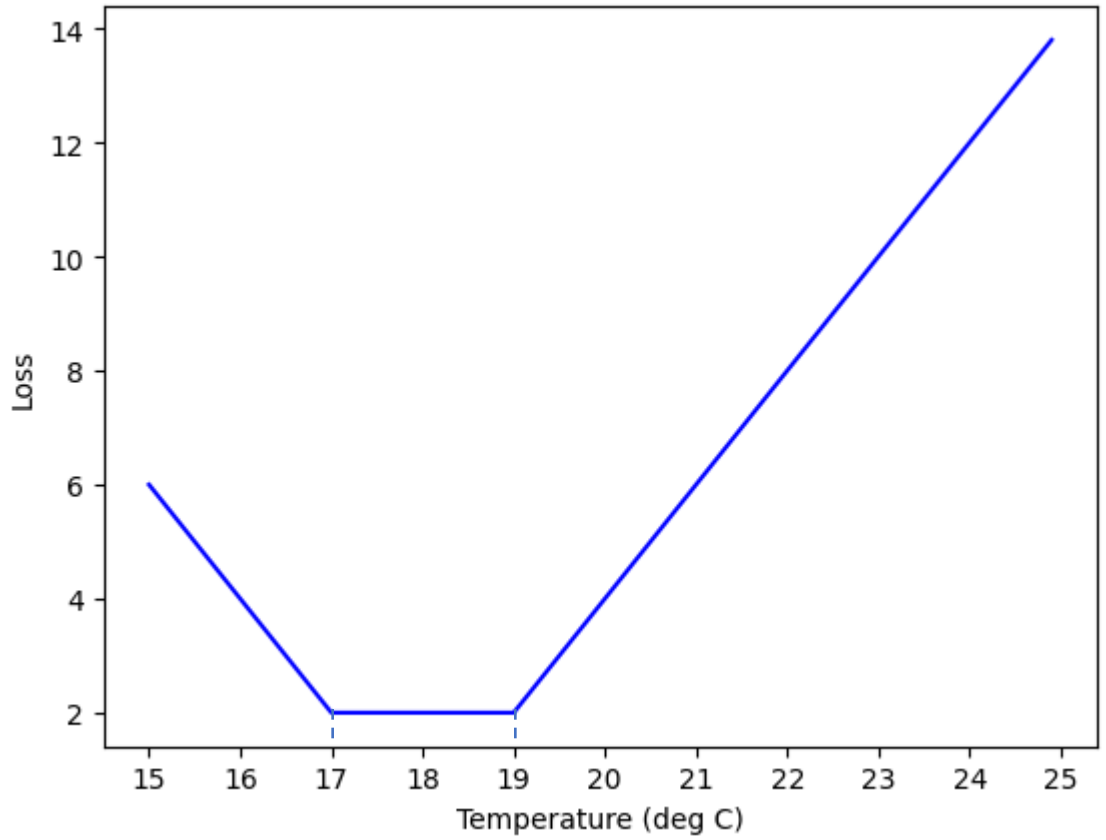
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Optimisation Problems – Absolute Error

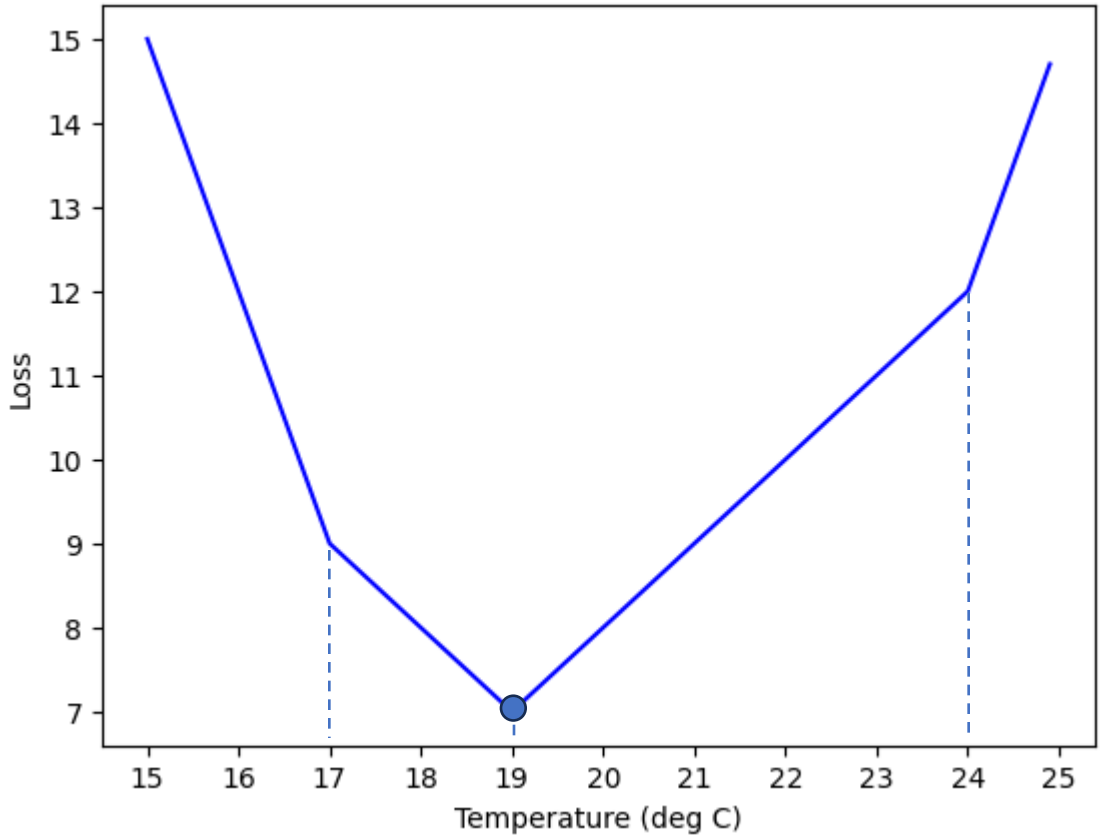
- Find estimate of true temperature
- Measurements:

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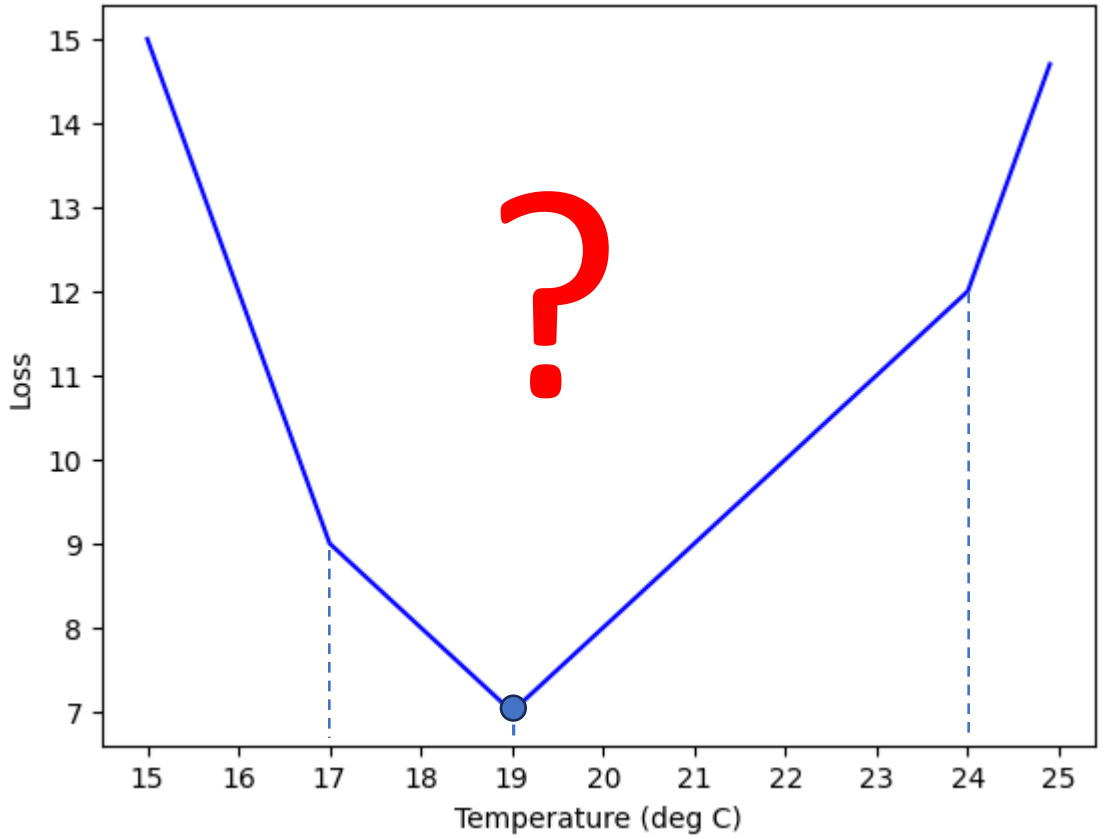
Optimisation Problems – Mean Absolute Error

$$L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$

Question:

What will happen if we use the mean instead of sum?

$$L_{\text{avg}}(\hat{x}, \boldsymbol{x}) = \frac{1}{n} \sum_{i=1}^{n} |x_i - \hat{x}|$$



Optimisation Problems – Mean Absolute Error

$$L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$

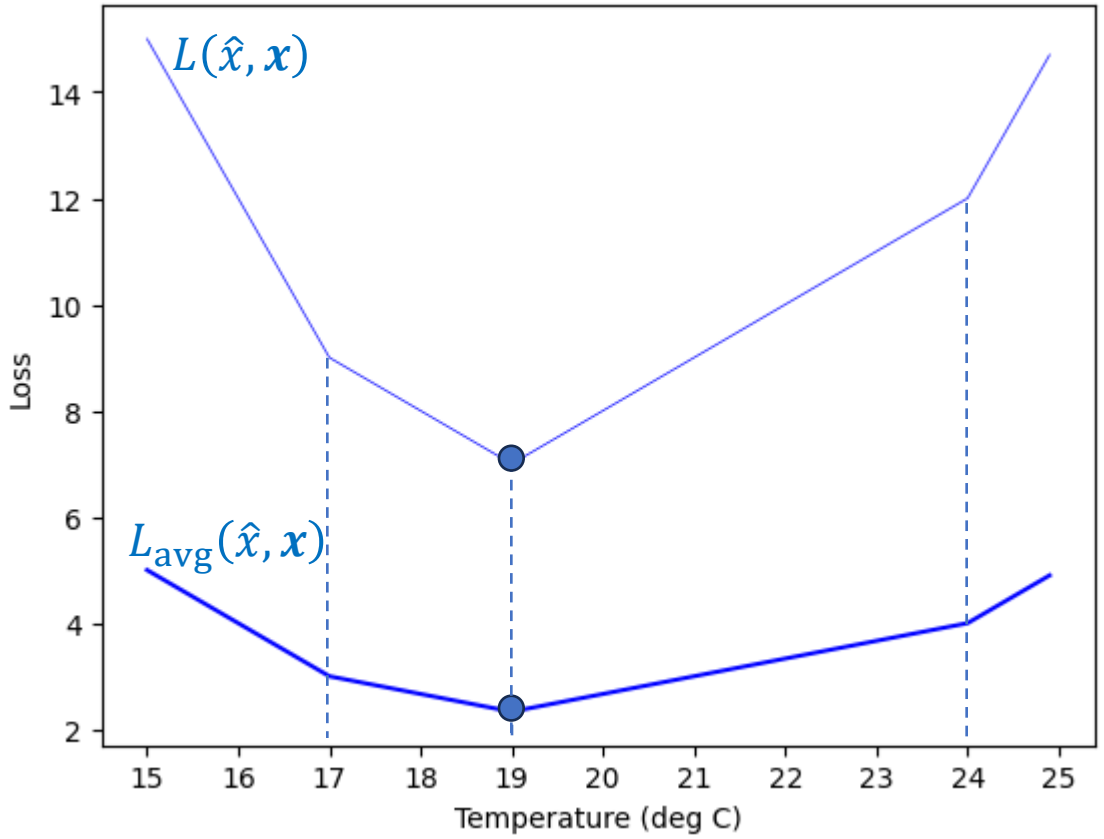
Question:

What will happen if we use the mean instead of sum?

$$L_{\text{avg}}(\hat{x}, \boldsymbol{x}) = \frac{1}{n} \sum_{i=1}^{n} |x_i - \hat{x}|$$

Answer:

Just scaling; optimum will stay the same!



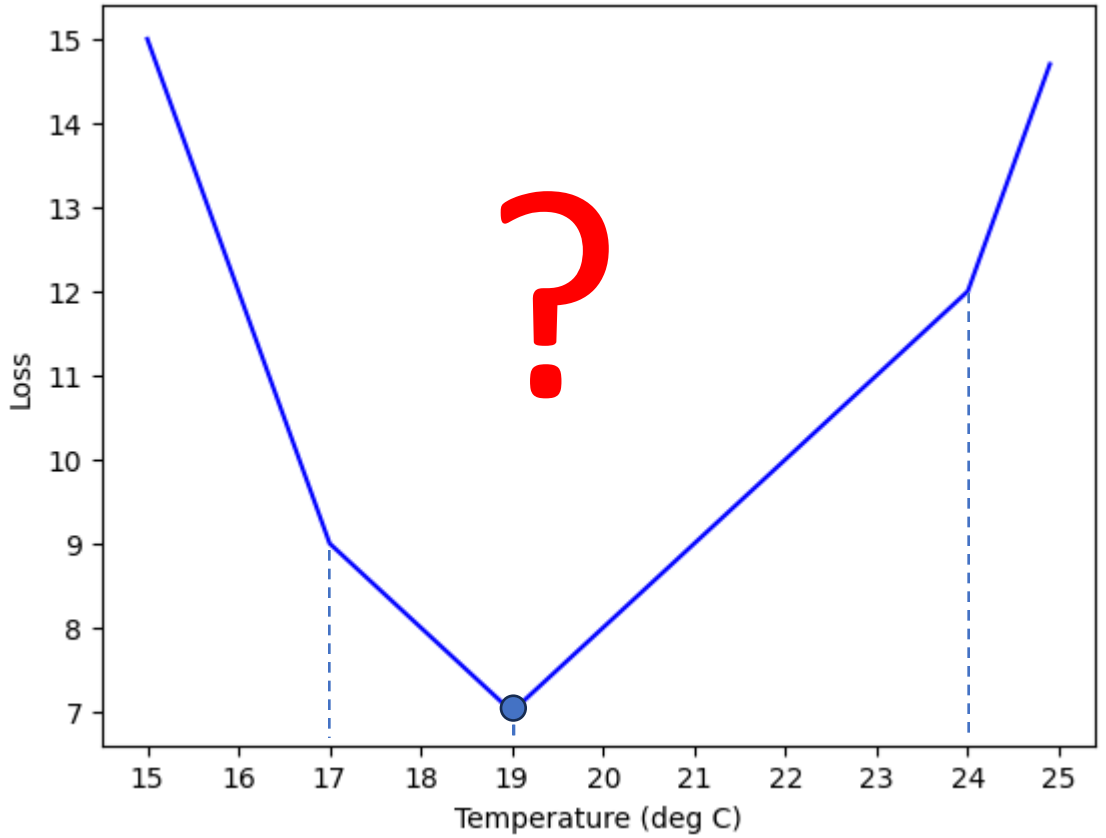
Optimisation Problems – Mean Absolute Error

$$L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$

Question:

What will the logarithm change the optimum?

$$L_{\log}(\hat{x}, \boldsymbol{x}) = \log_{\mathbf{10}} \sum_{i=1}^{n} |x_i - \hat{x}|$$



Optimisation Problems

$$L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} |x_i - \hat{x}|$$

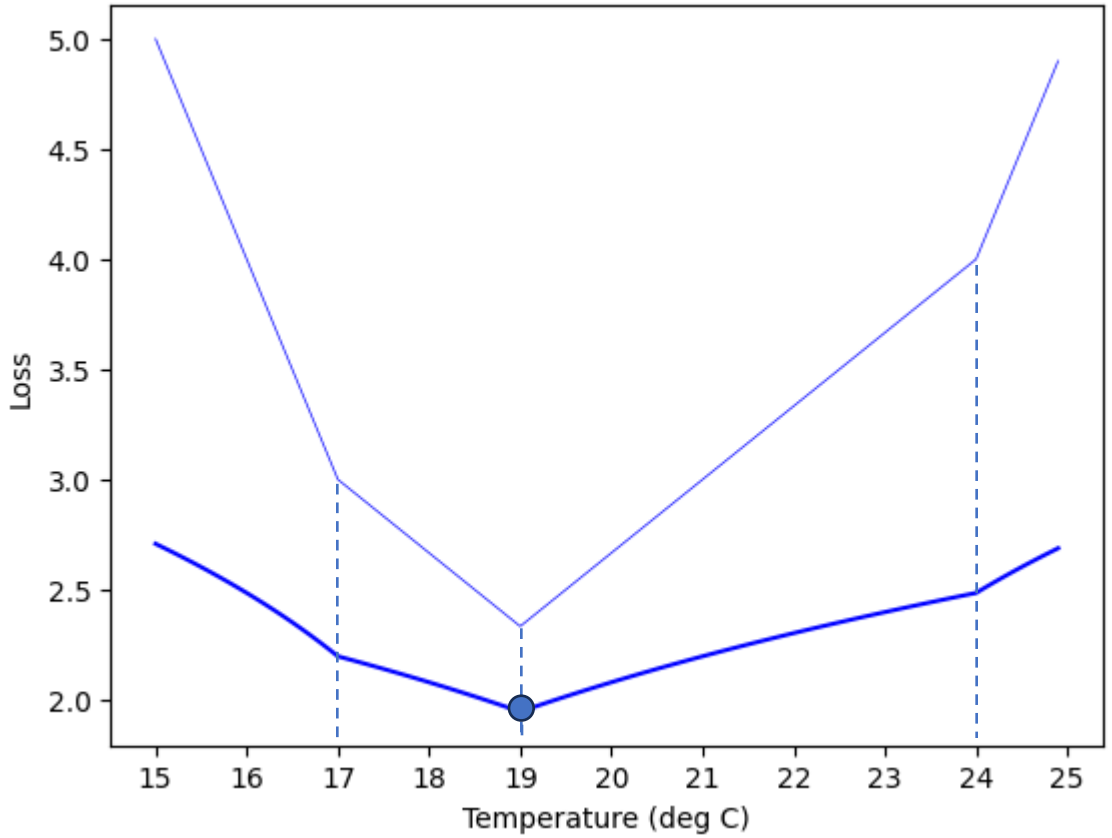
Question:

What will the logarithm change the optimum?

$$L_{\log}(\hat{x}, \boldsymbol{x}) = \ln \sum_{i=1}^{n} |x_i - \hat{x}|$$

Answer:

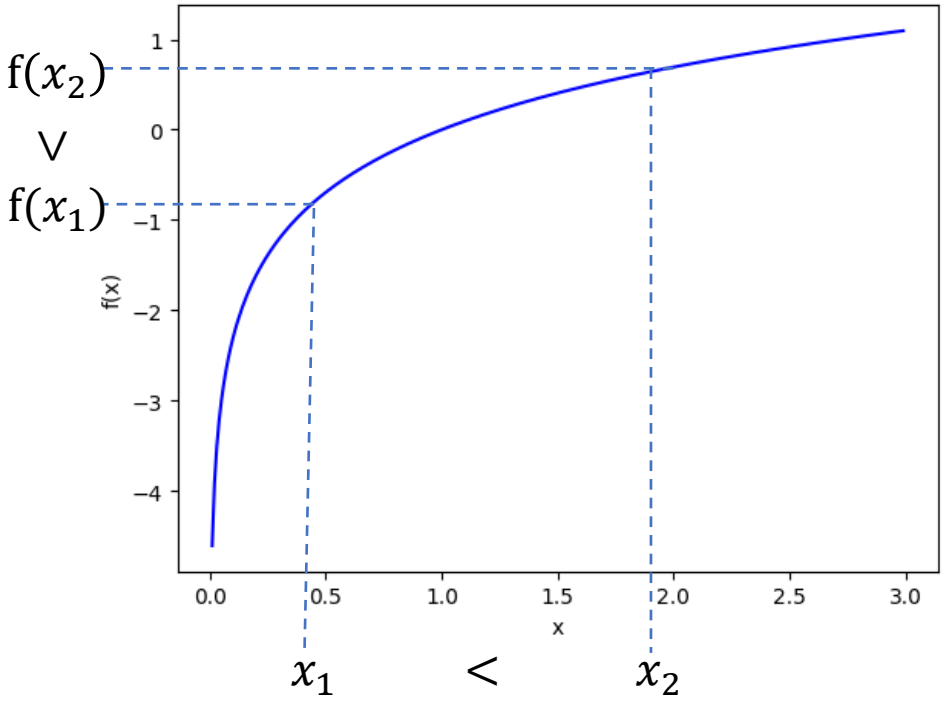
Again, does not change the optimum!



Monotonically Increasing Functions

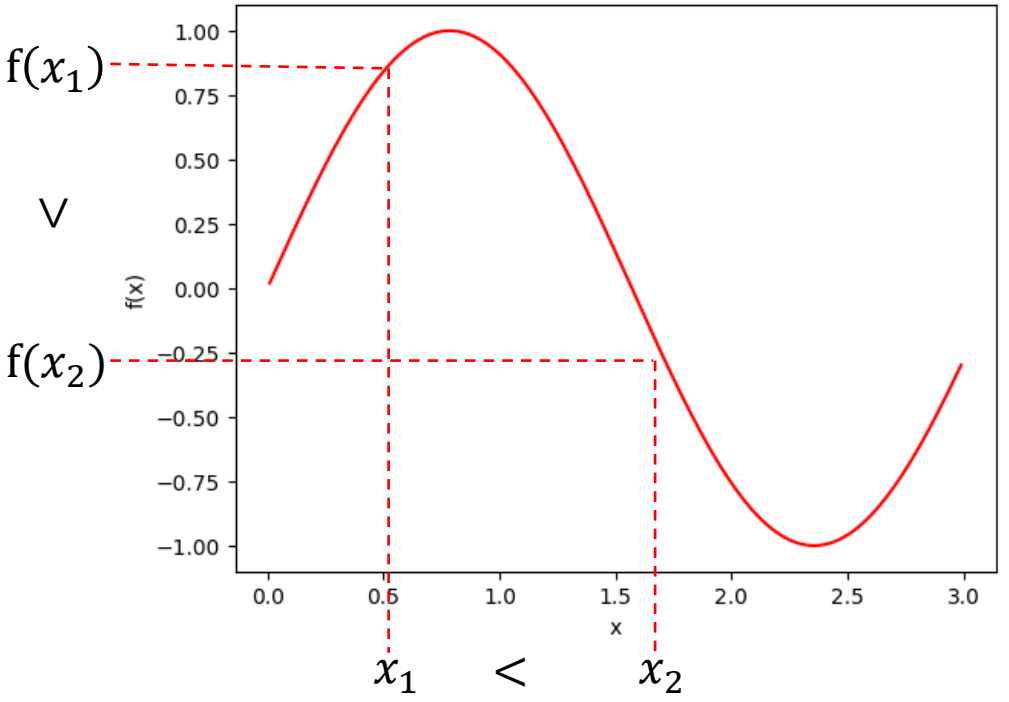
If $x_1 < x_2$ then $\mathrm{f}(x_1) < \mathrm{f}(x_2)$

$$\underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x}) = \underset{\hat{x}}{\operatorname{argmin}}\, \mathrm{f}(L(\hat{x}, \boldsymbol{x}))$$



Counterexample:

- not monotonically increasing



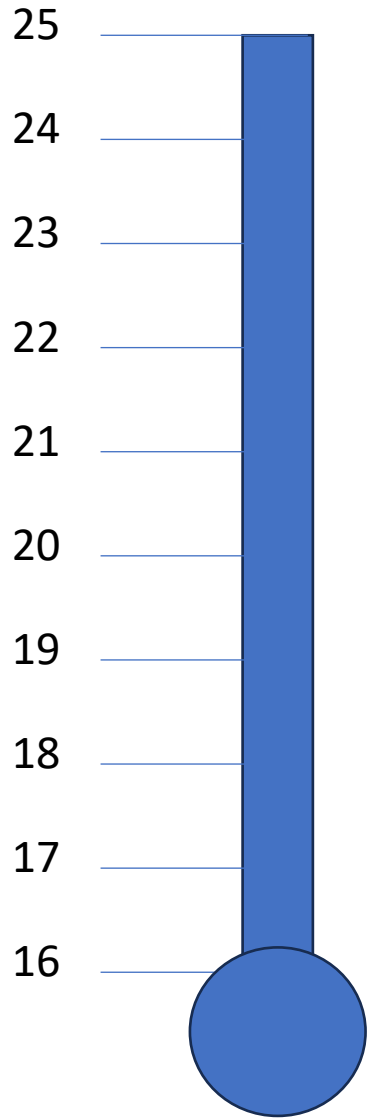
Maximum Likelihood

- Our measurements are noisy and follow a normal distribution:

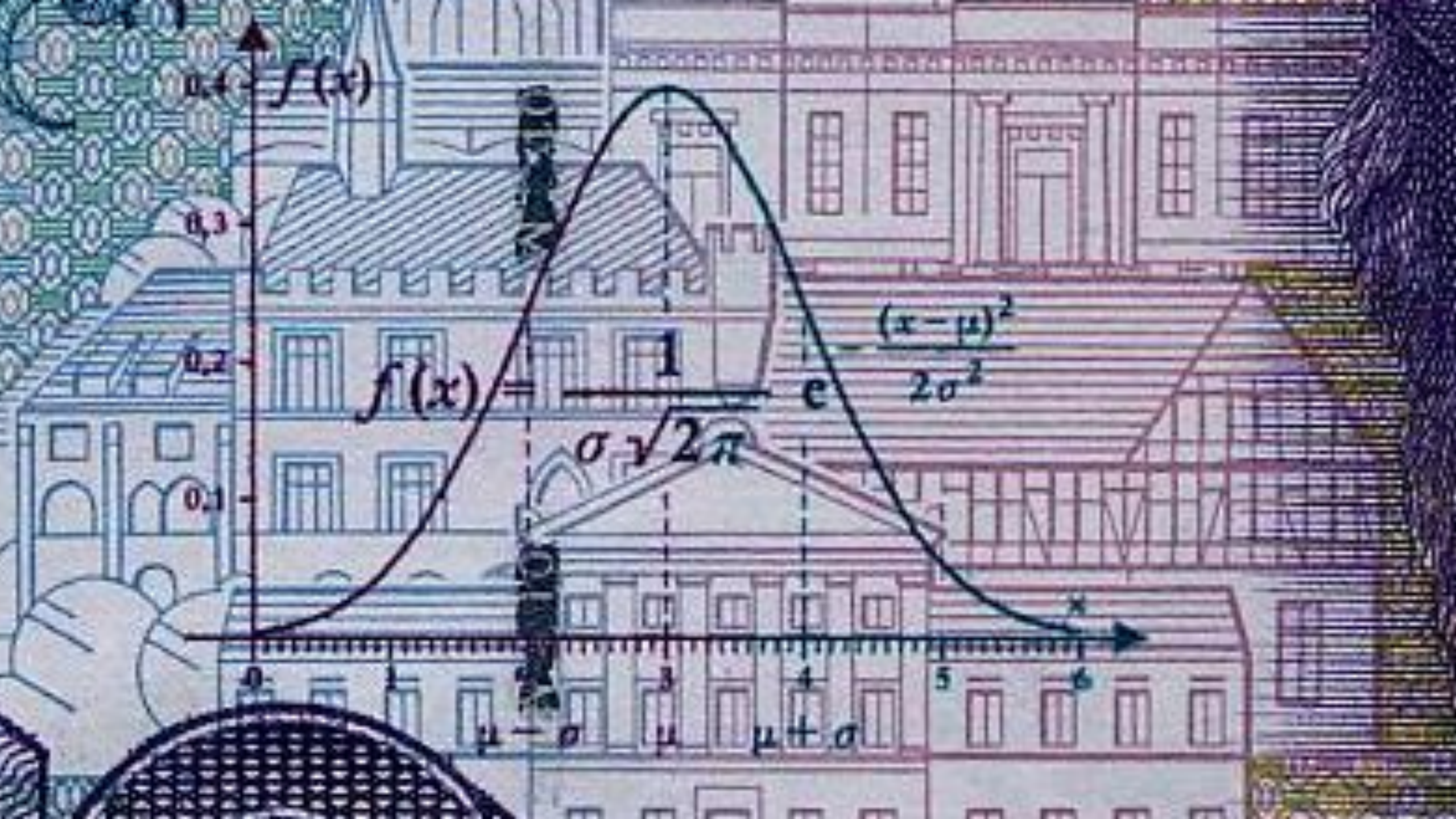
$$p(x_i; \mu, \sigma) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\mu - x_i)^2}{2\sigma^2}\right)$$

- Measurements are cond. Independent:

$$p(\boldsymbol{x}; \mu, \sigma) = \prod_{i=1}^{n} p(x_i; \mu, \sigma)$$




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Deutsche Bundesbank
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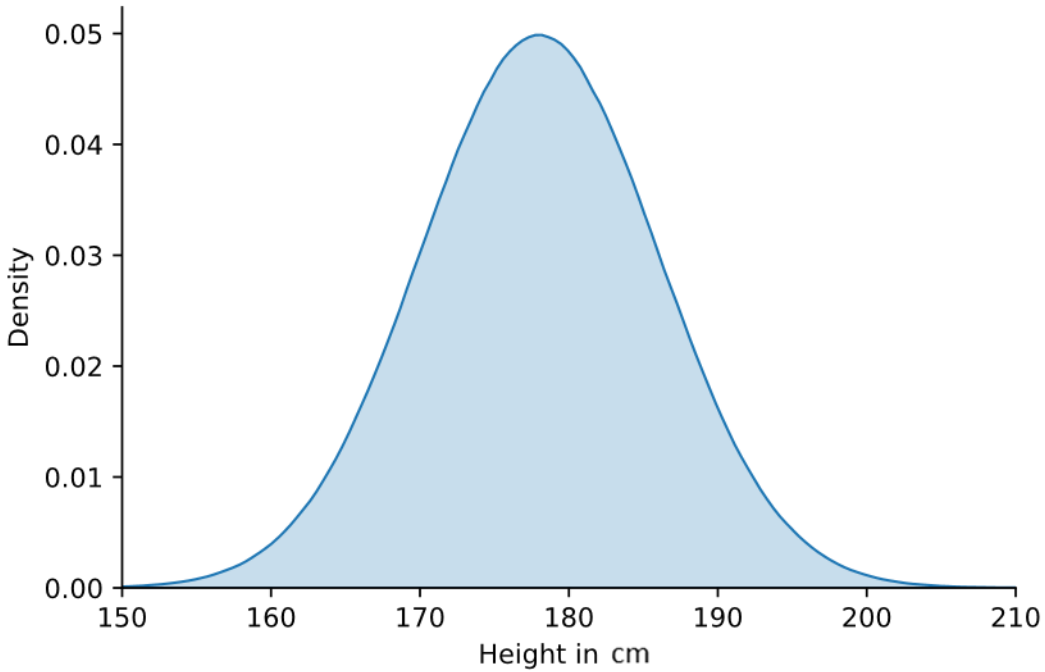

f(x)
0,4
0,3
0,2
0,1
0
1
3
4
5
6
x
$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
$\mu - \sigma$
μ
$\mu + \sigma$

The Normal/Gaussian Distribution

- Many things are approx. normal distributed, e.g. height.
- Go-to-distribution for many modelling applications

Question:

Read what is the probability of being 180cm tall?



Probability Density Function (PDF)

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mean

Standard deviation

Variance

The Normal/Gaussian Distribution

- Many things are approx. normal distributed, e.g. height.

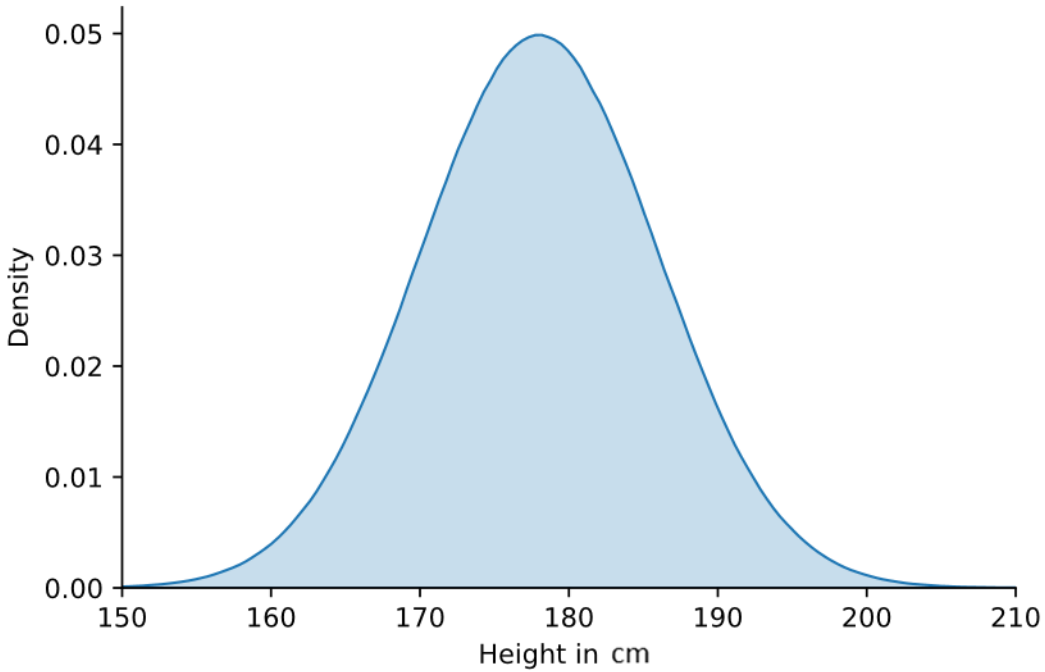
- Go-to-distribution for many modelling applications

Question:

Read what is the probability of being 180cm tall?

Answer:

The probability is zero. The probabilities only make sense for intervals.



Probability Density Function (PDF)

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mean

Standard deviation

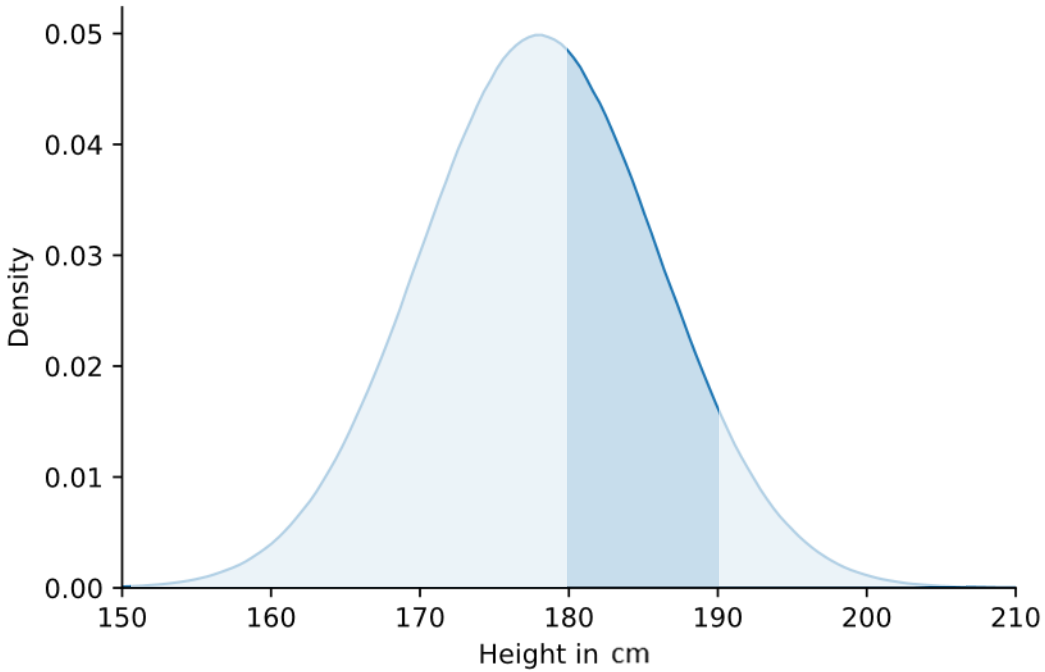
Variance

The Normal/Gaussian Distribution

- $\int_{-\infty}^{\infty} f(x)\,dx = 1$

- Large values -> large sample density
- Completely describes distribution

- $\int_{a}^{b} f(x)\,dx$ is probability $a \leq x_i \leq b$



Probability Density Function (PDF)

$$f(x;\mu,\sigma) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Mean

Standard deviation

Variance

Maximum Likelihood

- Our measurements are noisy and follow a normal distribution:

True temp.

$$p(x_i|\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(\mu - x_i)^2}{2\sigma^2}\right)$$

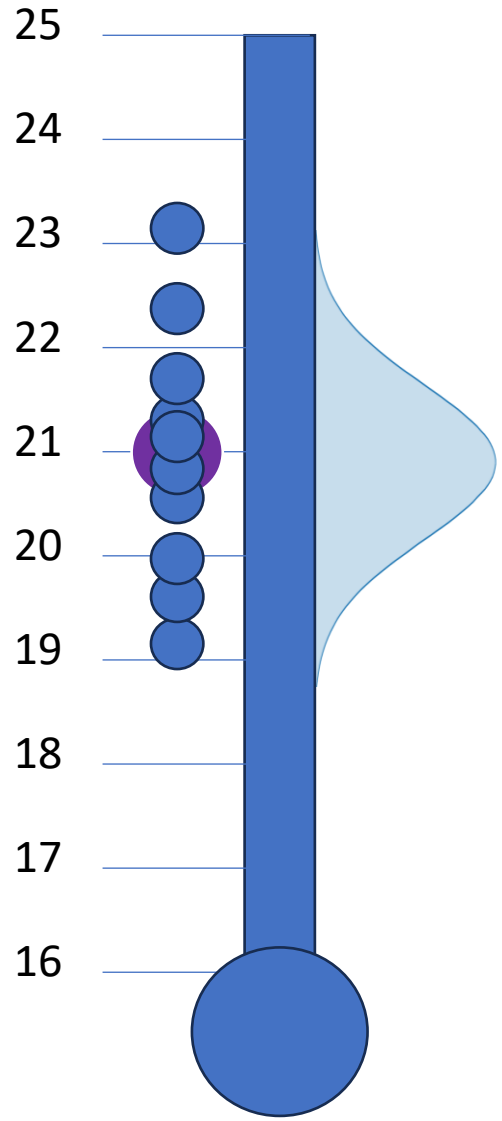
Amount of Noise

- Measurements are cond. Independent:

$$p(\boldsymbol{x}|\mu,\sigma) = \prod_{i=1}^{n} p(x_i|\mu,\sigma)$$

$\mu = 21$

$\sigma = 1$



Maximum Likelihood

- Our measurements are noisy and follow a normal distribution:

True temp.

$$p(x_i|\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(\mu - x_i)^2}{2\sigma^2}\right)$$

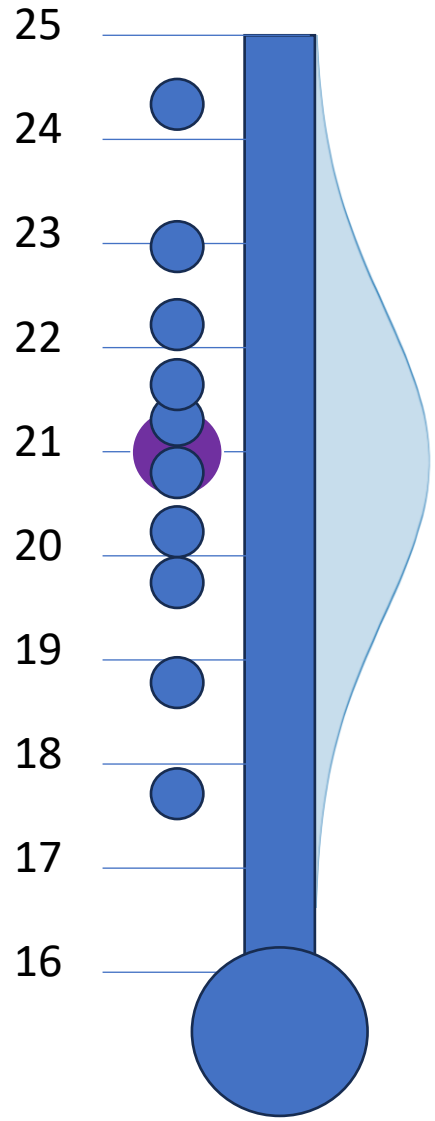
Amount of Noise

- Measurements are cond. Independent:

$$p(\boldsymbol{x}|\mu,\sigma) = \prod_{i=1}^{n} p(x_i|\mu,\sigma)$$

$\mu = 21$

$\sigma = 2$



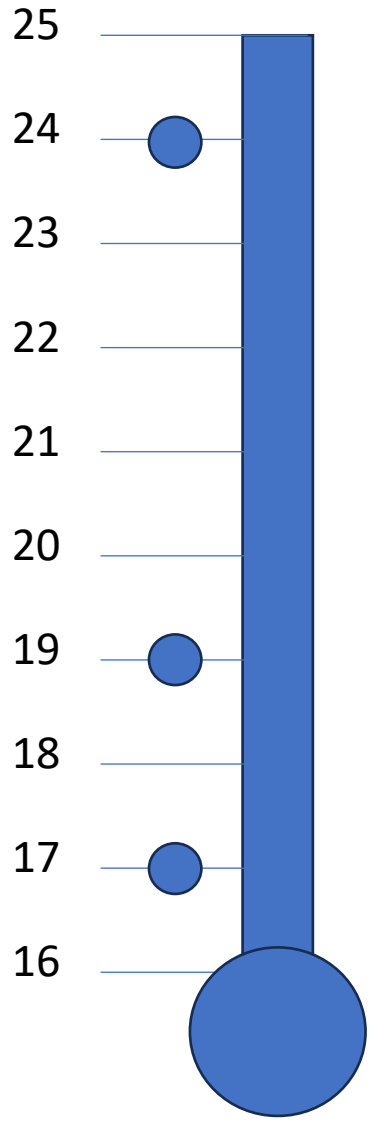
Maximum Likelihood

$$p(x_i|\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma^2}}\exp\left(-\frac{(\mu - x_i)^2}{2\sigma^2}\right)$$

$$p(\boldsymbol{x}|\mu,\sigma) = \prod_{i=1}^{n} p(x_i|\mu,\sigma)$$

What is true temp. most likely to produce measurements? I.e., highest probability density.

$$x^* = \underset{\hat{x}}{\operatorname{argmax}}\, p(\boldsymbol{x}|\mu = \hat{x},\sigma) \qquad x_1 = 19, x_2 = 17, x_3 = 24$$



Maximum Likelihood (optional derivation)

$$x^* = \underset{\hat{x}}{\operatorname{argmax}}\, p(\boldsymbol{x}|\mu = \hat{x}, \sigma)$$

$$= \underset{\hat{x}}{\operatorname{argmax}} \prod_{i=1}^{n} p(x_i|\mu = \hat{x}, \sigma)$$

$$= \underset{\hat{x}}{\operatorname{argmax}} \ln\left(\prod_{i=1}^{n} p(x_i|\mu = \hat{x}, \sigma)\right)$$

$$= \underset{\hat{x}}{\operatorname{argmax}} \sum_{i=1}^{n} \ln p(x_i|\mu = \hat{x}, \sigma)$$

$$= \underset{\hat{x}}{\operatorname{argmax}} \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(\hat{x} - x_i)^2}{2\sigma^2}\right)\right)$$

$$= \underset{\hat{x}}{\operatorname{argmax}} \sum_{i=1}^{n} \ln\left(\frac{1}{\sqrt{2\pi\sigma^2}}\right) - \frac{(\hat{x} - x_i)^2}{2\sigma^2}$$

$$= \underset{\hat{x}}{\operatorname{argmax}} \sum_{i=1}^{n} -\frac{(\hat{x} - x_i)^2}{2\sigma^2}$$

$$= \underset{\hat{x}}{\operatorname{argmax}} \sum_{i=1}^{n} -(\hat{x} - x_i)^2 \quad = \underset{\hat{x}}{\operatorname{argmin}} \sum_{i=1}^{n} (\hat{x} - x_i)^2$$

Optimisation Problems – Quadratic Error

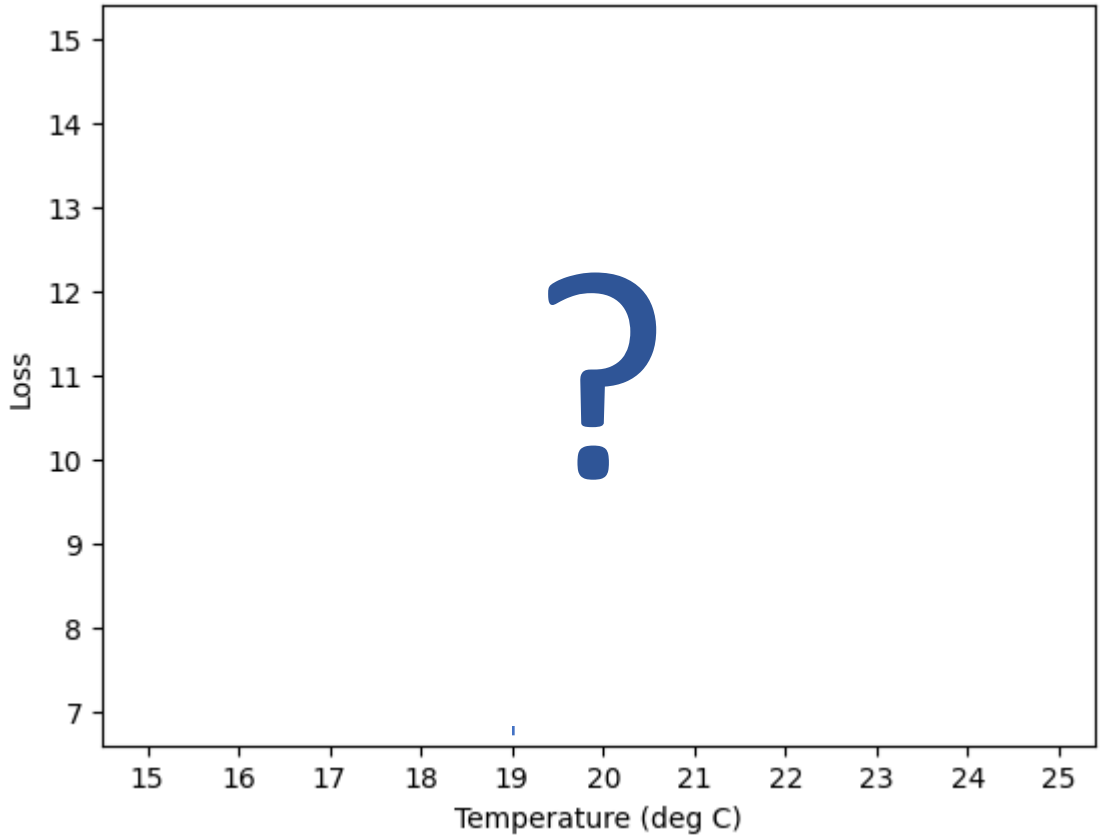
- Find estimate of true temperature
- Measurements:

$$x_1 = 19$$

$$\boldsymbol{x} = (x_1)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x}) \qquad L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} (x_i - \hat{x})^2$$



Optimisation Problems – Quadratic Error

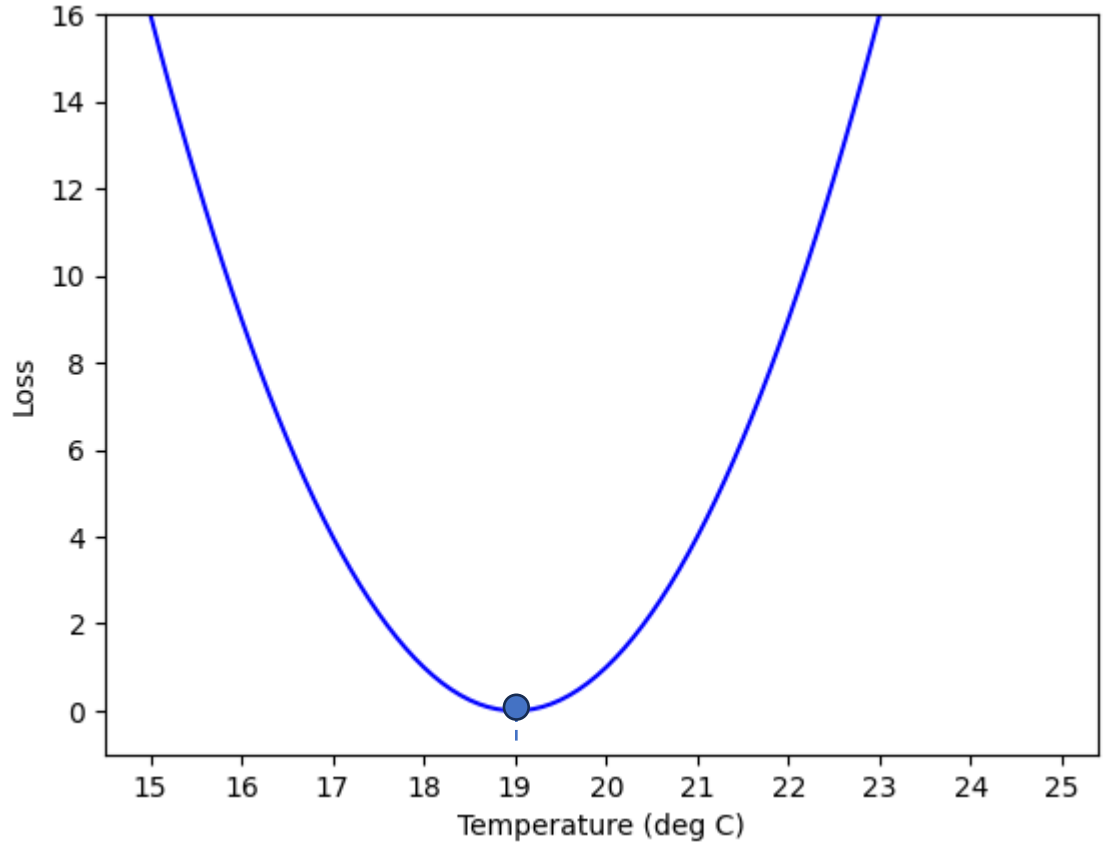
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Optimisation Problems – Quadratic Error

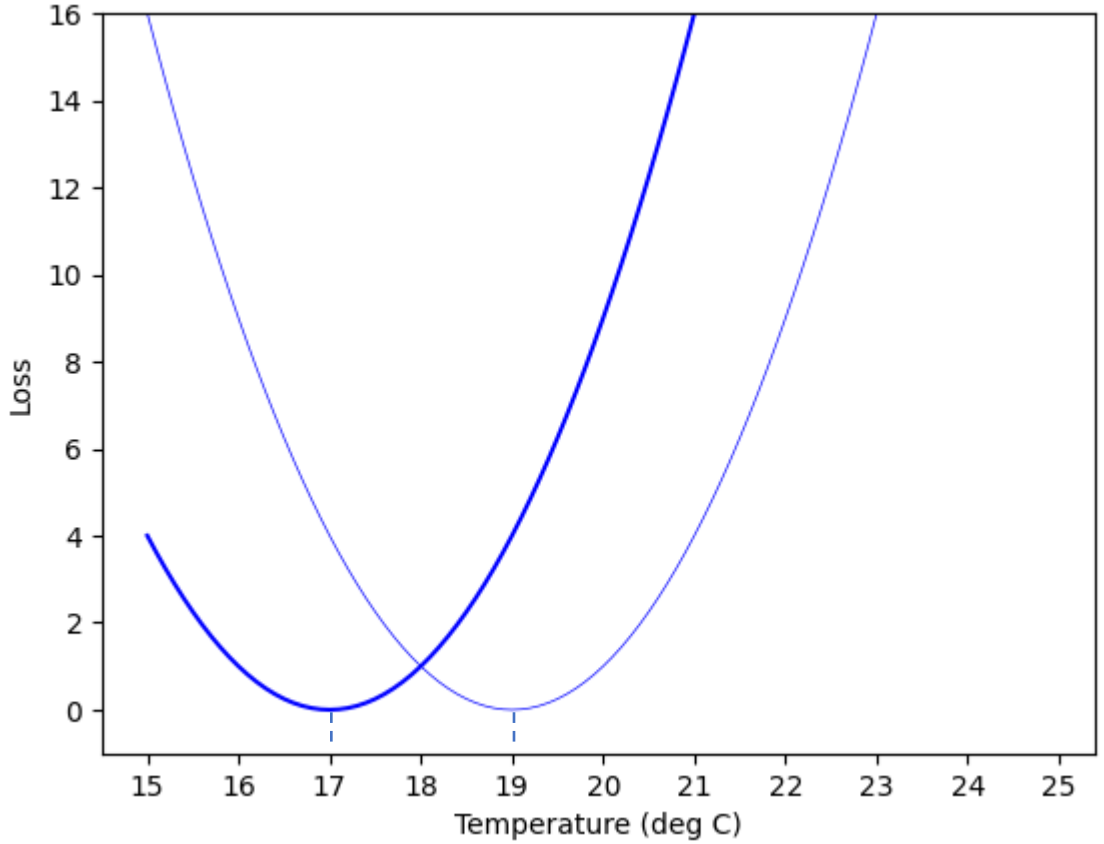
- Find estimate of true temperature
- Measurements:

$$x_1 = 19, x_2 = 17$$

$$\boldsymbol{x} = (x_1, x_2)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x}) \qquad L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} (x_i - \hat{x})^2$$



Optimisation Problems – Quadratic Error

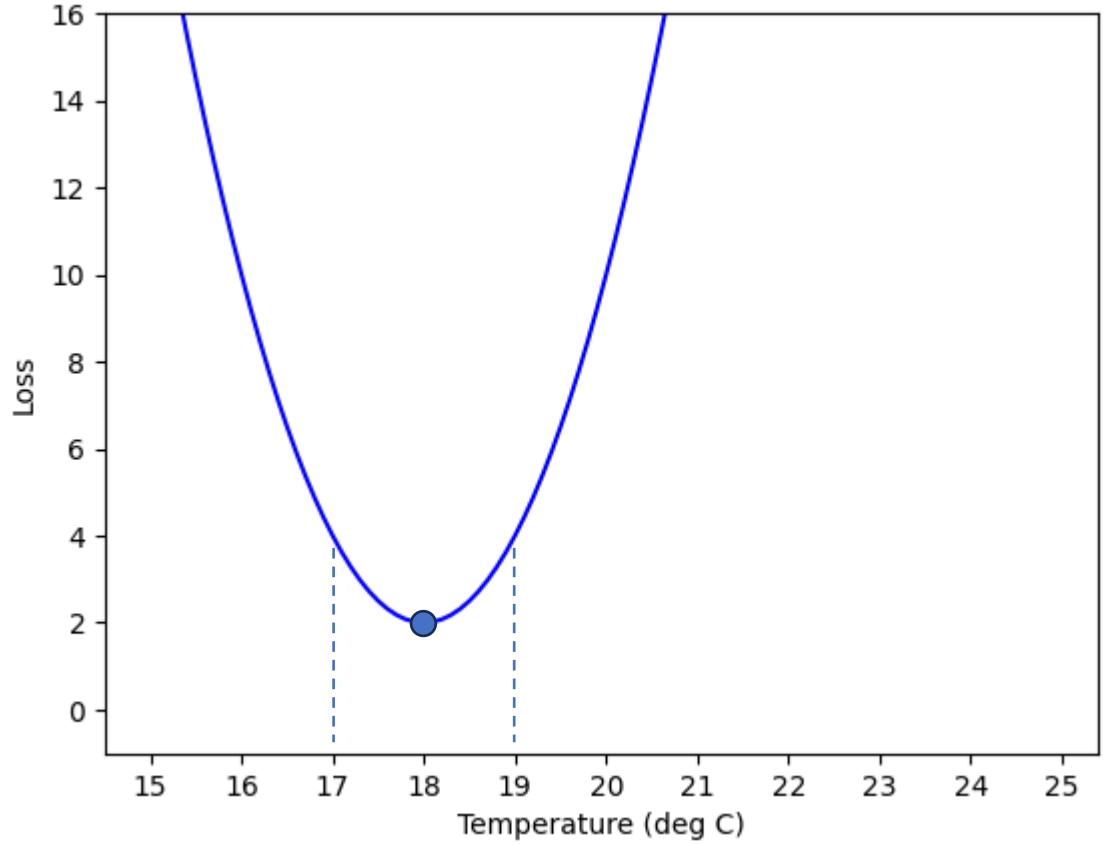
- Find estimate of true temperature
- Measurements:

$$x_1 = 19, x_2 = 17$$

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Optimisation Problems – Quadratic Error

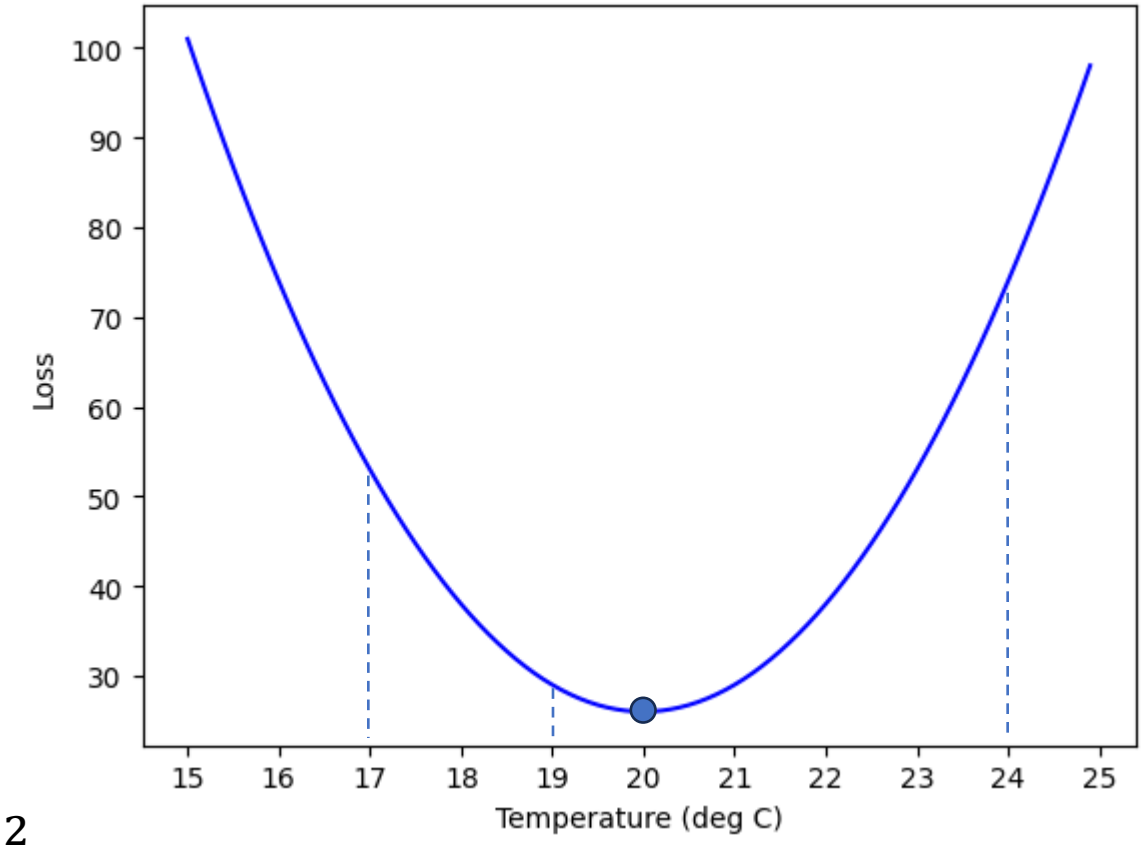
- Find estimate of true temperature
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$$x_1 = 19, x_2 = 17, x_3 = 24$$

$$\boldsymbol{x} = (x_1, x_2, x_3)$$

- Strategy: find estimate that minimises loss.

$$x^* = \underset{\hat{x}}{\operatorname{argmin}}\, L(\hat{x}, \boldsymbol{x}) \qquad L(\hat{x}, \boldsymbol{x}) = \sum_{i=1}^{n} (x_i - \hat{x})^2$$

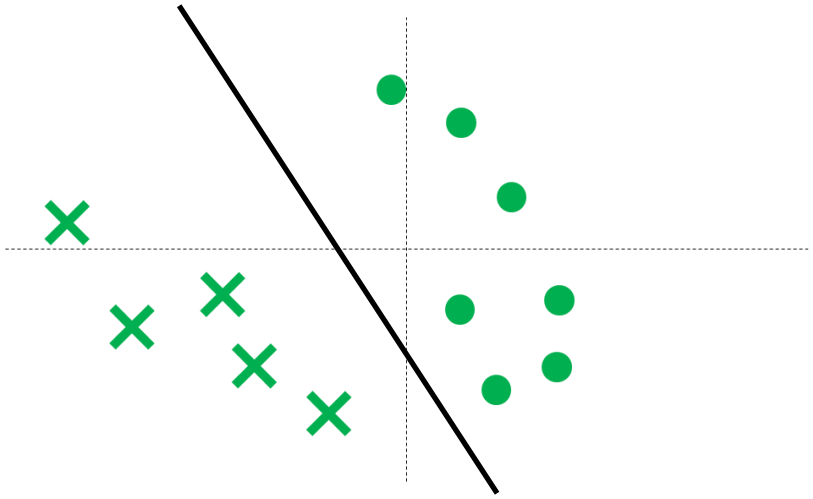


Summary

- Estimation problems can be framed by:
  - Defining a loss function quantifying the quality of parameter
  - Finding the argmax
- Argmax/argmin is not changed by
  - Adding a constant
  - Multiplying a positive constant
  - Applying any monotonically increasing function

- We can use Maximum Likelihood to derive loss functions
  - Finding the parameter most likely to produce observed data.

Next Week - Learning



- How do we find w and b?
  - Adjusting parameters from data to solve task
  - Called supervised learning/training (week 2)
  - For regression and classification

